

**VERBAL & NON VERBAL**

***Reasoning***

# **Complete Handwritten Notes**

**USEFUL FOR**

SSC CGL, CHSL, CPO, MTS, RRB, DSSSB,  
SSC, & Other Competitive Exam



**By GAJANAND KUMAWAT**

## Content

1. Calender
2. Clock
3. Dice
4. Cube & Cuboid
5. Direction
6. Syllogism
7. Ranking order
8. Sitting Arrangement
9. Symbols & Notations
10. Series
11. Missing Number
12. Classification
12. Coding - Decoding
14. Ven Diagrams
15. Blood Relation
16. Counting figure
17. Analogy & Similarity
18. Arithmetical Problem
19. Arrangement of Words in Logical Order
20. Word Formation
21. Analogy and Similarity
22. Mirror Image
23. Water Image
24. Completion of Figure
25. Paper Cutting & Folding

# 1. CALENDER

## Types of Years

| Normal Year (N.Y)                                                | leap Year (L.Y)                                     |
|------------------------------------------------------------------|-----------------------------------------------------|
| (i) 1 Year = 365 days                                            | 1 Year = 366 days                                   |
| (ii) 1 Year = 52 week 1 Day<br>Feb = 28 days                     | 1 Year = 52 week 2 Days<br>Feb = 29 days            |
| How to check:-                                                   |                                                     |
| (iii) IF $\frac{\text{Year}}{4} = \text{Remainder}^{\checkmark}$ | $\frac{\text{Year}}{4} = \text{Remainder}^{\times}$ |
| Ex. $\frac{1981}{4} = \text{Rem.} = 1$                           | Ex. $\frac{1680}{4} = \text{Rem.} = 0$              |
| Ex. 1982, 1186,                                                  | Ex. 1988, 1984, 2012                                |
| (iv) IF $\frac{\text{Century}}{400} = \text{Rem.}^{\checkmark}$  | $\frac{\text{Century}}{400} = \text{Rem.}^{\times}$ |
| Ex. $\frac{1700}{400} = \text{Rem.} = 1$                         | Ex. $\frac{1600}{400} = \text{Rem.} = 0$            |
| Ex. 2100, 1800, $\rightarrow$ Normal Year                        | Ex. 2000, 1200 $\rightarrow$ leap Year              |

## Extra Days / Odd days:-

Most of questions from this topic depends on Extra days.

(a) Extra days in days :-

$$\frac{\text{days}}{7} = \text{Remaining days are extra days.}$$

Ex.  $\frac{77 \text{ days}}{7} = 0 \text{ (Rem)} \quad \text{so Extra/odd days} = 0$

Ex.  $\frac{67 \text{ days}}{7} = 4 \text{ (Rem)} \quad \text{so Extra days} = 4$

(b) Extra days in months :-

$$28 \text{ days} = 0$$

$$29 \text{ days} = 1$$

$$30 \text{ days} = 2$$

$$31 \text{ days} = 3$$

(c) Extra days in Years :-

$$\text{Odd days in a Normal year} = 1$$

$$\text{Odd days in a Leap Year} = 2$$

$$\text{Formula : } \frac{\text{Year} + \text{no. of Leap Year in these Year}}{7} = \text{Remainder}$$

Ex Find Odd days in 10 Years.

$$\Rightarrow \text{Leap Year in 10 Years} = \frac{10}{4} = 2 \text{ (Quotient)}$$

$$\Rightarrow \frac{10 + 2}{7} = \frac{12}{7} = 5 \text{ (Rem.)}$$

$$\text{so odd days} = 5 \text{ Ans}$$

Ex. Find odd days in 99 Years

$$\Rightarrow \text{leap year} = \frac{99}{4} = 24 \text{ (Quotient)}$$

$$\text{Odd days :- } \frac{99 + 24}{7} = \frac{123}{7} = \underline{\underline{4}} \text{ (Rem.)}$$

Odd days in 100 Years:

$$\begin{array}{c} \text{99 Year} + \text{100th Year (Normal Year)} \\ \downarrow \quad \quad \downarrow \\ \text{odd days} = 4 + 1 = 5 \text{ days} \end{array}$$

Odd days in 200 Years:

$$\begin{array}{c} \text{100 Year} \longrightarrow 5 \text{ days} \\ \downarrow \\ \text{200 Year} \longrightarrow \underline{10 \text{ days}} \Rightarrow \frac{10}{7} = \underline{\underline{3}} \text{ (Rem)} \end{array}$$

odd days in 300 Years

$$\begin{array}{c} \text{200 Years} + \text{99 Years} + \text{300th (Normal Year)} \\ \downarrow \quad \quad \downarrow \quad \quad \downarrow \\ 3 + 4 + 1 = 8 \text{ days} \\ = \frac{8}{7} = \underline{\underline{1}} \text{ (Rem)} \\ = \text{odd day} \end{array}$$

odd days in 400 Years:

$$\begin{array}{c} \text{300 Years} + \text{99 Years} + \text{400th Year (leap Year)} \\ \downarrow \quad \quad \downarrow \quad \quad \downarrow \\ 1 + 4 + 2 = 7 \\ \text{hence} = \frac{7}{7} = 0 \text{ odd day} \end{array}$$

Note [Every multiple of 400 have odd days = 0]

Now we can find odd days in any no. of years

Ex. odd days in 1700 Years

$$\Rightarrow 1600 + 100$$

$$\downarrow \quad \downarrow$$

$$0 + 5 = 5 \text{ days}$$

Ex odd days in 2100 Years

$$\Rightarrow 2000 + 100$$

$$\downarrow \quad \downarrow$$

$$0 + 5 = 5 \text{ days}$$

Ex. Odd days in 1322 Years

$$\Rightarrow 1200 + 100 + 22$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$0 + 5 + 6 = 11 \text{ days}$$

$$\text{Leap Year } \frac{22}{4} = 5 \text{ (Quotient)}$$

$\rightarrow \text{odd} = 4 \text{ days}$

$$\text{odd days} = \frac{22 + 5}{7} = \frac{27}{7} = 6 \text{ (Rem)}$$

Type 1:-

|      | Normal Year                      |      | Leap Year                          |
|------|----------------------------------|------|------------------------------------|
| IF   | 1 Jan 2001 $\rightarrow$ Monday  |      | 1 Jan 2004 $\rightarrow$ Mond.     |
| then | 31 Dec 2001 $\rightarrow$ Mond.  | } +1 | 31 Dec. 2004 $\rightarrow$ Tuesday |
|      | 1 Jan 2002 $\rightarrow$ Tuesday |      | 1 Jan 2005 $\rightarrow$ wednesday |

Ex. 15 July 2006  $\rightarrow$  Monday

then 15 July 2007  $\rightarrow$  ?

$\downarrow +1$   
Tuesday

Ex. 26 Jan 2008  $\rightarrow$  Monday

26 Jan 2009  $\rightarrow$  ?

$\downarrow +2$   
wednesday

(Leap Year)  
 $\frac{2008}{4} = \text{Rem}^x$

Ex. 15 Aug. 2003  $\rightarrow$  Monday

15 Aug 2004  $\rightarrow$  ?

$\downarrow +2$   
wednesday

(Leap Year 2004)  
29 Feb is  
Include

Ex. 15 Feb 2007  $\rightarrow$  Monday  $+1$  (29 Feb. 2008 is not include)  
 15 Feb 2008  $\rightarrow$  ? Tuesday

Note:- If date is same, month is same, and we cross (28 February)  $\rightarrow +1$

If 29 February  $\rightarrow +2$

Ex. The Republic day in 1996 was celebrated on Wednesday  
 On what day was it celebrated in the year 1997?

$\Rightarrow$  1996  $\rightarrow$  Wed.  $+2$  (we cross 29 Feb 1996)  
 1997  $\rightarrow$  Friday

## Type 2 Difference of Days :-

Ex. 10 Jan 2001  $\rightarrow$  Monday  $+1$   
 then 15 March 2001  $\rightarrow$  ? Tuesday Ans

Soln  
 Jan  $\rightarrow 21$   
 Feb  $\rightarrow 28$   
 March  $\rightarrow 15$   
 $\hline 64$   
 $\frac{64}{7} = 1 \text{ (Rem)}$

Ex. 15 Feb 2005  $\rightarrow$  Monday  
 5 May 2005  $\rightarrow$  ?  $+2$   
 Wednesday Ans

Soln  
 Feb  $\rightarrow 13$   
 March  $\rightarrow 31$   
 April  $\rightarrow 30$   
 May  $\rightarrow 5$   
 $\hline + 79$   
 $\frac{79}{7} = 2 \text{ (rem)}$

★ we take

|        |   |   |
|--------|---|---|
| 28 Day | → | 0 |
| 29 Day | → | 1 |
| 30 Day | → | 2 |
| 31 Day | → | 3 |

Ex. 26 March 2001 → Monday  
 5 Oct 2001 → ?  $\downarrow +4$

⇒ march → 5  
 April<sub>(30)</sub> → 2  
 May<sub>(31)</sub> → 3  
 June<sub>(30)</sub> → 2  
 July → 3  
 Aug. → 3  
 Sept → 2  
 Oct → 5  
25

Friday one

$\frac{25}{7} = 4 \text{ (rem)}$

Ex. 20 April 2004 → Monday  
 15 Nov 2004 → ?  $\downarrow +6$  OR  $(-1)$   
 Sunday

Soln  
 (30) April →  $\frac{10}{7} \rightarrow 3$   
 (31) May → 3  
 Jun → 2  
 July → 3  
 Aug. → 3  
 Sept → 2  
 Oct → 3  
 Nov →  $\frac{15}{7} \rightarrow 1$   
20

$\frac{20}{7} = +6 \text{ (rem)}$   
 OR  
 $\frac{20}{7} = -1 \text{ (Rem)}$

Ex.★ 4 July 2005 → Monday  
 25 March 2005 → ?

Diagram: A horizontal line from Monday to a point. From this point, a vertical line goes down to Friday, labeled -3. Another vertical line goes up to a point, labeled +4. A curved arrow from Monday to this upper point is labeled OR and (7-3)=4.

Soln:-  
 March → 6  
 April → 2  
 May → 3  
 Jun → 2  
 July → 4  
 17

$\frac{17}{7} = +3 \text{ (rem)}$

Ex 10 Oct 2004 → Monday  
 5 May 2004 → ?

Diagram: A horizontal line from Monday to a point. From this point, a vertical line goes down to Thursday, labeled -4. Another vertical line goes up to a point, labeled +3. A curved arrow from Monday to this upper point is labeled OR and (7-4)=3.

Soln May →  $\frac{26}{7} \rightarrow 5$   
 Jun → 2  
 July → 3  
 Aug → 3  
 Sept → 2  
 Oct →  $\frac{10}{7} \rightarrow 3$   
 18

$\frac{18}{7} = 4 \text{ (Rem)}$

Maths with Gajanand

### Type 3:- Difference of Years:-

Ex.1 10 March 2001  $\rightarrow$  Monday  
10 March 2010  $\rightarrow$  ?

+4  
Friday Ans

Soln  $(2010 - 2001) = 9$

Normal Year  
7

Leap Years  
2

$$7 \times 1 + 2 \times 2 = 7 + 4 = 11$$

$$\frac{11}{7} = 4 \text{ (rem.)}$$

Note:- [ For N.Y multiply by 1  
and For L.Y multiply by 2 ]

Ex.2 15 Aug. 1997  $\rightarrow$  Monday  
15 Aug 2013  $\rightarrow$  ?

+6  
Sunday Ans

Soln:-  $(2013 - 1997) = 16$

N.Y  
 $12 \times 1$

+  
L.Y  
 $4 \times 2$

$= \frac{20}{7} = 6 \text{ (rem)}$

★★  
Ex.3. 10 Jan 2002  $\rightarrow$  Monday  
15 March 2011  $\rightarrow$  ?

Soln:- Firstly Find for  
10 Jan 2011

Maths with Gajanand

10 Jan 2002 → Monday  
 10 Jan 2011 → Friday

$$(2011 - 2002) = 9$$

$$7 \times 1 + 2 \times 2 = 11$$

M.y      L.y

$$\frac{11}{7} = 4 \text{ (rem)}$$

According to  
 Type 3

Now 10 Jan 2011 → Friday  
 15 March 2011 → ?

According to  
 Type 2

Jan →  $\frac{21}{7} \rightarrow 0$

Feb → 0

March →  $\frac{15}{7} \rightarrow 1$   
1

Saturday

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**Type 4** To find the day on a Particular date

month Code :-

| Month | Jan | Feb | March | Apr. | May | Jun | July | Aug. | Sept | Oct | Nov | Dec |
|-------|-----|-----|-------|------|-----|-----|------|------|------|-----|-----|-----|
| Code  | 1   | 4   | 4     | 0    | 2   | 5   | 0    | 3    | 6    | 1   | 4   | 6   |
| L.y   | 0   | 3   | "     |      |     |     |      |      |      |     |     |     |

century Code

| Century | 1800 | 1900 | 2000 |
|---------|------|------|------|
| Code    | 2    | 0    | 6    |

Formula :-  $\left[ \frac{\text{Day} + \text{Month code} + \text{Century code} + \text{Year} + \text{L.y}}{7} \right]$

If Remainder is :-

|   |             |
|---|-------------|
| 0 | → Saturday  |
| 1 | → Sunday    |
| 2 | → Monday    |
| 3 | → Tuesday   |
| 4 | → Wednesday |
| 5 | → Thursday  |
| 6 | → Friday    |

Ex. 1:- What was the day of week on  
19 Sept. 2018

Soln:-  $\frac{\text{Day} + \text{Month code} + \text{Century code} + \text{Year} + \text{L.y}}{7}$

$$\Rightarrow \frac{19 + 6 + 6 + 18 + 4}{7}$$

$$\Rightarrow \frac{53}{7} = 4(\text{rem})$$

└→ wednesday

Ans

|            |                                |
|------------|--------------------------------|
| Month code | → 6                            |
| (Sept)     |                                |
| Century    | → 6                            |
| Code       |                                |
| (2000)     |                                |
| Year       | → 18                           |
| L.y        | → $\frac{18}{4} \Rightarrow 4$ |

Ex. 10 July 2000 → ? day

Soln  $\frac{10 + 0 + 6 + 0 + 0}{7}$

$$= \frac{16}{7} = 2(\text{rem})$$

└→ monday

|              |     |
|--------------|-----|
| month code   | → 0 |
| (July)       |     |
| Century Code | → 6 |
| (2000)       |     |

Ex 26 Jan 2004 → ? day

Soln  $\frac{26 + 0 + 6 + 4 + 1}{7}$

$$= \frac{37}{7} = 2(\text{rem})$$

└→ monday

Maths with Gajanand

### Type 5 Same month :-

In month code table which have same code will be same month.

0  $\rightarrow$  April, July  $\left\{ \begin{smallmatrix} \text{It} \\ \text{will be same} \end{smallmatrix} \right\}$

4  $\rightarrow$  February, March, November

6  $\rightarrow$  Sept., Dec.

1  $\rightarrow$  Jan. October

### Type 6 Same calendar :-

IF Leap Year  $\longrightarrow +28$

Leap Year + 1 Year  $\longrightarrow +6$

other year  $\longrightarrow +11$

Ex.1 Calendar of 2004 could be used again ?

$\Rightarrow$  2004  $+28 \Rightarrow 2032$  Ans  
 $\searrow$  Leap Year so

Ex.2 Calendar of 1985 could be used again ?

(a) 1990      (b) 1991      (c) 1996      (d) 1992

$\Rightarrow$  1985  
 $\rightarrow$  (Leap year + 1)  $\Rightarrow +6 \Rightarrow 1991$

Ex.3 Calendar of 1991 could be used again ?

$\Rightarrow 1991 + 11 \Rightarrow 2002$

## Type 7 Century Day:-

⇒ First days of century years may be :-

only ⇒ Monday, Saturday, Thursday, Tuesday

⇒ Last days of century years may be :-

only ⇒ Monday, Wednesday, Friday, Sunday.

Ex. First day and last day of century years may be?

⇒ Monday

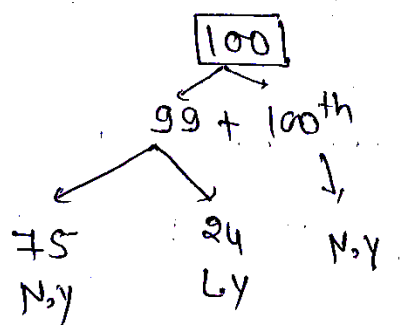
Ex. First day of century years may be?

(a) Friday (b) Wednesday (c) Sunday ~~(d) Saturday~~

### Example

Q.1 How many Leap Year in 100 Years?

Sol?



Leap Year

$$\frac{99}{4} = 24 \text{ (Quotient)}$$

So. N.Y =  $75 + 1 = 76$

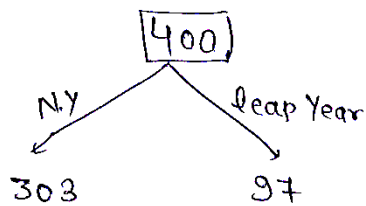
Leap Year = 24 Ans

Q.2 How many Leap years in 400 Years

OR

How much times 29 Feb comes in 400 Years?

Soln



OR

$$24 \times 4 = 96 + 1^{400th}$$

$$\text{Leap Year} = \underline{97} \quad \underline{\text{Ans}}$$

$$\begin{array}{l} \text{Leap Yr} \\ 99 \rightarrow 24 - 100^{th} \times \text{N.Y} \\ 99 \rightarrow 24 - 200^{th} \times \text{N.Y} \\ 99 \rightarrow 24 - 300^{th} \times \text{N.Y} \\ 99 \rightarrow 24 - 400^{th} \checkmark \text{LY} \\ \hline 96 \end{array}$$

Type.8:- Table for birthday, anniversary, Republic, Independence celebrated on same day  $\Rightarrow$

|                             | Leap Year | (L.Y+1)Yr | (L.Y+2)Yr | (L.Y+3)Yr |
|-----------------------------|-----------|-----------|-----------|-----------|
| Till 28 <sup>th</sup> Feb.  | + 5 Yr    | + 6 Yr    | + 6 Yr    | + 11 Yr   |
| After 28 <sup>th</sup> Feb. | + 6 Yr    | + 6 Yr    | + 11 Yr   | + 5 Yr    |

Ex Rohit birth day was on 16 January 1991 on Monday.

When would he celebrated his birth day again on Monday.

$$\Rightarrow 1991 \text{ is } (L.Y+3)Yr$$

$$\text{So add } +11Yr \Rightarrow 1991 + 11 = 2002 \text{ for}$$

Ex Independence day in 1984 was celebrated on Tuesday. In which Year was it celebrated on same day?

$$\Rightarrow 1984 + 6 \Rightarrow 1990 \quad \underline{\text{Ans}}$$

L.Y



# CLOCK

Maths with Gajanand

## 2. CLOCK

Ring :-

$$\left[ \frac{n(n+1)}{2} \right]$$

$$(1+2+3+\dots+11+12) = \frac{12 \times 13}{2} = 78$$

12 hr  $\longrightarrow$  78 times

1 Day  $\longrightarrow$   $78 \times 2 = 156$  times

Ex. 10 hr  $\longrightarrow \frac{n(n+1)}{2} = \frac{10 \times 11}{2} = 55$  times

Ex. In 15 hours



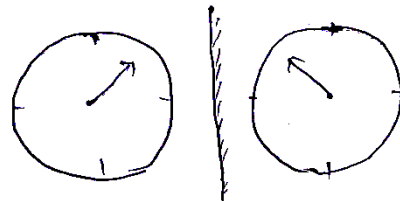
$$\frac{12 \times 13}{2} + \frac{3 \times 4}{2} \Rightarrow 78 + 6 = 84 \text{ times}$$

Mirror Image (M.I) :-

$\Rightarrow$  [Subtract from 11:60 OR 23:60 to know clock mirror image]

eg. mirror image of 2:25 = ?

$$\begin{array}{r} \Rightarrow \quad 11:60 \\ - \quad 2:25 \\ \hline 9:35 \quad \text{Ans} \end{array}$$



eg. m.i of 12:25 = ?

$$\begin{array}{r} 23:60 \\ - 12:25 \\ \hline 11:35 \quad \text{Ans} \end{array}$$

Eg. m.i of 11:20 = ?

Soln:-

$$\begin{array}{r} 11:60 \\ 11:20 \\ \hline 00:40 \end{array} \Rightarrow 12:40 \text{ Ans}$$

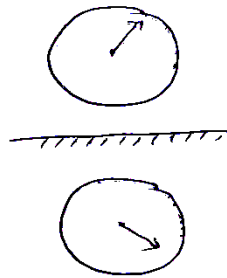
Water Image:- (W.I)

$\Rightarrow$  [Subtract from 6:30 OR 18:30]

Eg. w.i of 2:15 = ?

Soln:-

$$\begin{array}{r} 6:30 \\ - 2:15 \\ \hline 4:15 \text{ Ans} \end{array}$$



Eg. W.I of 8:45 = ?

Soln:-

$$\begin{array}{r} 18:30 \\ - 8:45 \\ \hline \end{array} \Rightarrow \begin{array}{r} 17:30 \\ - 8:45 \\ \hline 9:45 \text{ Ans} \end{array}$$

Eg. M.I and W.I of 2:15 = ?

Soln:- Mirror Image

$$\begin{array}{r} 11:60 \\ - 2:15 \\ \hline 9:45 \text{ Ans} \end{array}$$

Water Image

$$\begin{array}{r} 6:30 \\ - 2:15 \\ \hline 4:15 \text{ Ans} \end{array}$$

## Degree (डिग्री) :-

### (i) Minute Hand (मिनट की सुई) :-

$$\begin{aligned} 60 \text{ min.} &\longrightarrow 360^\circ \\ [1 \text{ min} &\longrightarrow 6^\circ] \end{aligned}$$

Eg. In 50 min  $\longrightarrow 50 \times 6^\circ$   
 $= 300^\circ$  Ans

### (ii) Hour Hand (घण्टे की सुई) :-

$$\begin{aligned} 12 \text{ hour} &\longrightarrow 360^\circ \\ [1 \text{ hour} &\longrightarrow 30^\circ] \\ [1 \text{ min.} &\longrightarrow \frac{1^\circ}{2}] \end{aligned}$$

Eg. From 2 P.M - 4 P.M  
 $\Rightarrow$   2 hr  $\longrightarrow 2 \times 30^\circ = 60^\circ$  Ans

Eg In 50 minute

$$\begin{aligned} \Rightarrow 1^\circ \text{ in } 1 \text{ min.} &\longrightarrow \frac{1^\circ}{2} \\ 50 \text{ min.} &\longrightarrow 50 \times \frac{1}{2} = 25^\circ \text{ Ans} \end{aligned}$$

Eg. 12:00 Noon to 3:00:45 min [SSC CGL-2015]

$\Rightarrow$   2 hr 45 min  $\Rightarrow 225 \text{ min}$

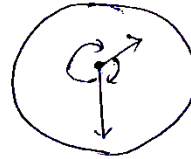
$$\begin{aligned} 1 \text{ min.} &\longrightarrow \frac{1^\circ}{2} \\ 225 \text{ min} &\longrightarrow 225 \times \frac{1}{2} = 112.5^\circ \text{ Ans} \end{aligned}$$

Imp. Angle between Both Hand :-

$$\text{Angle (कोण)} : \boxed{\text{min.} \times \frac{11}{2} - \text{hr} \times 30}$$

hr to min  $\rightarrow \oplus$

min to hr  $\rightarrow \ominus$



Eg. 2:30 Find Angle ?

$$\Rightarrow \text{angle} : 30 \times \frac{11}{2} - 2 \times 30$$

$$= 165 - 60$$

$$\text{hr to min} = 105^\circ$$

$$\begin{aligned} \text{min to hr} &= 360^\circ - 105^\circ \\ &= 255^\circ \end{aligned}$$

Eg. 7:30 Find Angle ?

$$\Rightarrow 30 \times \frac{11}{2} - 7 \times 30$$

$$= 165 - 210$$

$$= -45^\circ \leftarrow \text{min to hr}$$

$$360^\circ - 45^\circ = 315^\circ \leftarrow \text{hr to min}$$

Eg. 3:15 Find Angle ?

$$= 15 \times \frac{11}{2} - 3 \times 30$$

$$= 82.5 - 90$$

$$= -7.5^\circ \text{ Ans (min to hr)}$$

$$\text{OR } 360 - 7.5 = 352.5^\circ \text{ Ans (hr to min)}$$

eg.  $7 : 46\frac{4}{11}$  find Angle?

$$\Rightarrow 46\frac{4}{11} = 46 + \frac{4}{11} = \frac{46 \times 11 + 4}{11} = \frac{510}{11}$$

$$= 7 : \frac{510}{11}$$

$$\text{Angle: } \frac{510}{11} \times \frac{11}{2} - 7 \times 30$$

$$= 255 - 210$$

$$= 45^\circ \text{ Ans}$$

$\Rightarrow 0^\circ / \text{Overlap (अतिव्यापन):-}$

1 hr  $\rightarrow$  1 Time

12 hr  $\rightarrow$  11 Time

1 Day  $\rightarrow$  22 Time



[ 11:00 - 1:00 ]  
1 time

eg. 9:00 AM - 2:00 PM

$$\underbrace{9-10}_1, \underbrace{10-11}_1, \underbrace{11-1}_1, \underbrace{1-2}_1 = 4 \text{ times}$$

Overlapping time

$$\boxed{x : x \times \frac{60}{11}}$$

eg. 3pm - 4pm

$\downarrow$   
 $x$

$$3 : 3 \times \frac{60}{11} = 3 : \frac{180}{11}$$

$$= 3 : 16\frac{4}{11} \text{ Ans}$$

eg. 2am - 3Am

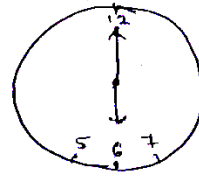
$\downarrow$   
 $x$

$$2 : 2 \times \frac{60}{11} = 2 : 10\frac{10}{11} \text{ Ans}$$

Maths with Gajanand

⇒ 180° / Opposite (विपरीत) :-

1 hr → 1 Time  
 12 hr → 11 Time [5-7]  
 1 Day → 22 Time



$$x : (x \pm 6) \times \frac{60}{11}$$

we take

$$x = 6 \rightarrow (-)$$

$$x > 6 \rightarrow (-)$$

$$x < 6 \rightarrow (+)$$

e.g. 2pm - 3pm

$\frac{1}{x}$

$$2 : (2 + 6) \times \frac{60}{11}$$

$$2 : 8 \times \frac{60}{11} = \left[ 2 : 43 \frac{7}{11} \text{ PM} \right] \underline{\underline{\text{Ans}}}$$

e.g. 6Am - 7Am

$$6 : (6 - 6) \times \frac{60}{11} = 6 : 00 \underline{\underline{\text{Ans}}}$$

e.g. At what time between 8:00 and 9:00 clock are the opposite to each ?

$$\Rightarrow 8 - 9$$

$$= 8 : (8 - 6) \times \frac{60}{11}$$

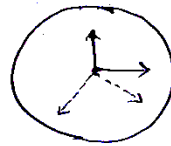
$$= 8 : \frac{120}{11}$$

$$= 8 : 10 \frac{10}{11} \underline{\underline{\text{Ans}}}$$

Maths with Gajanand

⇒ 90°/Right Angle (समकोण) :-

1 hr → 2 Times  
12 hr → 22 Times  
1 Day → 44 Times



$\left. \begin{array}{l} 2-3 \\ 3-4 \end{array} \right\} \text{ 3 times}$   
 $\left. \begin{array}{l} 8-9 \\ 9-10 \end{array} \right\} \text{ 3 times}$

$$x : (x \pm 3) \times \frac{60}{11}$$

If  $(x \pm 3) \geq 12 \Rightarrow$  Subtract 12

2 If  $(x \pm 3) < 0 \Rightarrow$  Add 12

Eg.  $10 - 11$   
↓  
 $x$

⊕  
 $10 : (10 + 3) \frac{60}{11}$

$10 : (13) \frac{60}{11}$

$10 : (13 - 12) \frac{60}{11}$

$10 : 1 \times \frac{60}{11}$

$10 : 5 \frac{5}{11}$  Ans

⊖  
 $10 : (10 - 3) \frac{60}{11}$

$10 : 7 \times \frac{60}{11}$

$10 : \frac{420}{11}$

$10 : 38 \frac{2}{11}$  Ans

Eg.  $2 - 3$   
↓  
 $x$

⊕  
 $2 : (2 + 3) \frac{60}{11}$

$2 : 5 \times \frac{60}{11}$

$2 : 27 \frac{3}{11}$  Ans

⊖  
 $2 : (2 - 3) \frac{60}{11}$

$2 : (-1) \times \frac{60}{11}$

$2 : (-1 + 12) \times \frac{60}{11}$

$2 : 11 \times \frac{60}{11}$

$2 : 60$

$3 : 00$  Ans

Maths with Gajanand

# Clock Error

## Type-1

If in a clock minute hand cross hour hand in 65 minute, then how many times clock in 24 hour goes slow down or fast.

Soln:-

$$\left[ \frac{65\frac{5}{11} - \text{Time}}{\text{Time}} \times \text{Total time (Hr/day)} \right]$$

$$= \frac{65\frac{5}{11} - 65}{65} \times 24$$

$$= \frac{5}{11 \times 65} \times 24$$

$$= + \frac{24}{143} \text{ Hr. } \underline{\text{Fast}} \underline{\text{Ans}}$$

E.g. 2:- If in a clock minute hand cross hour in 60 minute, then how many times clock in 22 hr goes slow down or fast.

$$= \frac{65\frac{5}{11} - 60}{60} \times 22$$

$$= \frac{5\frac{5}{11}}{60} \times 22$$

$$= \frac{60}{11 \times 60} \times 22$$

$$= \frac{22}{2} = +2 \text{ Fast.}$$

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Type 2:- A clock which moves continuously fast, it lags 5 minute on Sunday 8am, it is ahead 7 minute on Tuesday 8am then find when clock shows right time?

Soln:-

$$\begin{array}{lcl} \text{Sunday - 8 Am} & \longrightarrow & -5 \\ \text{Tuesday 8 Am} & \longrightarrow & +7 \end{array} \quad \begin{array}{l} \text{total time} \\ 48 \text{ hr} \end{array}$$

$$\left[ \text{Formula:- } \frac{\text{1st Slow/fast}}{\text{Slow + fast}} \times \text{total time} \right]$$

$$= \frac{5}{12} \times 48 = 20 \text{ hr}$$

$$\text{Sunday 8 Am} + 20 \text{ hr} \Rightarrow \underline{\text{Monday 4 a.m. Ans}}$$

eg 2 A clock which moves continuously fast, It lags 10 min. on Sunday 7am, it is ahead 5 minute on Tuesday 7pm then find when clock shows right time?

Soln:-

$$\begin{array}{lcl} \text{Sunday - 7am} & \longrightarrow & -10 \text{ min} \\ \text{Tuesday - 7pm} & \longrightarrow & -5 \text{ min} \end{array} \quad \begin{array}{l} \\ \end{array} \quad \begin{array}{l} \\ 48 + 12 = 60 \text{ hr} \end{array}$$

$$\frac{10}{15} \times 60$$

$$= 40 \text{ hr}$$

$$\text{Sunday 7am} + 40 \text{ hr} \Rightarrow \underline{\text{Monday 11 pm Ans}}$$

Type 3:- A clock fast 1 hour in each 24 hours. if clock shows correct time on Sunday 9 am then what is the correct time, when clock shows 11:00 am Tuesday?

Soln:-

|       |       |
|-------|-------|
| Right | wrong |
| 24 hr | 25 hr |

Sun - 9:00 AM  
Tues - 11:00 AM  $\rightarrow$  50 hr

$$\frac{24}{25} \times 50 = 48 \text{ hr (Right)}$$

9:00 AM + 48 hr = 9 AM Tuesday Ans

Eg.

24 hr  $\rightarrow$  4 hr slow

Soln:-

|       |             |
|-------|-------------|
| Right | wrong       |
| 24    | (24-4) = 20 |

$$\frac{24}{20} \times 60 = 72 \text{ hr}$$

Sun 7 AM show correct time  
Tues 7 AM = ?

Sun 7 AM  
Tues 7 AM  $\rightarrow$  60 hr

= Sun 7 am + 72 hr

= wed 7 AM Ans

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# DICE

Maths with Gajanand

### 3. DICE (पासा )

There are 4 Cases in this topic

Case 1 :- when only one position is given

Case 2 :- when only two Position are given

Case 3 :- When more than two position are given,

Case 4 :- Opening of a Dice.

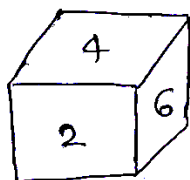
#### Case 1

When only one position is Given.

When only one position is Given ~~with~~ in a question then we will check our dice is standard or general.

Standard Dice:-

If every opposite surface sum is equal to 7 then the dice is called standard dice.



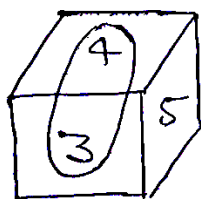
$$4 \longleftrightarrow (7-4) = 3 \text{ (opposite)}$$

$$6 \longleftrightarrow (7-6) = 1 \text{ (opposite)}$$

$$2 \longleftrightarrow (7-2) = 5 \text{ (opposite)}$$

General Dice:-

If any two adjacent surface have Sum 7 then the dice is called general dice.

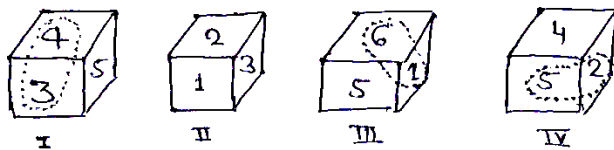


$$3 + 4 = 7$$

★ In General dice visible numbers are opposite to hidden 1

$$3/4/5 \longleftrightarrow 1/6/2$$

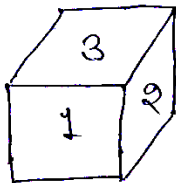
eg.1 which of the following is a standard dice.



I, III, IV  $\rightarrow$  General Dice

II  $\rightarrow$  standard dice.

eg.2

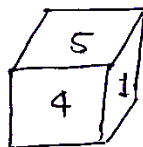
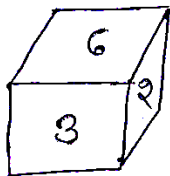


Opposite to 5 = ?

Soln:- it is standard dice

$$\text{So } (7 - 5) \rightarrow (7 - 5) = 2 \text{ Ans}$$

eg.3.

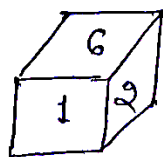
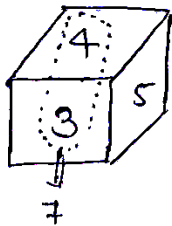


Opposite to 6 = ?

$\Rightarrow$  It is standard dice because there is no two adjacent surface have sum 7

$$\text{So } 6 \rightarrow (7 - 6) = 1 \text{ Ans}$$

eg.4.



Opposite to 4 = ?

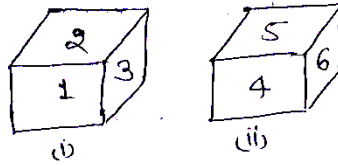
$\Rightarrow$  It is General dice so  $4 \leftrightarrow 1/6/2$  Ans

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## Case 2

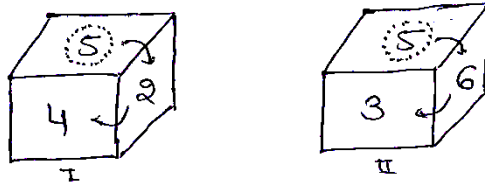
When only two position are given

(a) No Common :-

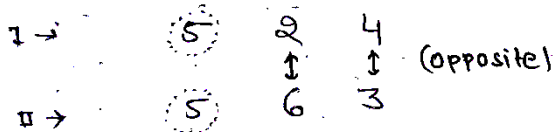


If there is nothing common in (i) and (ii) then we will check dice is standard or general.

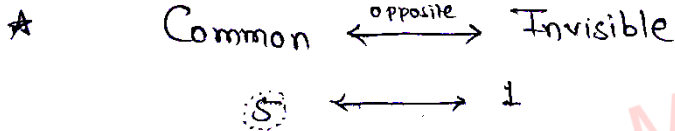
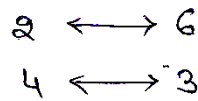
(b) Only one Common :-



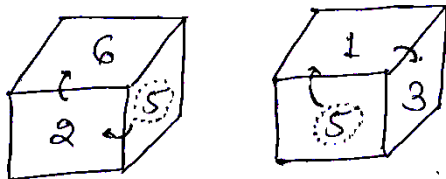
(In clockwise direction)



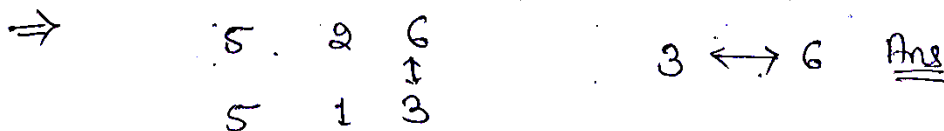
Opposite:-



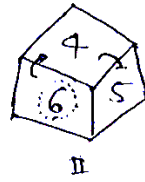
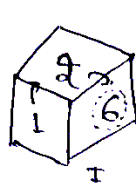
Ex. 1



Opposite to 3 = ??



eg. 2:- What is the no. at bottom surface if '1' is on top surface.



(A) 3

(B) 5

~~(C) 2~~

(D) 6

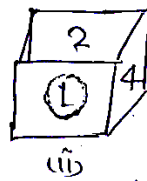
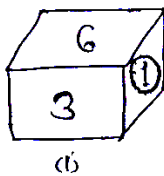
Soln

Common

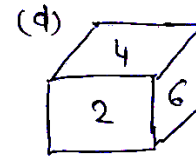
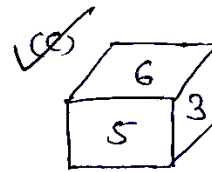
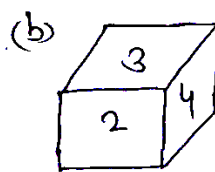
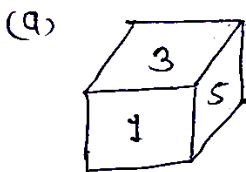
I → 6      1      2  
II → 6      4      5

1 ↔ 4      Ans

eg. 3



which is correct?



Soln:-

Common  
1      3      6  
1      2      4

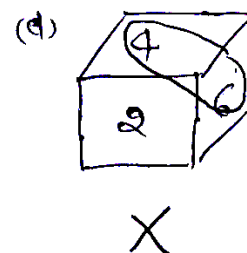
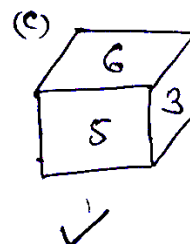
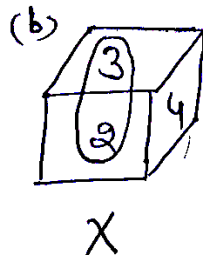
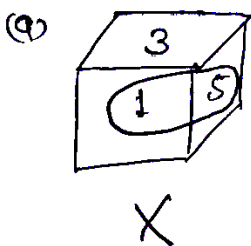
Opposite:-

3 ↔ 2

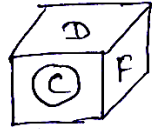
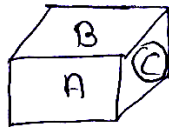
6 ↔ 4

(Common) 1 ↔ 5 (Invisible)

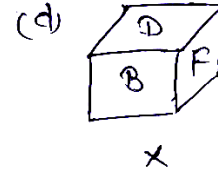
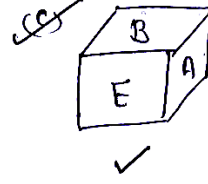
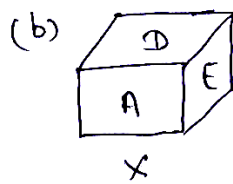
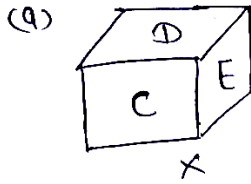
Note:- [Two opposite number can never be seen at a time.]



eg. 4.



Which is correct



Soln

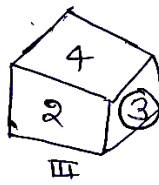
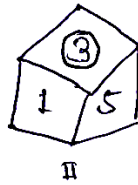
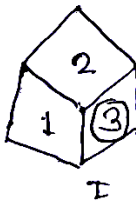
|   |   |   |
|---|---|---|
| C | A | B |
|   | ↓ | ↓ |
| C | D | F |

Opposite

|   |   |   |
|---|---|---|
| A | ↔ | D |
| B | ↔ | F |
| C | ↔ | E |

eg. 5.

Opposite of 3 = ??



(a) 1

~~(b) 6~~

(c) 4

(d) 5

⇒

Common

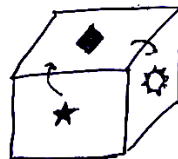
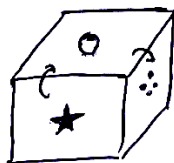
(3) ↔

Invisible

6

Ans

eg 6



○  $\xrightarrow{\text{opposite}}$  ??

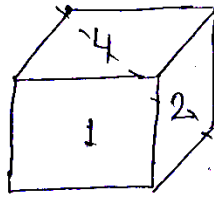
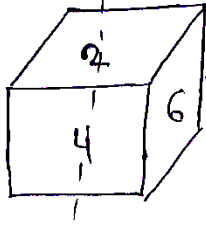
⇒

Common

|   |   |   |
|---|---|---|
| ★ | ○ | ⋯ |
| ★ | ◆ | ⚙ |

○ ↔ ◆ Ans

(c) Two Common :-

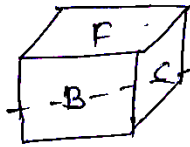
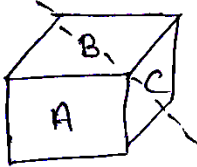


Remaining number will be opposite to each other

$$6 \longleftrightarrow 1$$

and Common  
 $2/4 \longleftrightarrow$  Hidden  $3/5$

eg. 1



$$E \longleftrightarrow ??$$

(a) A

(b) F

~~(c) B~~

(d) D

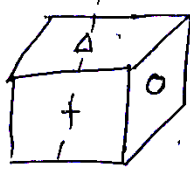
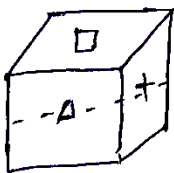
$\Rightarrow$

$$A \longleftrightarrow F$$

and  $B/C \longleftrightarrow D/E$

$$E \longleftrightarrow B/C \text{ Ans}$$

eg. 2

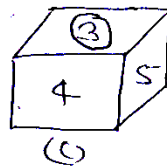
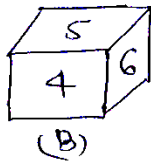
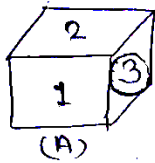


$$\square \longleftrightarrow ??$$

$$\square \longleftrightarrow O \text{ Ans}$$

### Case - III

More than two positions are Given

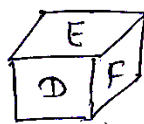
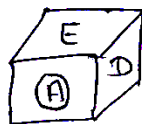
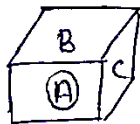


⇒ we solve the question by taking any two position of only one common.

From A & C

|     |   |   |   |                       |
|-----|---|---|---|-----------------------|
| (A) | ② | 1 | 2 | $1 \leftrightarrow 2$ |
|     |   | ↑ | ↑ |                       |
| (C) | ③ | 5 | 4 | $5 \leftrightarrow 4$ |
|     |   |   |   | $3 \leftrightarrow 6$ |

Eg. 1.



$D \leftrightarrow ??$

⇒

From (i) & (ii)

|   |   |   |
|---|---|---|
| ① | B | C |
| ② | E | D |

$B \leftrightarrow E$

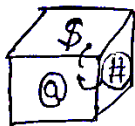
$C \leftrightarrow D$

$A \leftrightarrow F$

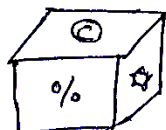
$D \leftrightarrow C$

Ans

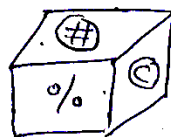
Eg. 2.



(i)



(ii)



(iii)

Opposite to \$ → ??

⇒

From (i) & (iii)

|   |   |    |
|---|---|----|
| # | @ | \$ |
| # | © | %  |

$\$ \leftrightarrow \%$

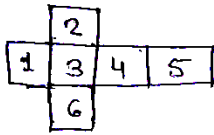
Ans

Maths with Gajanand

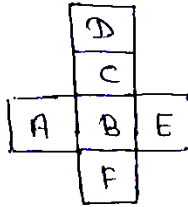
Case IV

Opening of a Dice,

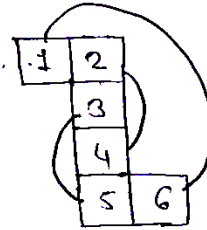
⇒ Each alternate surface are opposite to each other.



$$\begin{aligned} 1 &\leftrightarrow 4 \\ 3 &\leftrightarrow 5 \\ 2 &\leftrightarrow 6 \end{aligned}$$

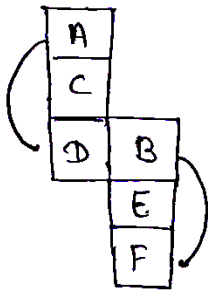


$$\begin{aligned} A &\leftrightarrow E \\ D &\leftrightarrow B \\ C &\leftrightarrow F \end{aligned}$$



$$\begin{aligned} 2 &\leftrightarrow 5 \\ 3 &\leftrightarrow 4 \\ 1 &\leftrightarrow 6 \end{aligned}$$

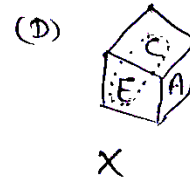
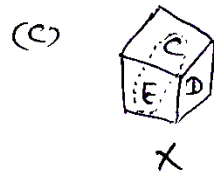
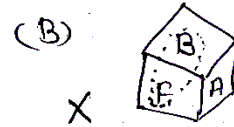
eg In a given figure of open dice which dice we can make?



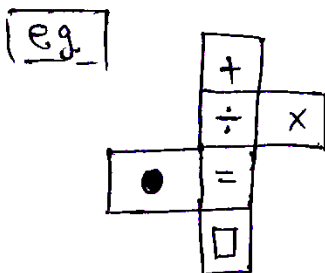
Sol<sup>n</sup>:-

$$\begin{aligned} A &\leftrightarrow D \\ B &\leftrightarrow F \\ C &\leftrightarrow E \end{aligned}$$

check



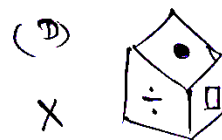
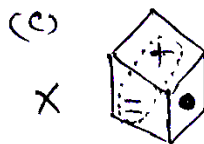
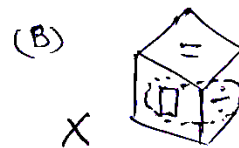
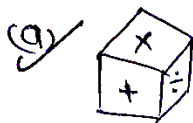
Maths with Gajanand



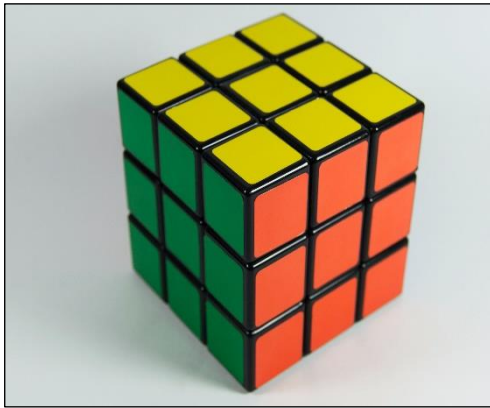
Sol<sup>n</sup>:-

$$\begin{aligned} + &\leftrightarrow = \\ \div &\leftrightarrow \square \\ \times &\leftrightarrow \bullet \end{aligned}$$

check



## 4. Cube & Cuboid



$$n = \frac{\text{Big Side}}{\text{Small Side}}$$

$$\text{Total cube} = n^3$$

$$\text{Corner Cubes (3 surface painted)} = 8$$

$$\text{middle Cubes (2 Surface Painted)} = 12(n-2)$$

$$\text{Central Cubes (One surface Painted)} = 6(n-2)^2$$

$$\text{Inner central cubes (Colourless)} = (n-2)^3$$

[e.g.] A 6cm coloured cube <sup>is cut</sup> into 3cm x 3cm smaller cubes, then how many cubes are formed which have

(i) Only one face painted

$$\begin{aligned} \text{Ans: } n &= \frac{6}{3} = 2 \rightarrow 6(n-2)^2 \\ &= 6(2-2)^2 = 0 \text{ Ans.} \end{aligned}$$

(ii) 2 surface Painted :-

$$\begin{aligned} &= 12(n-2) \\ &= 12(2-2) = 0 \text{ Ans} \end{aligned}$$

(iii) more than 3 Surface Painted :-

$$= 0 \text{ Ans}$$

Eg. 2 - A Coloured cube is cut into 64 smaller cubes then how many smaller cubes are formed which have minimum two face painted.

Soln:-  $n^3 = 64$

$n = 4$

$$\begin{aligned}\text{Two surface Painted} &= 12(n-2) \\ &= 12(4-2) \\ &= 24 \text{ Ans}\end{aligned}$$

At least two

②

③

$$24 + 8 = 32 \text{ Ans}$$

Eg. 3 - Coloured cube is cut into 125 smaller cubes of length 2cm.

(i) what is the length of bigger cube

(ii) How many smaller cubes are there which have <sup>at most</sup> ~~one~~ one surface painted.

Soln (i)  $n^3 = 125$

$n = 5$

$n = \frac{\text{Big side}}{\text{Small side}}$

$$\begin{aligned}\text{Big side} &= n \times \text{small side} \\ &= 5 \times 2\text{cm} = 10\text{cm} \text{ Ans}\end{aligned}$$

(ii) at most one

$$\begin{array}{c} \downarrow \quad \quad \downarrow \\ 0 \quad + \quad 1 \end{array}$$

$$(n-2)^3 + 6(n-2)^2$$

$$(5-2)^3 + 6(5-2)^2 \Rightarrow 27 + 54 = 81 \text{ Ans}$$

## CUBOID

- Corner Cubes (3 Surface Painted) = 8
- Middle cubes (2 surface painted) =  $4(n_1-2) + 4(n_2-2) + 4(n_3-2)$
- Central cubes (One surface Painted) =  
 $= 2(n_1-2)(n_2-2) + 2(n_2-2)(n_3-2) + 2(n_3-2)(n_1-2)$
- Inner central cubes (Colourless) =  $(n_1-2)(n_2-2)(n_3-2)$
- Total no. of smaller cubes =  $n_1 \times n_2 \times n_3$

**eg 1** A cuboid of size  $4 \times 8 \times 10 \text{ cm}^3$  is cut into smaller cubes of 2cm length.

Soln:-  $n_1 = \frac{4}{2} = 2, \quad n_2 = \frac{8}{2} = 4, \quad n_3 = \frac{10}{2} = 5$

Total cubes =  $2 \times 4 \times 5 = 40$

(i) 3 surface :- 8 Ans

(ii) 4 Surface :- 0 Ans

(iii) 1 Surface :-  $2(2-2)(4-2) + 2(4-2)(5-2) + 2(5-2)(2-2)$   
 $= 12 \text{ Ans}$

(iv) Colourless :-  $(2-2)(4-2)(5-2)$   
 $= 0 \text{ Ans}$

**eg.2** A cuboid of  $9 \times 15 \times 18 \text{ cm}^3$  is cut into 3cm length of smaller cubes, how many smaller cubes?

Soln:-

$$\frac{9}{3}, \frac{15}{3}, \frac{18}{3}$$

$$n_1 = 3, n_2 = 5, n_3 = 6$$

$$3 \times 5 \times 6 = 90 \text{ Ans}$$

**eg.3** A cuboid of  $2 \times 6 \times 10 \text{ cm}^3$  is cut into 2cm length of smaller cubes, then many smaller cube are there which have 4 surface painted?

Soln:- Total :  $\frac{2}{2}, \frac{6}{2}, \frac{10}{2}$

$$n_1 = 1, n_2 = 3, n_3 = 5$$

$$\text{Total} = n_1 \times n_2 \times n_3 = 1 \times 3 \times 5 = 15 \text{ Ans}$$

4 Surface are a

Corner :- 4

Middle :- 3

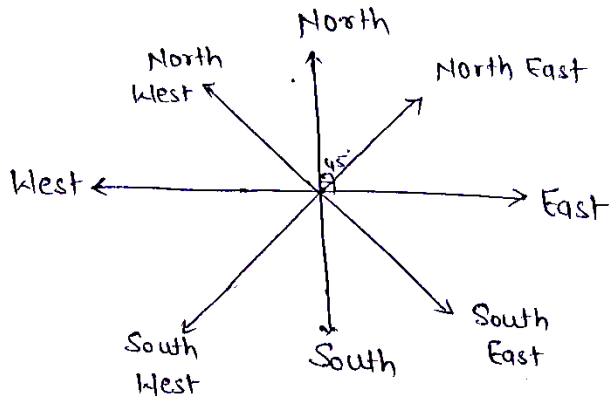
Central :- 2

One :- 0

Colourless :- 0

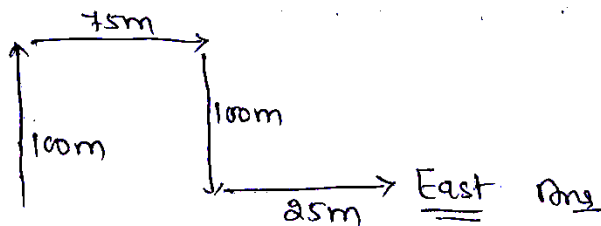
Maths with Gajanand

## 5. Direction



**Type-1** A man walking towards north after walking 100m turns right and walk 75m again he turns right and walk 100m and last movement he turns left and walk 25m now which direction he is walking?

Soln:-



Trick:-

North  
 Right → East Ans  
 Right X  
 Left X

Note:- Cancel all right and left Pair

Eg. 2

East  
Left x  
Right x  
Right x  
Left x  
Left

→ North Ans

North  
↑ Left  
→ East

Eg. 3

South  
Left x  
R x  
R  
R  
—  
?

> 2R

South  $\xrightarrow[\text{2R}]{\text{opposite}}$  North

Note :- IF 2R / 2L → opposite direction

Eg. 4

South  $\xrightarrow{\text{opposite}}$  North  
L  
L  
R x  
L x  
—  
?

> 2L

Maths with Gajanand

Eg. 5

West  
R  
R  
R  
R  
—  
?

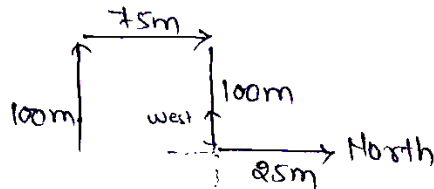
> 4R → same direction

= West Ans

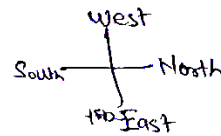
## Type 2

A man walks in straight 100m and turns his right and walk 75m. Again he turns his right and walk 100m. And last he turn his left and walk 25m. if now he is walking in north direction. Then find from which direction he started?

Soln:-



West Ans



Tricks

? → West

R

R

L

North

Maths with Gajanand

Eg. 2

? → South Ans

L

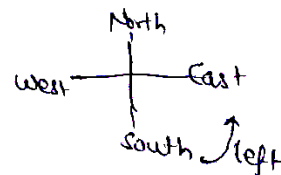
R

L

R

L

East



Eg. 3

? → North east Ans

R

L

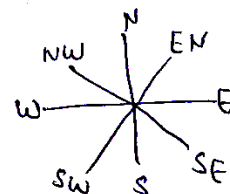
R

R

R

South west

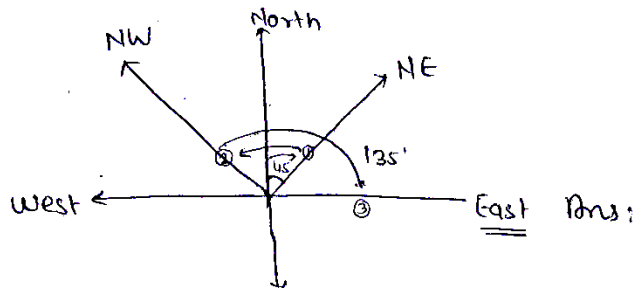
> 2R → opposite



### Type-3

A man faces towards north first he turns in  $45^\circ$  clockwise then he turns  $90^\circ$  Anticlockwise. At last he turns  $135^\circ$  clockwise. Now his face on which side?

Soln:-



Trick

$$\begin{array}{l}
 \text{North} \\
 45^\circ \text{ Clockwise} \\
 90^\circ \text{ Anti-clockwise} \\
 135^\circ \text{ Clockwise}
 \end{array}
 \rightarrow
 \begin{array}{r}
 \oplus 180^\circ \text{ clockwise} \\
 - 90^\circ \text{ Anticlockwise} \\
 \hline
 90^\circ \text{ clockwise}
 \end{array}$$

North  $\xrightarrow{90^\circ \text{ CW}}$  East Ans

eg. 2

$$\begin{array}{l}
 \text{East} \\
 45^\circ \text{ CW} \\
 90^\circ \text{ C.W} \\
 180^\circ \text{ A.C.W} \\
 135^\circ \text{ C.W}
 \end{array}
 \rightarrow
 \begin{array}{r}
 \oplus 270^\circ \text{ C.W} \\
 - 180^\circ \text{ A.C.W} \\
 \hline
 90^\circ \text{ CW} \rightarrow \text{South Ans}
 \end{array}$$

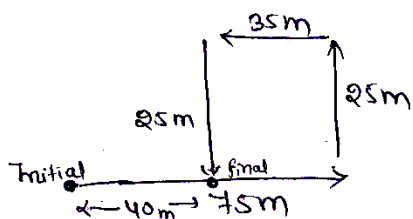
eg. 3

$$\begin{array}{l}
 \text{East} \\
 45^\circ \text{ CW} \\
 90^\circ \text{ A.C.W} \\
 45^\circ \text{ A.C.W} \\
 90^\circ \text{ CW} \\
 45^\circ \text{ C.W}
 \end{array}
 \rightarrow
 \begin{array}{r}
 180^\circ \text{ A.C.W} \\
 - 135^\circ \text{ A.C.W} \\
 \hline
 45^\circ \text{ CW} \\
 \text{East } 45^\circ \text{ CW} \rightarrow \text{North-east Ans}
 \end{array}$$

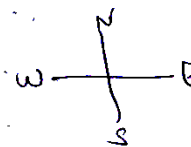
**Eg 4** A man is walking towards east. He walks 75m and then left and again walks 25m. Again he turn left he walks 35m. At last he walk 25m and turn left.

- Now, he is going in which direction?
- Now he is on which direction from the initial point?
- Now he is how far distance from the initial point?
- Now he is how far and in which direction from the initial point?

Soln:-



South



i) South

m-n

East

$\begin{array}{l} L \\ L \\ L \end{array} > 2L \xrightarrow{\text{opposite west}} \text{Left} \xrightarrow{\text{South}}$   
South

(ii) East

(iii)  $(75 - 35) = 40m$

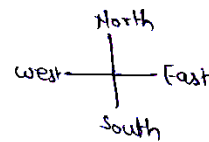
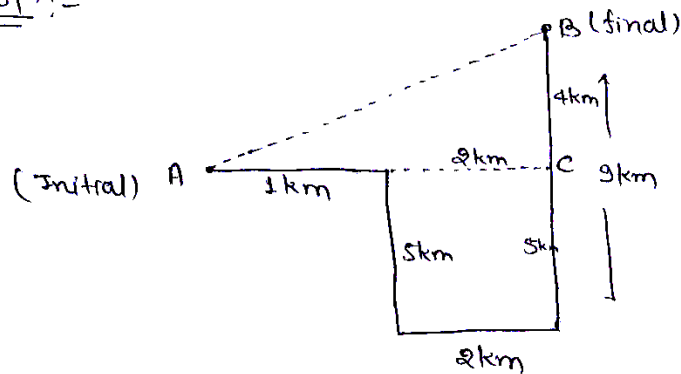
(iv) 40m, east

Maths with Gajanand

### Type - IV

1. A man walk in 1 km East. Now he turns south and walk 5 km again he turns East and walk 2 km. At last he turns north and walk 3 km. Now find that he in which direction end how far from the initial point?

Soln:-



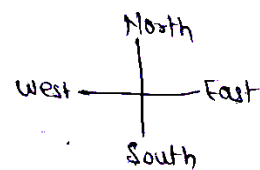
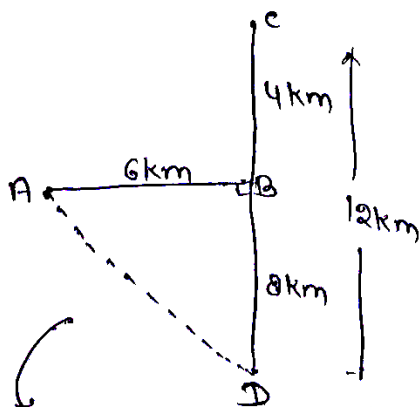
in  $\triangle ABC$

$$AB = \sqrt{(3)^2 + (4)^2} = 5 \text{ km}$$

Hence 5 km, North-east Ans

- eg. 2] A is 6 km West from B. And C is 4 km north to B. D is 12 km South from C. now D on which direction and how far from A.

Soln:-

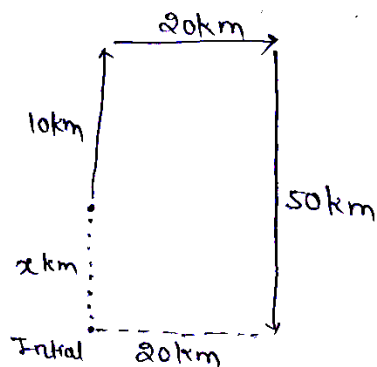


$$AD = \sqrt{6^2 + 8^2}$$

= 10 km, South-East Ans

**eg 8** A man walks 10km in north and turns his right. And walk 20km. Again he turn right and walk 50km. if he is 20km west from the initial point. Now find that he how much walk initially?

Soln:-

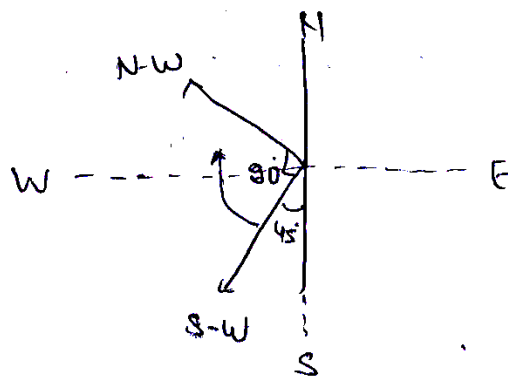


$$x + 10 = 50$$

$$x = 40 \text{ km} \quad \text{Ans}$$

**eg. 4** A man walks in south direction. After he walks 1km he turn his right  $45^\circ$  and walk 2 km again he turn his right  $90^\circ$  and walk same distance. Now find that in which direction is he going?

Soln:-

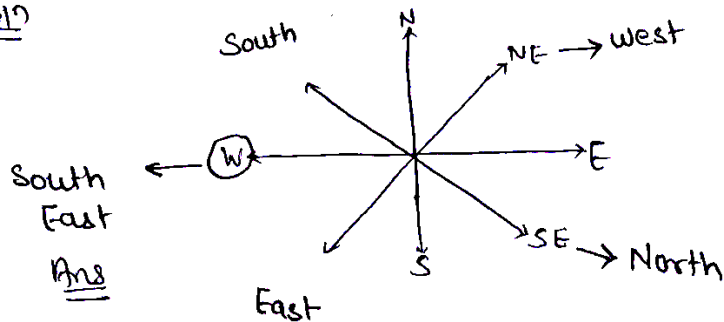


North-west Ans

## Type - 5

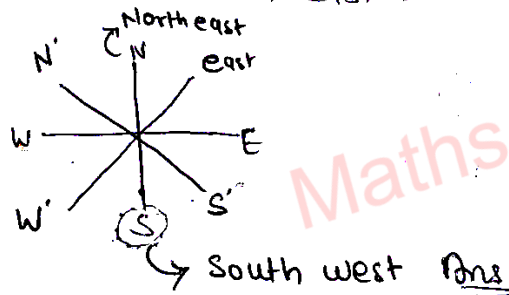
**Eg. 1** If South-east becomes North, North-east becomes west and so on that what will west becomes?

Soln

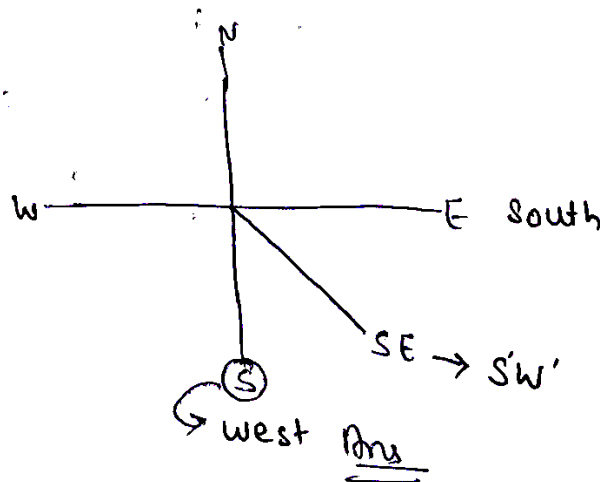


**Eg. 2** If North becomes North-east - south becomes ??

Soln

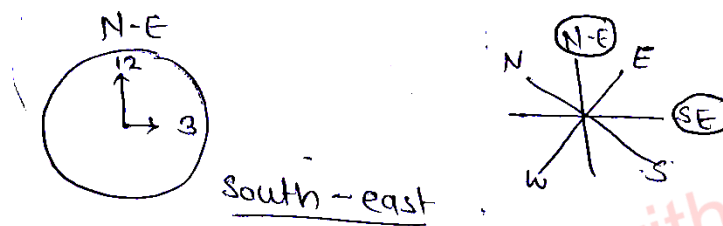


**Eg. 3** If South-east becomes South-west then South becomes ?



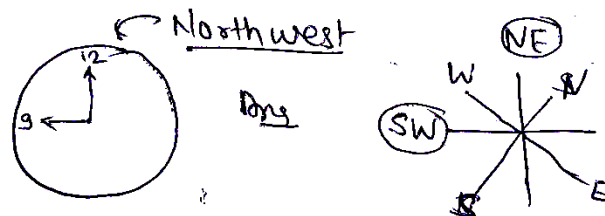
**eg 4** If it is 3 o'clock if at into time the minute hand points towards North-east then in which hours would be?

Soln



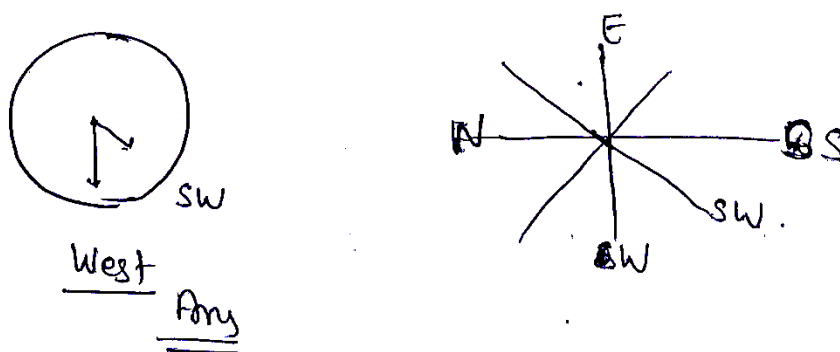
**eg. 5** It is 9 o'clock, if at this time the hours hand point towards south-west then in which direction Minute would be?

Soln



**eg. 6** It is 4:30 o'clock if at this time hours hand point towards south-west then in which direction minute would be?

Soln:-

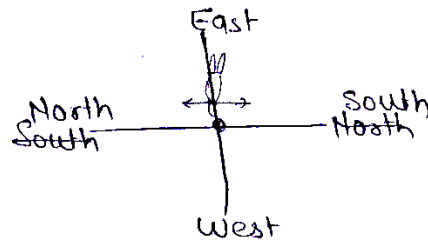


### Type - VI

In the morning A man was performing Yoga with his Head down and legs up, if at this time his face towards west then which direction his left hand would be?

Sol?

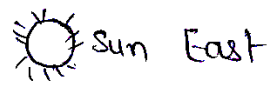
North Ans



### Type - VII

Shadow (Opposite to sun) E/W

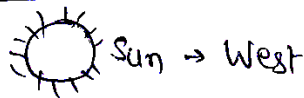
Morning :-



Shadow → West

Evening :-

Shadow → East

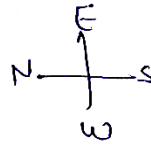
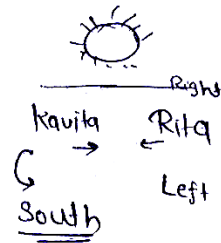


If

1. Right hand → west  $\Rightarrow$  face → North South
2. Left hand → East West  $\Rightarrow$  face → North
3. Right hand → East  $\Rightarrow$  face → North
4. Left hand → East  $\Rightarrow$  face → South

**Ex. 1** A morning two friend Reeta and kavita talks to each other. If at the time kavita shadow from the left side of Reeta, now find that kavita face on which direction?

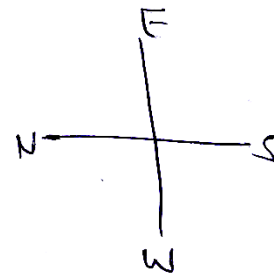
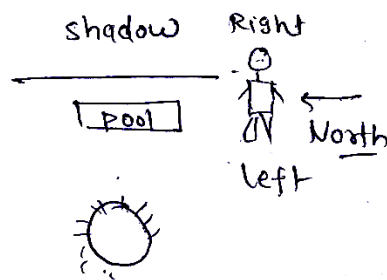
Soln:-



**Ex. 2**

An evening Gopal saw a pool. If at the time a pool shadow made his right, now find that Gopal face which direction?

Soln



face → North Ans

## 6. Syllogism

Rule 1 :-

If statement is (+ve) then conclusion must be (+ve)  
and if statement is (-ve) then conclusion must be (-ve).

(ii) 'All' can be written as 'some' but 'some' cannot be written as 'all'

Rule 2 :- Two negative statements can not be included in a Conclusion.

Ex. Statement

(i) No Pen is Copy.

(ii) No Copy is Book.

Conclusion:

(i) No Pen is Book (X)

(ii) No book is pen (X)

(iii) All pens are books (X)

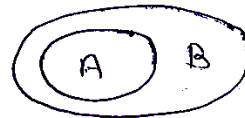
(iv) All books are pens (X)

Rule 3 :-

one +ve and One -ve statement concludes as -ve,

Types of statement:

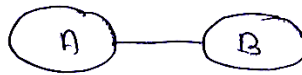
(i) All A are B



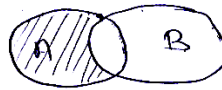
(ii) Some A are B



(iii) No A are B



(iv) Some A are not B



Ex. Statement

(i) All B are C. (true)

(ii) No C are D. (-ve)

Conclusion:-

(i) All B are D (x)

(ii) Some B are D (x)

(iii) Some D are B (x)

(iv) All D are B (x)

(v) No B is D (✓)

(vi) No D is B (✓)

(vii) Some D are not B (✓)

(viii) Some B are not D (✓)



1. Statement (i) All A are B

Conclusion:-

(i) No A are B (x)

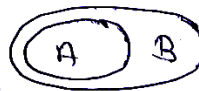
(ii) Some A are B (✓)

(iii) Some A are not B (x)

(iv) All B are A (x)

(v) No B are A (x)

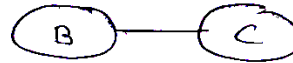
(vi) Some B are A (✓)



2. Statement : No B are C

Conclusion :-

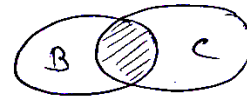
- (i) All B are C (X)
- (ii) Some B are C (X)
- (iii) Some B are not C (✓)
- (iv) All C are B (X)
- (v) No C are B (✓)
- (vi) Some C are B (X)
- (vii) Some C are not B (✓)
- (viii) Atleast some B are C (X)



3. Statement :- Some B are C

Conclusion :-

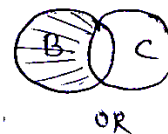
- (i) All B are C (X)
- (ii) No B are C (X)
- (iii) Some B are not C (X)
- (iv) All C are B (X)
- (v) No C are B (X)
- (vi) Some C are not B (X)
- (vii) Atleast some B are C (✓)



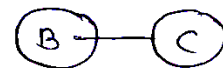
4. Statement : Some B are not C

Conclusion :-

- (i) All B are C (X)
- (ii) No B are C (✓)
- (iii) Some B are not C (X)
- (iv) All C are B (X)
- (v) No C are B (✓)
- (vi) Some C are B (X)
- (vii) Some C are not B (X)
- (viii) Atleast some B are C (X)



OR



## Complimentary pair

Either, or

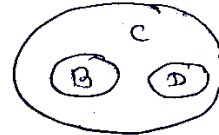
In Conclusion

|      |      |     |         |
|------|------|-----|---------|
| (i)  | Some | +ve | Same    |
| (ii) | No   | -ve | Subject |

in Both Conclusion

Ex. Statement :-

- (i) All B are C
- (ii) All D are C



Case-1

Conclusion :-

- |      |      |         |       |                               |
|------|------|---------|-------|-------------------------------|
| (i)  | Some | B are D | ← +ve | } either (i) or (ii) follow ✓ |
| (ii) | No   | B are D | ← -ve |                               |
- Same Subject in Both Conclusion

Case 2

Conclusion

- |                  |                               |
|------------------|-------------------------------|
| (i) All B are D  | } Neither I nor (ii) follow x |
| (ii) No D are B  |                               |
| (iii) No B are D |                               |

Case 3

Conclusion:-

- |                       |                               |
|-----------------------|-------------------------------|
| (i) All B are D       | } either (i) or (ii) follow ✓ |
| (ii) Some B are not D |                               |

Case 4:

Conclusion:

- |                       |                               |
|-----------------------|-------------------------------|
| (i) Some B are D      | } either (i) or (ii) follow ✓ |
| (ii) Some B are not D |                               |

Eg. 1

Statement :-

All blades are sharp

Some blades are blunt

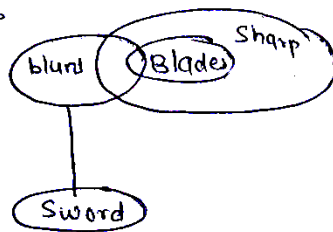
No blunt is sword

Conclusion:-

(I) All sword being blades is a possibility. ✓

(II) At least some blades may be swords. ✓

Soln



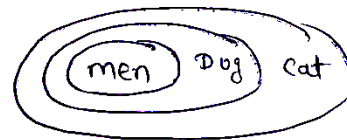
Maths with Gajanand

Eg. 2

Statement:-

All men are dogs.

All dogs are cats.



Conclusion:-

(I) All men are cats ✓

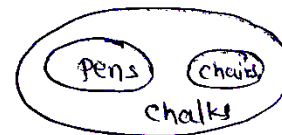
(II) All cats are men X

Eg. 3

Statement

All pens are chalks.

All chairs are chalks



Conclusion:-

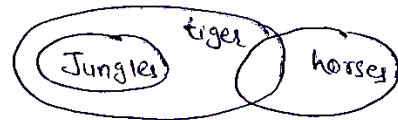
(I) Some pens are chairs X

(II) Some chalks are pens. ✓

**eg. 4** Statement :-

All jungles are tigers.

Some tigers are horses.



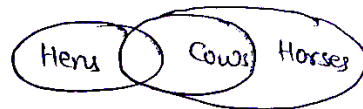
Conclusion:-

- (I) Some horses are jungles      +ve  
(II) No horses is jungles      -ve
- either I OR II follow ✓

**eg. 5** Statement:-

Some hens are cows.

All Cows are horses.



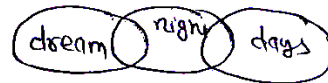
Conclusion:-

- (I) Some horses are hens ✓  
(II) Some hens are horses ✓

**eg. 6** Statement:-

Some dreams are nights

Some nights are days



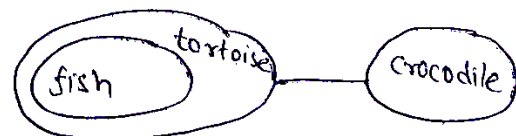
Conclusion:-

- (I) All days are either nights or dreams X  
(II) Some days are nights ✓

**eg. 7** Statement

All fish are tortoise

No tortoise is crocodile.

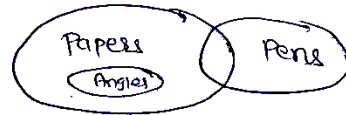


Conclusion:-

- (I) No crocodile is a fish. ✓  
(II) No fish is a crocodile ✓

**Eg.8** Statement:-

Some papers are pens.  
Angles is a paper

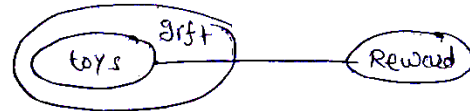


Conclusion:-

- (I) Some Angle is not a pen } -ve  
(II) Some Angle is a pen. } +ve
- either I or II

**Eg.9** Statement:-

All toys are gifts.  
No toy is a reward.

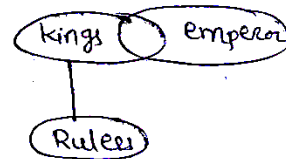


Conclusion:-

- (I) Atleast some rewards are gifts (X)  
(II) All gifts are toys. (X)

**Eg.10** Statement:-

Some kings are emperor.  
No ruler is a king.

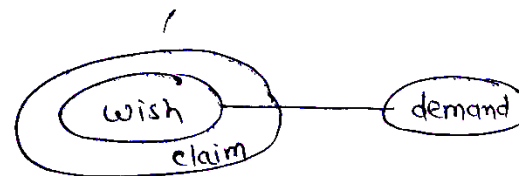


Conclusion,

- (I) No ruler is an emperor.  
(II) At least some rulers are emperor. }
- either I or II

**Eg.11** Statement:-

No wish is a demand.  
All wish are claims.

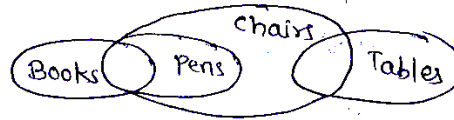


Conclusion:-

- (I) No demand is claim (X)  
(II) All claim are wishes (X)

**Eg. 12** Statement:-

Some books are pens.  
All pens are chairs,  
Some chairs are tables,

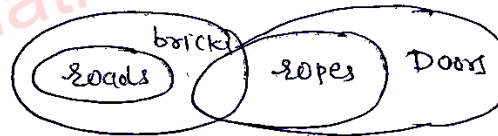


Conclusion:-

- (I) Some books are chairs. ✓
- (II) Some chairs are books. ✓
- (III) All tables are chairs. X
- (IV) Some tables are chairs. ✓

**Eg. 13** Statement:-

All roads are bricks.  
Some bricks are ropes.  
All ropes are doors.

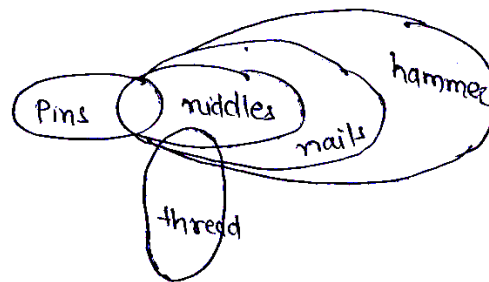


Conclusion

- (I) Some roads are doors. ✗
  - (II) Some doors are bricks. ✓
  - (III) Some roads are not doors. ✗
  - (IV) All doors are ropes. X
- either I or III follow ✓

**Eg. 14** Statement:-

Some pins are niddles.  
Some thread are niddles.  
All niddles are nails.  
All nails are hammers,



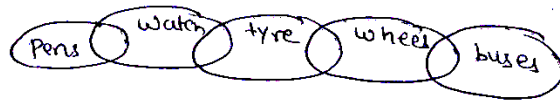
Conclusion:-

- (I) Some pins are hammer. ✓
  - (II) Some threads are nails. ✓
  - (III) Some pins are threads. ✓
  - (IV) No pins are thread. ✓
- either III or IV follow ✓

eg.15

Statements:

Some pens are watches.  
Some watch are tyre.  
Some tyre are wheel.  
Some wheels are buses.



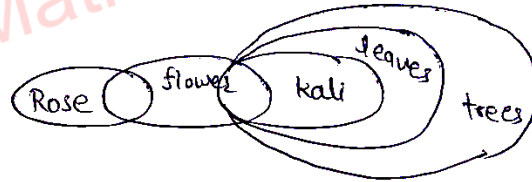
Conclusion:-

- (I) Some buses are tyre. X
- (II) Some wheels are watches, X
- (III) Some wheels are Pens, X
- (IV) Some buses are watches, X

eg.16

Statements:-

Some rose are flower  
Some flower is kali  
All kalis are leaves.  
All leaves are trees.



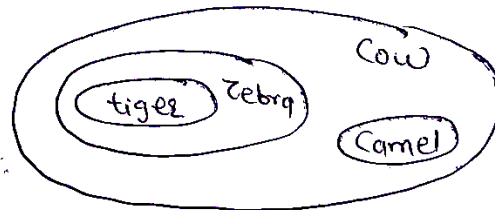
Conclusion

- (I) Some trees are flower. ✓
- (II) Some roses are kali. X
- (III) Some leaves are roses X
- (IV) Some roses are not kali. ✓

eg.17

Statement:-

All zebra are cows  
All camels are cows,  
All tigers are zebra.



Conclusion:

- (I) All tigers are cows. ✓
- (II) All camels being tigers is a possibility. ✓

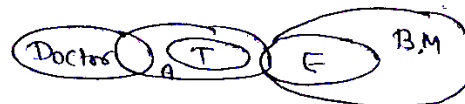
**Eg 18** Statement:-

Some doctors are advocate

All teachers are advocate

Some engineers are advocate.

All engineers are ~~business~~ business man.



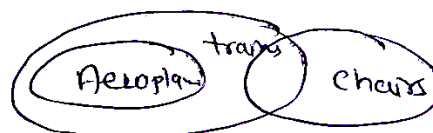
Conclusion:-

- (I) Some Teacher are Doctor. X
- (II) Some businessman are teacher. X
- (III) Some businessman are advocate. ✓
- (IV) Some Advocate are Teacher. ✓

**Eg 19** Statement:-

All aeroplanes are trains.

Some trains are chairs.



Conclusion:-

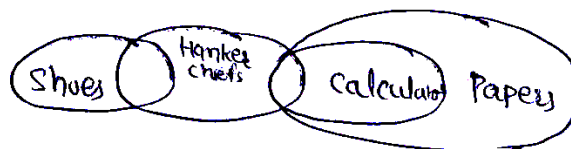
- (I) Some aeroplane are chairs.
- (II) Some chairs are aeroplane.
- (III) Some chairs are trains.
- (IV) Some trains are aeroplane.

**Eg. 20** Statement:

Some shoes are handkerchiefs

Some handkerchiefs are calculators

All calculators are papers.



Conclusion:-

- (I) No calculator is a shoe (X)
- (II) No shoe is a paper. (X)



# Ranking Order

Maths with Gajanand

## 7. Ranking Order

- ★  $\boxed{\text{Total} = \text{Left} + \text{Right} - 1}$
- ★  $\boxed{\text{Total} = \text{Top} + \text{Bottom} - 1}$
- ★  $\boxed{\text{Left} = \text{Total} - \text{Right} + 1}$
- ★  $\boxed{\text{Right} = \text{Total} - \text{Left} + 1}$

### Type-1

1. Ram's ranked seventh from the top and twenty sixth from the bottom in a class. How many students are there in the class?

Soln:-

$$\begin{aligned}\text{Total} &= \text{Top} + \text{Bottom} - 1 \\ &= 26 + 7 - 1 \\ &= 32 \text{ Ans}\end{aligned}$$

2. Shyam's ranked is 9<sup>th</sup> from the top and thirty-eight from the bottom in a class. How many students are there in the class?

Soln:-

$$\text{Total} = 9 + 38 - 1 = 46 \text{ Ans}$$

3. Sachin is fourteenth from the right end in a row of 40 boys. What is his position from left end?

Soln:-

$$\begin{aligned}\text{Left} &= \text{Total} - \text{Right} + 1 \\ \text{Left} &= 40 - 14 + 1 = 27 \text{ Ans}\end{aligned}$$

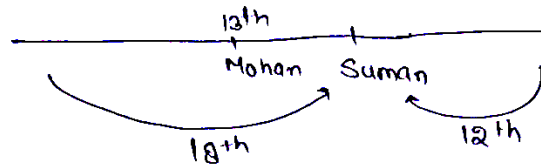
4. 49<sup>th</sup> students of a class Nitin gets 18<sup>th</sup> position from start. What is the rank of Nitin from end?

Soln:-

$$49 - 18 + 1 = 32 \text{ Ans}$$

5. In a row of students. Mohan is the 13<sup>th</sup> from the left end. Suman is 12<sup>th</sup> from Right end and 18<sup>th</sup> from left end. How many students of Mohan's right?

Soln



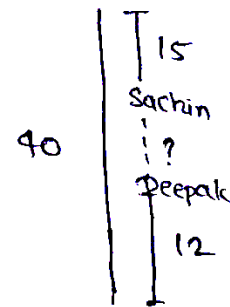
$$\begin{aligned}\text{Total} &= 18 + 12 - 1 \\ &= 29\end{aligned}$$

So  $29 - 13 = 16$  students of Mohan's right side.

6. In a class of 40 students Sachin is 15<sup>th</sup> from the top. Deepak is 12<sup>th</sup> from the bottom. How many boys are in between Sachin and Deepak.

Soln

$$\begin{aligned}&= 40 - (15 + 12) \\ &= 40 - 27 = 13 \text{ Ans}\end{aligned}$$

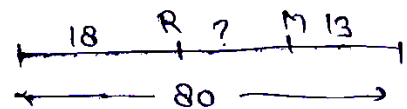


7.

- In a class of 80 students Mayank is 13<sup>th</sup> from the right and Raju is 18<sup>th</sup> from the left. how many students in between Mayank and Raju.

Soln

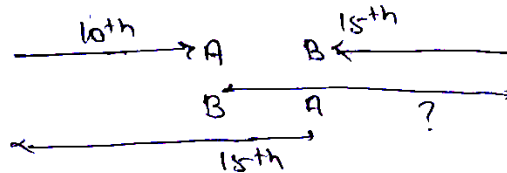
$$\begin{aligned}&= 80 - (18 + 13) \\ &= 49 \text{ Ans}\end{aligned}$$



## Type-2

Ex 1:- A is 10<sup>th</sup> from left and B is 15<sup>th</sup> from Right. When A and B exchange their position then A becomes 15<sup>th</sup> from the left. What is the new position of B from Right?

Soln Ans total =  $15 + 15 - 1$   
 $= 29$   
 $B^{\text{th}}$  from Right =  $29 - 10 + 1$   
 $= 20$  Ans

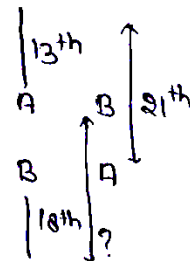


Trick difference between A's new and old position  
 $= 15 - 10 = 5$

B's new position =  $15 + 5 = 20$  Ans

Ex 2 A is 13<sup>th</sup> from top, B is 18<sup>th</sup> from bottom. When A and B change their position then A becomes 21<sup>th</sup> from top. What is the new position of B from bottom?

Soln  $A-B = 21 - 13 = 8$   
 $18 + 8 = 26$  Ans

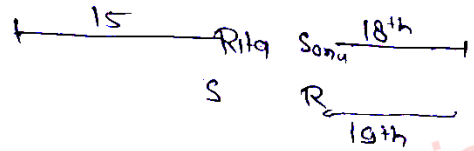


3. ~~दिए गए हैं~~ In a row of a students, Mohan is 10<sup>th</sup> from right. Sohan is 25<sup>th</sup> from left. When they change their position then Mohan becomes 22 from right. What is the new position of Sohan from left.

Soln  $22 - 10 = 12$ ,  $25 + 12 = 37$  Ans

4. In a class of students Rita is 15th from the left and Sonu is 18th from the right. When they exchange their position. Then Rita becomes 19th from right. What is the new position of Sonu from left?

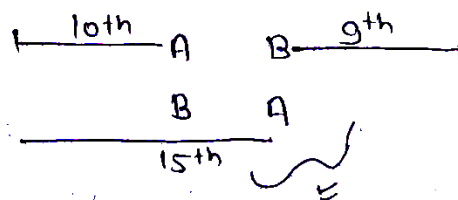
Soln: Can't be determined  
because Rita's position is different from Sonu's old position.



### Type-3

1. In a row of students, if A is 10th from left and B is 9th from the right and interchange their positions, A becomes 15th from the left. How many students are there in the row?

Soln Basic total =  $\frac{L}{15} + \frac{R}{9} - 1$   
= 23 Ans



Trick

total = A's New position + B's old position - 1

=  $15 + 9 - 1$   
= 23 Ans

2. In a row of girls, if Vimala is 9th from the left and Namrata is 16th from the right, interchanging their position, Vimala becomes 25th from the left. find total numbers of girls in row.

Soln  $25 + 16 - 1 = 40$  Ans

3. In a row of students Deepak is 17<sup>th</sup> from top. Sonu is 32<sup>th</sup> from bottom. interchange their positions. Deepak becomes 37<sup>th</sup> from top. What is the total number of students?

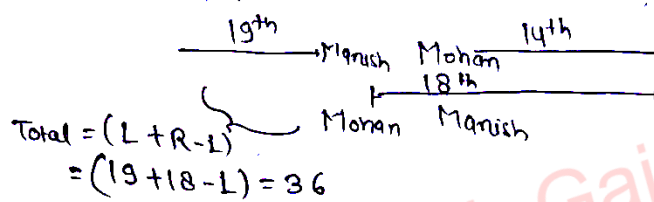
Soln

$$37 + 32 - 1 = \underline{68} \text{ Ans}$$

4. In a row of people, if Mohan stands 14<sup>th</sup> from right and Manish stands 19<sup>th</sup> from left, interchange their position then Mohan becomes 18<sup>th</sup> from right. How many people stands in row?

Soln

$$19 + 18 - 1 = 36$$

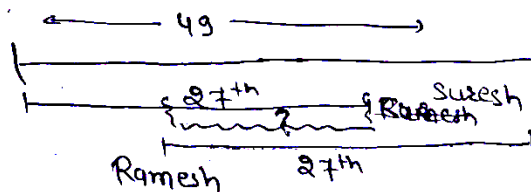


Type - 4

1. Suresh is 27<sup>th</sup> from left and Ramesh is 27<sup>th</sup> from Right. If total 49 people in row, how many people between Suresh and Ramesh?

Soln :- Basic

$$\begin{aligned} & 27 + 27 - 49 \\ &= 5 \\ &= 5 - 2 \text{ (Suresh \& Ramesh)} \\ &= \underline{3} \text{ Ans} \end{aligned}$$



Trick

Position of 1<sup>st</sup> person + Position of 2<sup>nd</sup> person - Total - 2

$$\begin{aligned} &= 27 + 27 - 49 - 2 \\ &= \underline{3} \text{ Ans} \end{aligned}$$

2. In a class of 40 students, if Kashish is 21<sup>st</sup> from right and Deepak is 23 from left. How many boys are there between them in row?

Sol<sup>n</sup>:-  $23 + 21 - 40 - 2 = 2$  Ans

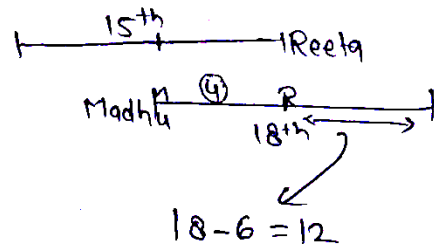
### Type-5

1. In a row, Rita is 15<sup>th</sup> from left and Madhu is 18<sup>th</sup> from right. If there are 4 girls between Madhu and Rita, then how many numbers of total girl in row minimum?

Sol<sup>n</sup>

Basic

$$15 + 18 = 27 \text{ Ans}$$



Trick

Total = Sum of both position - no. of person between both + both person

$$= 15 + 18 - 4 - 2$$

$$= 27 \text{ Ans}$$

2. In a class Deepak is 29<sup>th</sup> from top and Rana is 18<sup>th</sup> from bottom. If there are 6 boys between Deepak and Rana, Then how many students in class minimum?

Sol<sup>n</sup>

$$29 + 18 - 6 - 2$$

$$= 39 \text{ Ans}$$

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### Type-6

1. In a class of 49 students, A is the 17<sup>th</sup> from top and B is the 11<sup>th</sup> from bottom. If C is exact Middle of A and B how many students between B and C.

Soln Basic

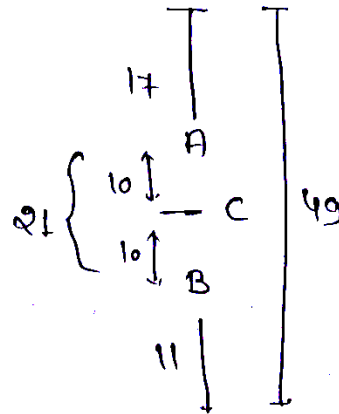
student between A and B

$$= 49 - (17 + 11)$$

$$= 21$$

then students between B & C

$$= 10$$



first we add that position after that subtract from total and then divide by 2

$$\Rightarrow \text{(i)} \quad 17 + 11 = 28$$

$$\text{(ii)} \quad 49 - 28 = 21$$

$$\text{(iii)} \quad \frac{21}{2} = 10.5 \Rightarrow \text{so ans will be } \underline{\underline{10}}$$

2. In a row of 44 people, Vinay is 12<sup>th</sup> from left and Ramu is 17<sup>th</sup> from right. If Sinu is exact middle of Vinay and Ramu. How many people between Vinay and Sinu?

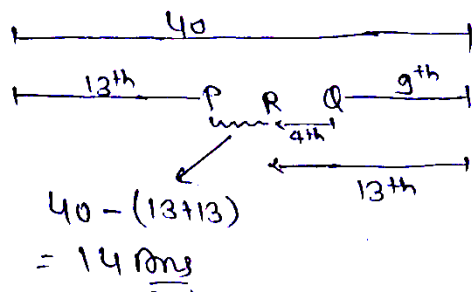
Soln:-  $44 - (12 + 17)$   
 $= 15 \Rightarrow \frac{15}{2} = 7.5 = 7 \underline{\underline{\text{Ans}}}$

3. In a class of 79 students A is 40<sup>th</sup> from left and Simran is 33<sup>th</sup> right. If Priya is exact middle of Rita and Simran, How many students between simran and Priya?

Soln  $79 - (40 + 33) = 6 \Rightarrow \frac{6}{2} = 3 \underline{\underline{\text{Ans}}}$

4. In a row of 40 children, P is 13<sup>th</sup> from the left and Q is 9<sup>th</sup> from the right end. How many children are there between P and Q. If R is 4<sup>th</sup> to the left of Q?

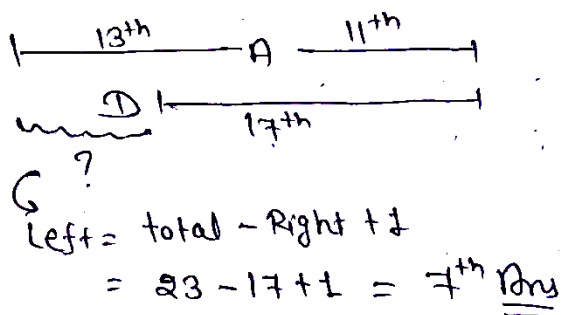
Soln:-



5. In a row of boys, A is 13<sup>th</sup> from left and D is 17<sup>th</sup> from right. if A is 11<sup>th</sup> from right in this line what is the position of D from left?

Soln

$$\begin{aligned}
 \text{total} &= L + R - 1 \\
 &= 13 + 11 - 1 \\
 &= 23
 \end{aligned}$$



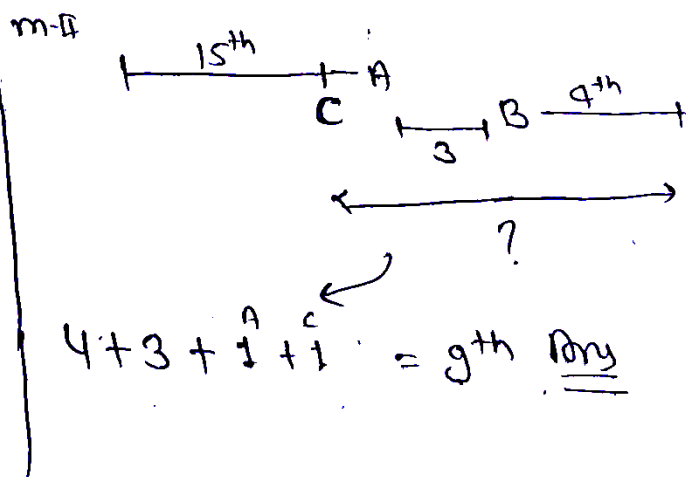
6. In a row of boys, A is 15<sup>th</sup> from left - B is 4<sup>th</sup> from right. 3 boys between A and B, - C is immediate left of A. what is the position of C, from right?

Soln:-

m-I

$$\begin{aligned}
 \text{total} &= 15 + 4 + 3 \\
 &= 22
 \end{aligned}$$

$$\begin{aligned}
 C \rightarrow \text{total} - \text{left} + 1 \\
 &= 22 - 14 + 1 \\
 &= 9 \text{ Ans}
 \end{aligned}$$



# Sitting Arrangement

Maths with Gajanand

## 8. Sitting Arrangement

Type - 1 = Circle Based

1. Six friend A, B, C, D and F seating in a circle facing towards center. E is the left of D. C seats between A and B and F is between E and A.

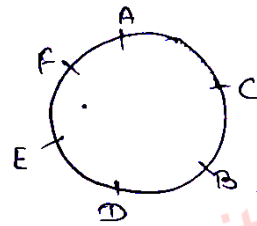
(a) Who is seating immediate right of C?

(b) Who is sitting immediate left of B?

Soln:-

(i) A

(ii) D

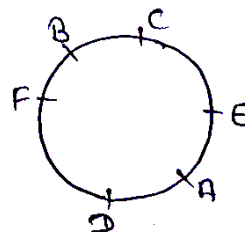


2.

- Six person A, B, C, D, E and F standing in a circular boundary facing towards center. B is between F and C. A is between E and D. F is to the left of D who is seating between A and F.

Soln:-

D Ans

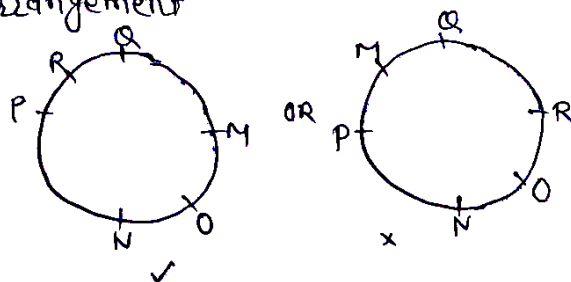


3. Six friends M, N, O, P, Q and R standing in a circle. N is between O and P, M is between O and Q. R and P is standing together. Who is standing between M and R?

Soln Here is possibility of two arrangement

Q is standing between M and R

Ans



## Type-2

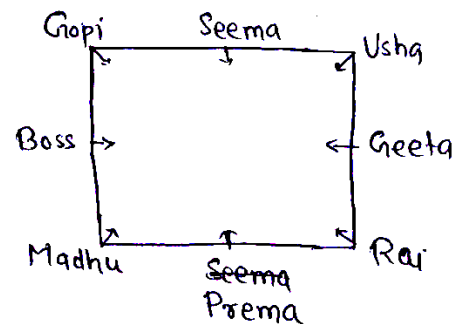
### Square based Question

**eg. 1**

Four girls and four boys sitting in a square facing towards center. Each of them sitting on the corner of square and middle of square. Madhu is diagonal front of Usha and middle of square. Geeta is right to the right of Geeta. Rai is Next to Geeta and and front of Gopi who is left to the boss. Seema is not right of Madhu and front of Prema. Who is seating in front of boss?

Soln:-

Geeta ans

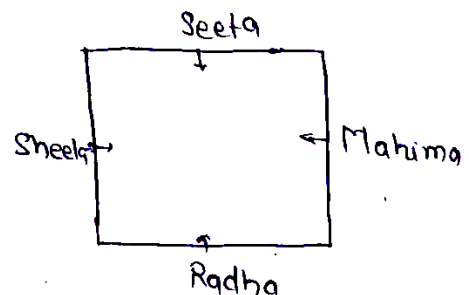


**eg. 2**

Radha, Sheela, Mahima, and Seeta sitting around a square Table. Radha is right to the Sheela. Mahima is left to the Seeta. Who is sitting in front of each other?

Soln:

- (a) Sheela - Seeta
- (b) Radha - Seeta
- (c) Radha - Sheela
- (d) Mahima - Radha



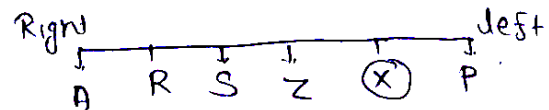
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### Type - 3

#### Row and Line Based Question

**eg. 1** A, P, R, X, S and Z are sitting in a row S and Z are in the center and A and P are at the ends. R is sitting on the left of A. Then who is sitting on the right of P?

Soln X Ans

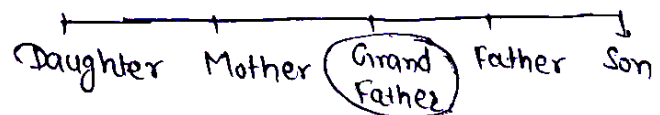


**eg. 2** Roshan is taller than Hardik. Hardik is smaller than Sushil. Nija is taller than Hazi but smaller than Roshan who is the tallest among them?

Soln: Roshan > Sushil > Hardik > Nija > Hazi  
Ans

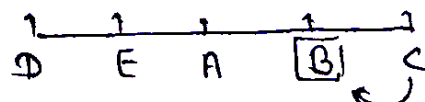
**eg. 3** While the group photo of a family was taken the father was found to be sitting to the left of the son and right to the left of grandfather Mother was sitting to the right of her daughter but left to the grandfather. Who is occupying the central place?

Soln: Grand father Ans



**eg. 4** Five school A, B, C, D and E whose face is toward north. A is between E and B while E is the right to the D. D and C are at the ends. Who is left to the C?

Soln B Ans



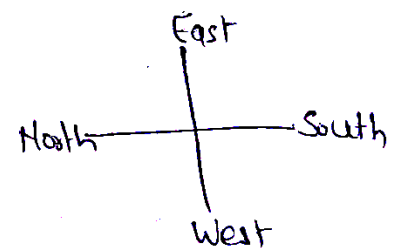
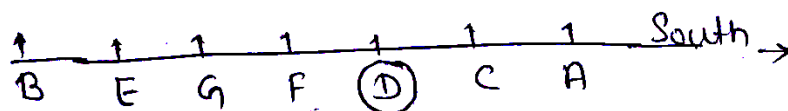
**Eg. 5** Five Friends Suresh, Kaushal, Madhur, Amit and Ramesh. Suresh is smaller than Kaushal but taller than Ramesh. Kaushal is taller than Madhur. Amit is smaller than Kaushal but taller than Suresh. If they are standing according to their height respectively who is become smallest among them.

Soln:-

Madhur > Kaushal > Amit > Suresh > Ramesh  
Ans

**Eg. 6** A, B, C, D, E, F and G sitting on a bench facing East. C is immediate right of D. B is sitting on any end and E is neighbour of G. G is between E and F. D is 3rd to the south where is D sitting.

Soln:-



D is between C and F Ans

Maths with Gajanand

## 9. Symbols & Notations

Rule

BODMAS

- B - Bracket
- O - of
- D - Divide ( $\div$ )
- M - Multiplication ( $\times$ )
- A - Add (+)
- S - Subtraction (-)

Type-1

eg. 1 If '+' stands for ' $\div$ ', ' $\times$ ' stands for '+', '-' stands for ' $\times$ ' and ' $\div$ ' stands for '-' then which of the following statement is correct?

- (a)  $36 \times 6 + 7 \div 2 - 6 = 20$       (b)  $36 \div 6 + 3 \times 5 - 3 = 45$   
 (c)  $36 + 6 - 3 \times 5 \div 3 = 24$       ✓ (d)  $36 - 6 + 3 \times 5 \div 3 = 74$

Sol<sup>n</sup> by checking option

(a)  $36 \times 6 + 7 \div 2 - 6 = 20$   
 $\Rightarrow 36 + 6 \div 7 - 2 \times 6 = 20$   
 $\Rightarrow 36 + \frac{6}{7} - 12 \neq 20$

$+$   $\longrightarrow$   $\div$   
 $\times$   $\longrightarrow$   $+$   
 $-$   $\longrightarrow$   $\times$   
 $\div$   $\longrightarrow$   $-$

(b)  $36 \div 6 + 3 \times 5 - 3 = 45$   
 $\Rightarrow 36 - 6 \div 3 + 5 \times 3 = 45$   
 $\Rightarrow 36 - 2 + 15 \neq 45$

(c)  $36 + 6 - 3 \times 5 \div 3 = 24$   
 $\Rightarrow 36 \div 6 \times 3 + 5 - 3 = 24$   
 $6 \times 3 + 2 \neq 24$

✓ (d)  $36 - 6 + 3 \times 5 \div 3 = 74$   
 $\Rightarrow 36 \times 6 \div 3 + 5 - 3 = 74$   
 $\Rightarrow 72 + 2 = 74$

(d) is correct

**eg. 2** If '-' stands for division, '+' stands for subtraction, ' $\div$ ' stands for multiplication, 'x' stands for addition, then which one of the following equation is correct?

- (a)  $70 - 2 + 4 \div 5 \times 6 = 44$   
 ✓ (b)  $70 - 2 + 4 \div 5 \times 6 = 21$   
 (c)  $70 - 2 + 4 \div 5 \times 6 = 341$   
 (d)  $70 - 2 + 4 \div 5 \times 6 = 36$

Soln:-  
 $70 - 2 + 4 \div 5 \times 6$   
 $\Rightarrow 70 \div 2 - 4 \times 5 + 6$   
 $\Rightarrow 35 - 20 + 6$   
 $\Rightarrow 41 - 20 = \underline{21}$  (b) Ans

**eg. 3** If 'x' stands for addition, 'v' stands for subtraction, 'u' stands for 'equal to', '^' stands for division, 'Σ' stands for multiplication, 'σ' stands for greater than and 'α' stands for 'less than'. State which expression is true.

- (a)  $3 \times 8 v 2 u 12 \wedge 3$  (b)  $13 v 12 \times 9 v 2 \sigma 51 \wedge 3$   
 ✓ (c)  $3 \Sigma 3 \Sigma 4 \sigma 51 \wedge 3$  (d)  $3 \Sigma 2 \Sigma 4 u 2 \times 7 v 3$

Soln:- (a)  $3 + 8 - 2 = 12 \div 3$   
 $9 \neq 4$

(b)  $13 - 12 + 9 - 2 > 51 \div 3$   
 $8 > 17$

✓ (c)  $3 \times 3 \times 4 > 51 \div 3$   
 $36 > 17$  ✓

(d)  $3 \times 2 \times 4 = 2 + 7 - 3$   
 $24 \neq 6$

x  $\rightarrow$  +  
 v  $\rightarrow$  -  
 u  $\rightarrow$  =  
 ^  $\rightarrow$   $\div$   
 Σ  $\rightarrow$  x  
 σ  $\rightarrow$  >  
 α  $\rightarrow$  <

**eg.4** If 'P' stands for '-', 'Q' stands for 'x', R for  $\div$  and S for '+', then what is the value of the given equation  
 $14Q3P12S4R2 = ?$

- (a) 17      ☒ (b) 32      (c) 28      (d) 6

Sol<sup>n</sup>  $\Rightarrow 14 \times 3 - 12 + 4 \div 2$   
 $\Rightarrow 42 - 12 + 2 = \underline{32} \text{ Ans}$

$P \rightarrow -$      $Q \rightarrow \times$   
 $R \rightarrow \div$      $S \rightarrow +$

**eg.5** If  $8+8=72$ ,  $5+5=30$  and  $7+7=56$ , what is  $6+6=?$

Sol<sup>n</sup>  $8+8=72$ ,  $5+5=30 \Rightarrow + \rightarrow \times$   
 $\Rightarrow 8 \times 9^{2+1} = 72$ ,  $\Rightarrow 5 \times 6^2 = 30$   
 $6+6$   
 $\Rightarrow 6 \times 7^{2+1} = \underline{42} \text{ Ans}$

**eg.6** If SENT is written as '+^\*- ' and ANT is written as 'X\*- ' then how is TEN written in that code

Sol<sup>n</sup>      SENT      ANT      Then      TEN       $(-^*) \text{ Ans}$   
 $\downarrow \downarrow \downarrow \downarrow$      $\downarrow \downarrow \downarrow$      $\downarrow \downarrow \downarrow$   
 $+^*-$        $X*-$        $-^*$

**eg.7** Find the correct answer for the unsolved equation on the basis

$5+7+2=725$ ,  $6+9+0=906$

then  $8+4+3=?$

- (a) 815      (b) 384      ☒ (c) 438      (d) 834

Sol<sup>n</sup>  $5+7+2$      $906$      $6+9+0$     then     $8+4+3$   
 $\swarrow \searrow$      $\swarrow \searrow$      $\swarrow \searrow$   
 $725$      $906$      $438 \text{ Ans}$

## Type-2

**Eg. 1** Which one of the following responses is correct?

$$25 * 2 * 6 = 4 * 11 * 0$$

☒ (a)  $\times, -, \times, +$

(b)  $+, -, \times, +$

(c)  $\times, +, \times, -$

(d)  $\times, +, +, \times$

Soln

(a)  $25 \times 2 - 6 = 4 \times 11 + 0$

$\Rightarrow 44 = 44 \checkmark$

**Eg. 2** Select the correct combination of mathematical signs to replace \* signs and to balance the given equation.

$$16 * 4 * 64 * 4$$

(a)  $\times, <, \div$     ☒ (b)  $\times, >, \div$     (c)  $\div, >, \times$     (d)  $\times, >, +$

Soln

(a)  $16 \times 4 < 64 \div 4$     (b)  $16 \times 4 > 64 \div 4$

$64 < 16$   
 $\times$

$\Rightarrow 64 > 16$   
 $\checkmark$

## Type-3

[Interchange and Replace type Question]

**Eg. 1** After interchanging  $\div$  and  $+$ , 12 and 18, which one of the following equation becomes correct?

(a)  $(90 \times 18) + 18 = 60$

(b)  $(18 + 6) \div 12 = 2$

(c)  $(72 \div 18) \times 18 = 72$

☒ (d)  $(12 + 6) \times 18 = 36$

Soln

(d)  $(12 + 6) \times 18 = 36$

$\Rightarrow (18 \div 6) \times 12 = 36$

$36 = 36 \checkmark$

**Eg. 2** Which of the following interchange of signs would make the given equation correct?

$$(12 \div 6) + 3 \times 7 = 42$$

- (a) + and  $\times$       (b) 6 and 7      (c)  $\div$  and +      (d) 12 and 3

Soln (c)  $(12 \div 6) + 3 \times 7 = 42$   
 $\Rightarrow (12 + 6) \div 3 \times 7 = 42$   
 $\Rightarrow 18 \div 3 \times 7 = 42$   
 $\underline{42 = 42}$

**Eg. 3** Which one of the four interchanges in signs and numbers would make the given equation correct?

$$6 \times 4 + 2 = 16$$

- (a) + and  $\times$ , 16 and 4      (b) + and  $\times$ , 2 and 4  
 $\checkmark$  (c) + and  $\times$ , 4 and 6      (d) None of these.

Soln (c)  $6 \times 4 + 2 = 16$   
 $\Rightarrow 4 + 6 \times 2 = 16$   
 $\Rightarrow 4 + 12 = 16$

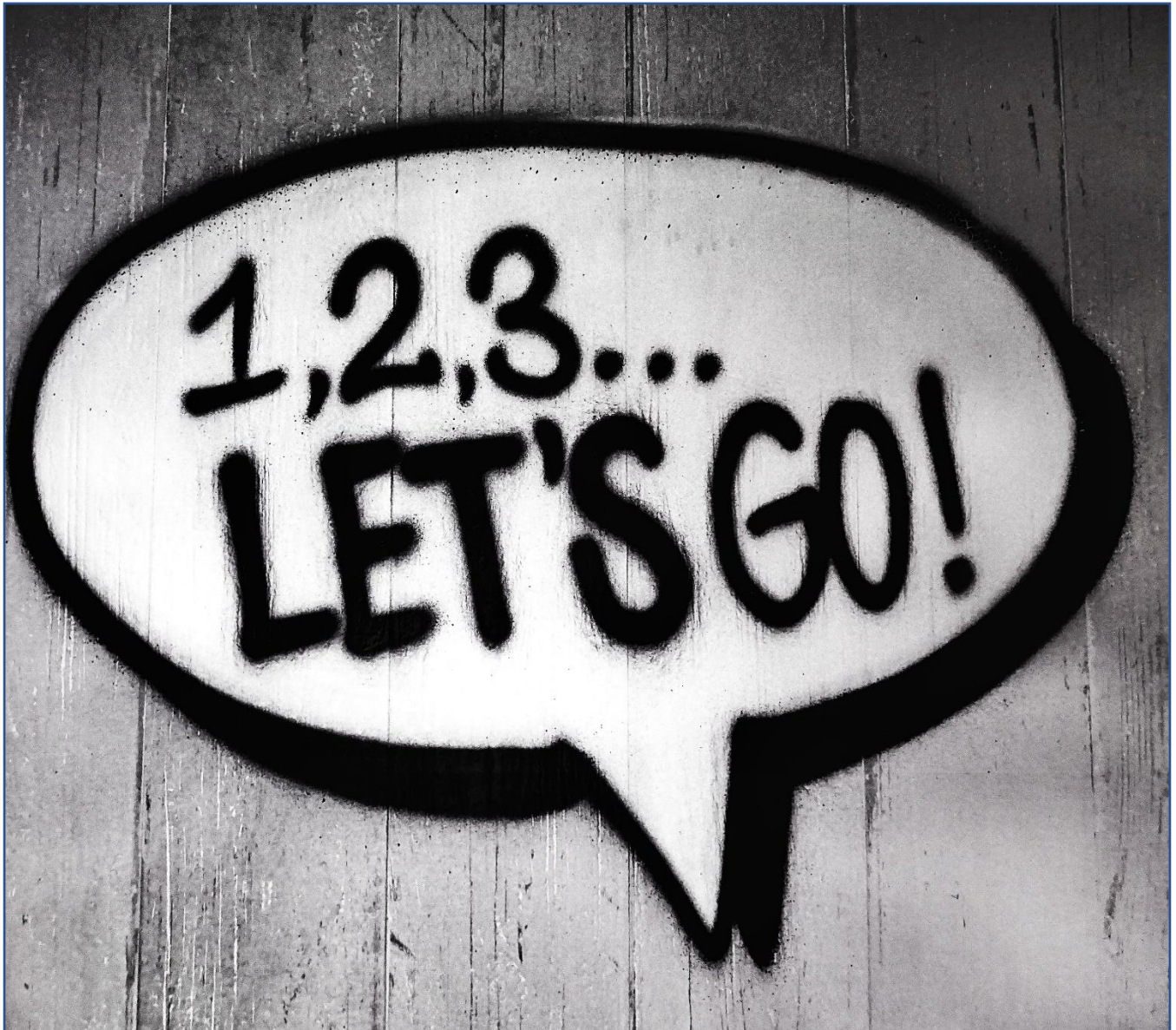
Maths with Gajanand

**Eg. 4** Interchange of signs and numbers would make the given equation correct.

$$6 + 2 - 3 = 16$$

- (a) + and  $-$ , 2 and 3      (b)  $\times$  and +, 3 and 2  
 (c)  $\div$  and  $-$ , 3 and 2      (d)  $\times$  and  $-$ , 2 and 3

Soln:- (b)  $6 + 2 - 3 = 16$   
 $\Rightarrow 6 \times 3 - 2 = 16$   
 $\Rightarrow 18 - 2 = 16$   
 $\Rightarrow \underline{16 = 16}$



# series

## 10. Series

There are two types of series:-

(i) Number series

(ii) Alphabet series

### Number series

Some Rules for number series:-

Add :-

$$2, 5, 8, 11, 14, 17, 20, \dots$$

$\underbrace{\quad\quad\quad}_{+3} \quad \underbrace{\quad\quad\quad}_{+3} \quad \underbrace{\quad\quad\quad}_{+3} \quad \underbrace{\quad\quad\quad}_{+3} \quad \underbrace{\quad\quad\quad}_{+3} \quad \underbrace{\quad\quad\quad}_{+3} \dots$

eg.

$$2, 4, 7, 11, 16, 22, 29, 37, \dots$$

$\underbrace{\quad\quad}_{+2} \quad \underbrace{\quad\quad}_{+3} \quad \underbrace{\quad\quad}_{+4} \quad \underbrace{\quad\quad}_{+5} \quad \underbrace{\quad\quad}_{+6} \quad \underbrace{\quad\quad}_{+7} \quad \underbrace{\quad\quad}_{+8} \dots$

Add + Subtract :-

$$5, 4, 6, 5, 7, 6, 8, \dots$$

$\underbrace{\quad\quad}_{-1} \quad \underbrace{\quad\quad}_{+2} \quad \underbrace{\quad\quad}_{-1} \quad \underbrace{\quad\quad}_{+2} \quad \underbrace{\quad\quad}_{-1} \quad \underbrace{\quad\quad}_{+2} \dots$

Difference of difference are equal :-

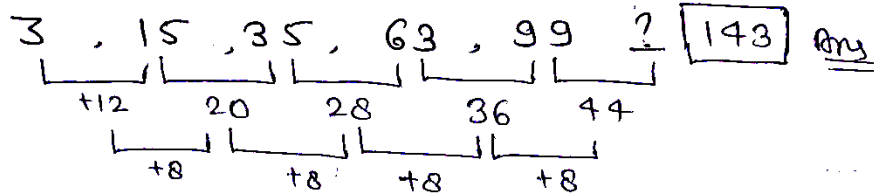
eg.

$$3, 10, 20, 33, 49, 68, ? \rightarrow \boxed{90} \text{ Ans}$$

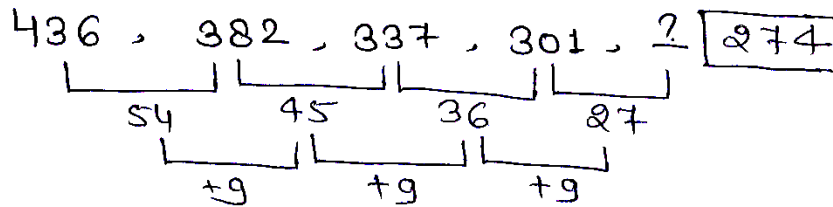
$\underbrace{\quad\quad}_{7} \quad \underbrace{\quad\quad}_{10} \quad \underbrace{\quad\quad}_{13} \quad \underbrace{\quad\quad}_{16} \quad \underbrace{\quad\quad}_{19} \quad \underbrace{\quad\quad}_{22}$

$\underbrace{\quad\quad}_{+3} \quad \underbrace{\quad\quad}_{+3} \quad \underbrace{\quad\quad}_{+3} \quad \underbrace{\quad\quad}_{+3} \quad \underbrace{\quad\quad}_{+3}$

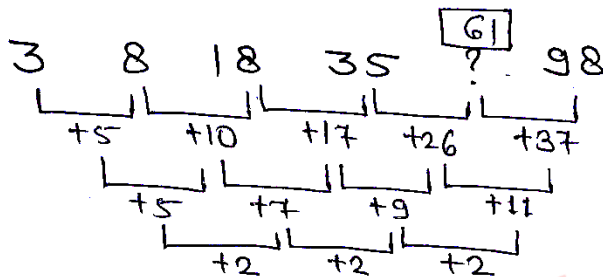
eg.



eg.

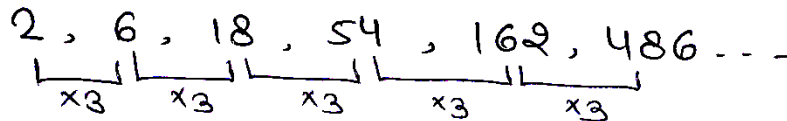


eg.

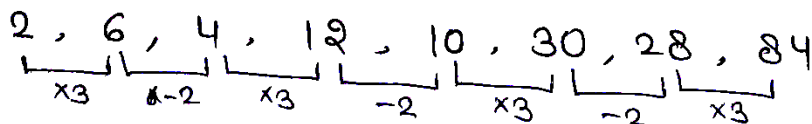


Multiplication:-

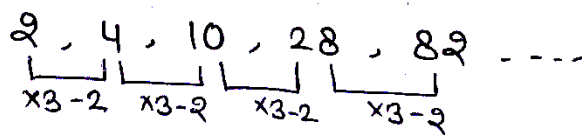
eg.



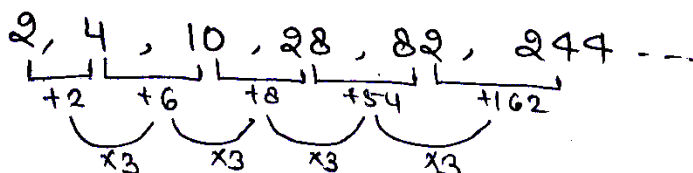
eg.



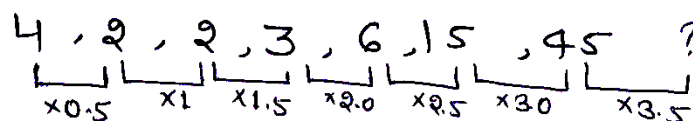
eg.



eg.



eg.



$45 \times 3.5 = 157.5$   
Ans

Square :- 4, 9, 16, 25, 36, 49, 64, 81, ---

[eg.1] 2, 6, 15, 31, 56, 92, 141, ---  
 $\begin{array}{cccccc} \hline & 2 & 6 & 15 & 31 & 56 & 92 & 141 \\ \hline & +4 & +9 & +16 & +25 & +36 & +49 \\ \hline \end{array}$

[eg.2] 2, 7, 14, 23, 34, 47, 62, ---  
 $\begin{array}{cccccc} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2^2-2 & 3^2-2 & 4^2-2 & 5^2-2 & 6^2-2 & 7^2-2 & 8^2-2 \end{array}$

[eg.3] 2, 7, 17, 34, 60, 97, 147, ---  
 $\begin{array}{cccccc} \hline & 2 & 7 & 17 & 34 & 60 & 97 & 147 \\ \hline & 5 & 10 & 17 & 26 & 37 & 50 \\ \hline 2^2+1 & 3^2+1 & 4^2+1 & 5^2+1 & 6^2+1 & 7^2+1 \end{array}$

[eg.4] 10, 40, 90, 161, 272, 423, 614, ---  
 01, 04, 09, 16, 25, 36, 49, ---

Cube :- 8, 27, 64, 125, 216, 343, 512, 729, ---

[eg.1] 6, 24, 60, 120, 210, 336, ---  
 $\begin{array}{cccccc} 2^3-2 & 3^3-3 & 4^3-4 & 5^3-5 & 6^3-6 & 7^3-7 \end{array}$

[eg.2] 4, 11, 30, 67, 128, 219, ---  
 $\begin{array}{cccccc} 1^3+3 & 2^3+3 & 3^3+3 & 4^3+3 & 5^3+3 & 6^3+3 \end{array}$

[eg.3] 1, 1, 4, 8, 9, 27, ?  
 $\begin{array}{cccccc} (1)^2 & (1)^3 & (2)^2 & (2)^3 & (3)^2 & (3)^3 & (4)^2 \end{array}$  16 Ans

[eg.4] 56, 72, 90, 110, 132, ?  
 $\begin{array}{cccccc} 7^2+7 & 8^2+8 & 9^2+9 & 10^2+10 & 11^2+11 & 12^2+12 \end{array}$  156 Ans

eg. 5  $336, 210, 120, \underline{\quad}, 24, 6, 0$   
 $7^3-7 \quad 6^3-6 \quad 5^3-5 \quad \underline{4^3-4} \quad 3^3-3 \quad 2^3-2 \quad 1^3-1$   
 $\rightarrow 60 \text{ Ans}$

eg. 6  $4, 18, \underline{\quad}, 100, 180, 294, 448$   
 $2^2 \times 1 \quad 3^2 \times 2 \quad \underline{4^2 \times 3} \quad 5^2 \times 4 \quad 6^2 \times 5 \quad 7^2 \times 6 \quad 8^2 \times 7$   
 $\rightarrow 48 \text{ Ans}$

eg. 7  $2, 3, 10, 29, 66, \underline{\quad}$   
 $0^3+2, 1^3+2, 2^3+2, 3^3+2, 4^3+2, 5^3+2 = \underline{127} \text{ Ans}$

Mathematical operation  $\rightarrow$  add, subtract, multiply, divide :-

eg. 1  $3, 7, 23, 95, \underline{\quad}$   
 $\underbrace{\quad}_{\times 2+1} \quad \underbrace{\quad}_{\times 3+2} \quad \underbrace{\quad}_{\times 4+3} \quad \underbrace{\quad}_{\times 5+4} \quad \underline{479} \text{ Ans}$

eg. 2  $80, 16, 4, \frac{4}{3}, \frac{2}{3}, \underline{\quad}$   
 $\underbrace{\quad}_{\div 5} \quad \underbrace{\quad}_{\div 4} \quad \underbrace{\quad}_{\div 3} \quad \underbrace{\quad}_{\div 2} \quad \underbrace{\quad}_{\div 1} \rightarrow \underline{\frac{2}{3}} \text{ Ans}$

eg. 3  $4, 10, \underline{28}, 82, 244, 730$   
 $\underbrace{\quad}_{\times 3-2} \quad \underbrace{\quad}_{\times 3-2} \quad \underbrace{\quad}_{\times 3-2} \quad \underbrace{\quad}_{\times 3-2} \quad \underbrace{\quad}_{\times 3-2}$

Miscellaneous (Puzzle) :-

eg. 1  $10, 10, 20, 15, 30, 20, 40, 25, 50, \dots$   
 $\underbrace{\quad}_{+10} \quad \underbrace{\quad}_{+10} \quad \underbrace{\quad}_{+20} \quad \underbrace{\quad}_{+10}$   
 $\quad \quad \quad \underbrace{\quad}_{+5} \quad \underbrace{\quad}_{+5} \quad \underbrace{\quad}_{+5}$

eg 2  $2, 3, 5, 8, 19, 21, 34, 55$

$(2+3) = +5$   $(3+5) = +8$   $(5+8) = +13$   $(8+19) = +27$   $(19+21) = +40$   $(21+34) = +55$

eg 3  $181, 191, 202, 206, 214, ?$  221 Ans

Soln

|                              |                |
|------------------------------|----------------|
| $181 \Rightarrow 1+8+1 = 10$ | $181+10 = 191$ |
| $191 \Rightarrow 1+9+1 = 11$ | $191+11 = 202$ |
| $202 \Rightarrow 2+0+2 = 4$  | $202+4 = 206$  |
| $206 \Rightarrow 2+0+6 = 8$  | $206+8 = 214$  |
| $214 \Rightarrow 2+1+4 = 7$  | $214+7 = 221$  |

eg 4  $6, 15, 35, 77, 143, 221$

$2 \times 3$   $3 \times 5$   $5 \times 7$   $7 \times 11$   $11 \times 13$   $13 \times 17$

Example

1.  $3, 10, 101, 10202$

$\times 3+1$   $\times 10+1$   $\times 101+1$

2.  $-1, 2, 7, 14, 23, 34, 47$

$+3$   $+5$   $+7$   $+9$   $+11$   $+13$

3.  $8, 29, 113, 449, ?$  1793 Ans

$\times 4-3$   $\times 4-3$   $\times 4-3$   $\times 4-3$

4.  $2, 8, 40, 240, 1680, 13440$

$\times 4$   $\times 5$   $\times 6$   $\times 7$   $\times 8$

5.  $7, 25, 61, 121, ?$

$(2^3-1)$   $(3^3-2)$   $(4^3-3)$   $(5^3-4)$   $(6^3-5) = 211$  Ans

## Alphabet Series

Type.1

A, C, F, J, O, ?, B

- (a) I      (b) H      (c) ~~U~~      (d) D

Soln:-

<sup>1</sup>A, <sup>3</sup>C, <sup>6</sup>F, <sup>10</sup>J, <sup>15</sup>O, <sup>21</sup>?, B

+2    +3    +4    +5    +6

21 → U Ans

2. AZ, CX, FU, ?

- (a) JQ      (b) KP      (c) IR      (d) IV

Soln

AZ, CX, FU, JQ Ans

+2    +3    +4

-2    -3    -4

3. KUZ, MOX, OIV, QET, ?

- (a) SUR      (b) SAR      (c) SAG      (d) SDR

Soln

KUZ, MOX, OIV, QET, SAR

-2    -2    -2    -2

+2    +2    +2    +2

Vowels: (U), (O), (I), (E), (A)

4. 1ZA, 3YB, 6XC, 10WD, ?

- (a) 14VE      (b) 15UE      (c) 12VE      (d) ~~15VE~~

Soln

1ZA, 3YB, 6XC, 10WD, 15VE Ans

+2    +3    +4    +5

-1    -1    -1    -1

5. PRIMARILY, RIMARILY, RIMARIL, ?

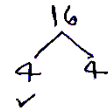
- (a) IMAR      (b) RIMARI      (c) ~~IMARIL~~      (d) RIMA

## Type 2

1.  $a\_b\_abb\_ab\_a\_bba$

- (a) bbaab (b) babba ☒ (c) baaba (d) aabba

Soln:  $a\_b\_a/$   
 $a\_b\_b\_a/ab\_b\_a/ab\_b\_a/a\_b\_b\_a$



2.  $\_a\_aaaba\_ \_ba\_ab\_$

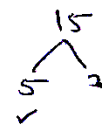
- ☒ (a) abaqa (b) abaaba (c) aababa (d) ababaa

Soln:  $a\_a\_b\_a/a\_a\_b\_a/a\_a\_b\_a/a\_a\_b\_a$  abaqa Ans

3.  $ab\_bcbca\_ \_c\_bab$

- (a) abcc (b) baqa (c) acbb ☒ (d) ccaa

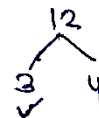
Soln:  $ab\_c\_bc/b\_c\_a\_c/a\_c\_a\_bab$



4.  $l\_n\_mlm\_n\_l$

- (a) mnmn ☒ (b) mnmm (c) mnmn (d) nmnm

Soln:  $l\_m\_n/n\_m\_l/l\_m\_n/n\_m\_l$



5.  $p\_qsp\_ \_sprq\_prqs$

- (a) srqs ☒ (b) rrrqs (c) rrrqs (d) rrrqs

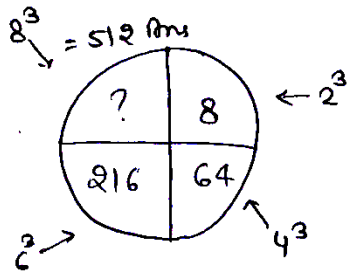
Soln:  $p\_r\_q\_s/p\_r\_q\_s/p\_r\_q\_s/p\_r\_q\_s$

rrqs Ans

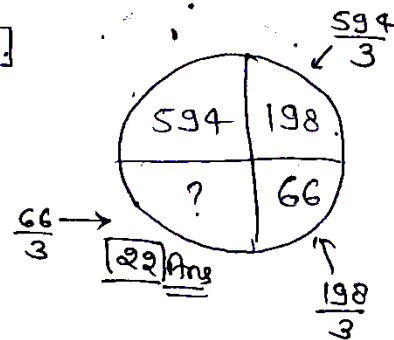
# Missing Number

## 11. Missing Number

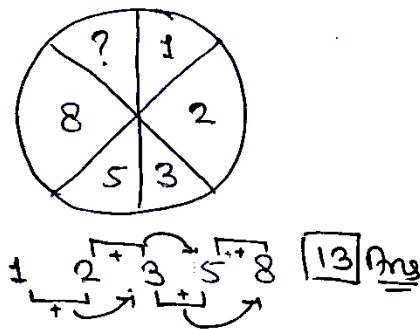
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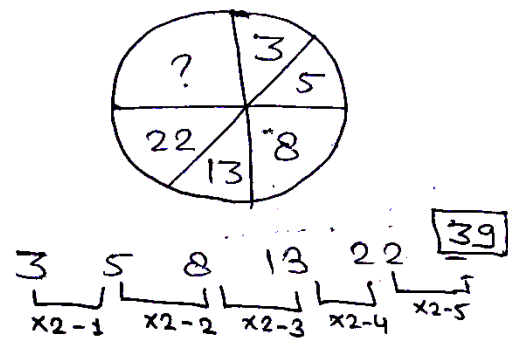
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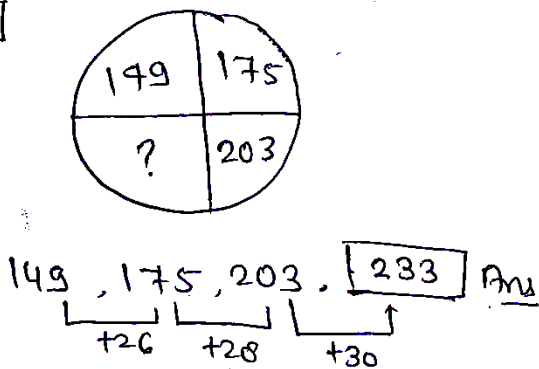
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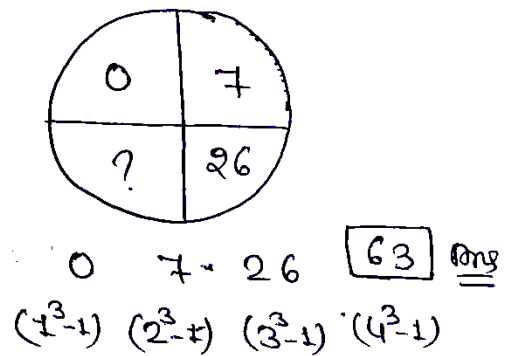
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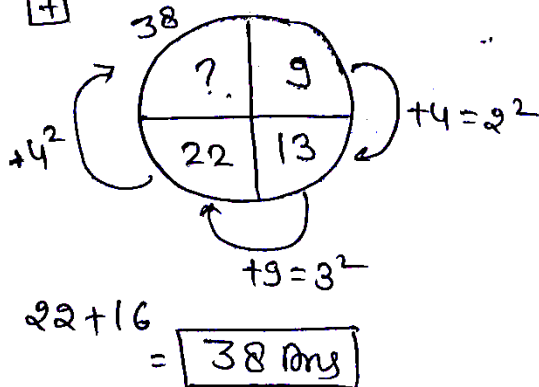
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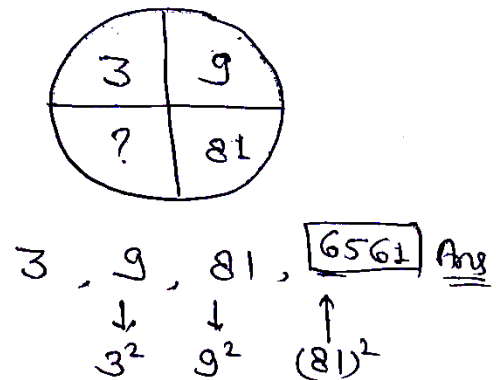
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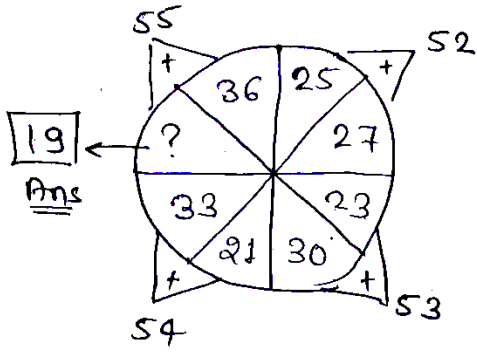
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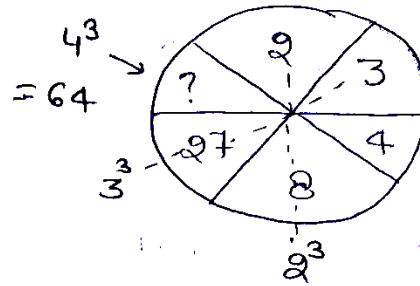
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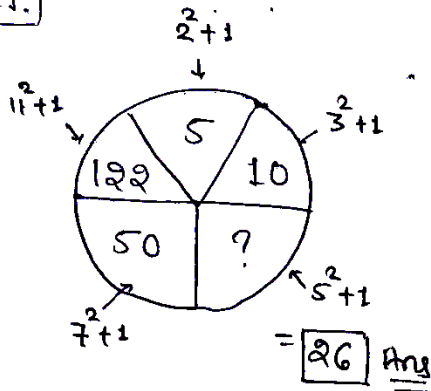
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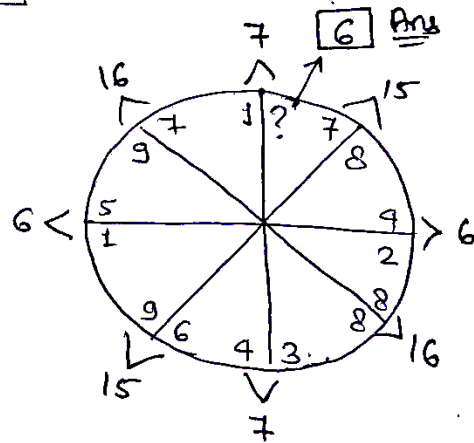
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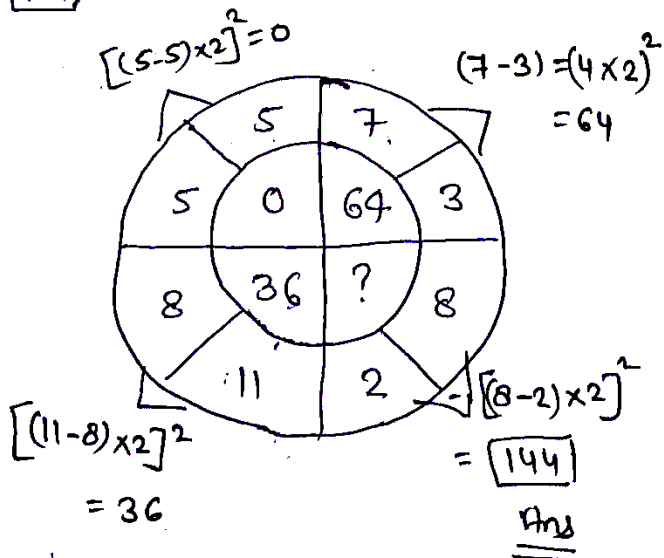
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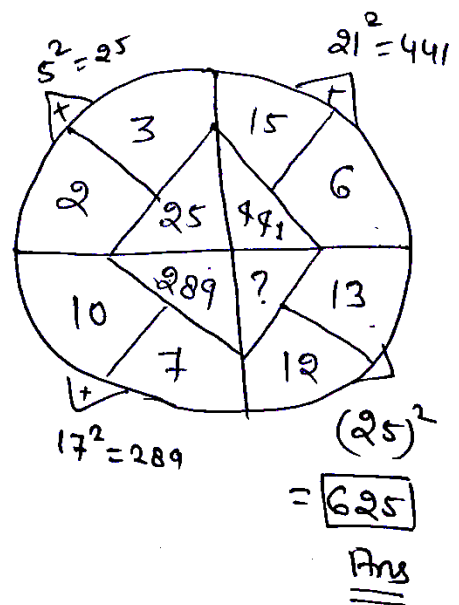
12.



13.



14.



15

|   |    |    |    |
|---|----|----|----|
|   | 6  | 15 | 20 |
| x | 8  | 4  | 5  |
| + | 3  | 5  | 20 |
| ↓ | 51 | 65 | ?  |

$$6 \times 8 + 3 = 51$$

$$20 \times 5 + 20 = \boxed{120} \text{ Ans}$$

16

|                      |    |    |    |
|----------------------|----|----|----|
|                      | 18 | 11 | 19 |
| (18-12) <sup>2</sup> | 12 | 13 | 16 |
| ↓                    | 36 | 4  | ?  |

$$(18-12)^2 = 36, (11-13)^2 = 4$$

$$(19-16)^2 = \boxed{9} \text{ Ans}$$

17

|   |   |    |    |
|---|---|----|----|
| 3 | 4 | 9  | 16 |
| 5 | 6 | 25 | 36 |
| 7 | 8 | ?  | 64 |

$$\text{row:- } 3 + 4 + 9 = 16$$

$$5 + 6 + 25 = 36$$

$$7 + 8 + x = 64$$

$$x = 49, \text{ Ans}$$

18

|     |    |   |
|-----|----|---|
| 85  | 20 | 5 |
| 126 | 24 | 6 |
| 175 | ?  | 7 |

$$\text{Row:- } (20 \times 5) - (20 - 5) = 85$$

$$(24 \times 6) - (24 - 6) = 126$$

$$(28 \times 7) - (28 - 7) = 175$$

$$\boxed{28}, \text{ Ans}$$

19

|    |    |    |    |
|----|----|----|----|
| 14 | 9  | 12 | 20 |
| 4  | 9  | 8  | 10 |
| 12 | 13 | 7  | 20 |
| 3  | 3  | 11 | ?  |
| 20 | 42 | 19 | 40 |

$$1^{\text{st}} \text{ Column:- } (14 \times 4) - (12 \times 3) = 20$$

$$4^{\text{th}} \text{ Column:- } (20 \times 10) - (20 \times x) = 40$$

$$x = 8, \text{ Ans}$$

20

|   |    |    |   |    |   |
|---|----|----|---|----|---|
| 3 | 8  | 10 | 2 | ?  | 1 |
| 6 | 56 | 90 | 2 | 20 | 0 |

$$1^{\text{st}} \text{ Column:- } 3^2 - 3 = 6$$

$$2^{\text{nd}} \text{ " :- } 8^2 - 8 = 56$$

$$x^2 - x = 20$$

$$x = 5, \text{ Ans}$$

15

|   |    |    |    |
|---|----|----|----|
|   | 6  | 15 | 20 |
| x | 8  | 4  | 5  |
| + | 3  | 5  | 20 |
| ↓ | 51 | 65 | ?  |

$$6 \times 8 + 3 = 51$$

$$20 \times 5 + 20 = \underline{120} \text{ Ans}$$

16

|                      |    |    |    |
|----------------------|----|----|----|
|                      | 18 | 11 | 19 |
| (18-12) <sup>2</sup> | 12 | 13 | 16 |
| ↓                    | 36 | 4  | ?  |

$$(18-12)^2 = 36, (11-13)^2 = 4$$

$$(19-16)^2 = \underline{9} \text{ Ans}$$

17

|   |   |    |    |
|---|---|----|----|
| 3 | 4 | 9  | 16 |
| 5 | 6 | 25 | 36 |
| 7 | 8 | ?  | 64 |

$$\text{row:- } 3 + 4 + 9 = 16$$

$$5 + 6 + 25 = 36$$

$$7 + 8 + x = 64$$

$$x = \underline{49} \text{ Ans}$$

18

|     |    |   |
|-----|----|---|
| 85  | 20 | 5 |
| 126 | 24 | 6 |
| 175 | ?  | 7 |

$$\text{Row:- } (20 \times 5) - (20 - 5) = 85$$

$$(24 \times 6) - (24 - 6) = 126$$

$$(28 \times 7) - (28 - 7) = 175$$

$$\underline{28} \text{ Ans}$$

19

|    |    |    |    |
|----|----|----|----|
| 14 | 9  | 12 | 20 |
| 4  | 9  | 8  | 10 |
| 12 | 13 | 7  | 20 |
| 3  | 3  | 11 | ?  |
| 20 | 42 | 19 | 40 |

$$1^{\text{st}} \text{ Column:- } (14 \times 4) - (12 \times 3) = 20$$

$$4^{\text{th}} \text{ Column:- } (20 \times 10) - (20 \times x) = 40$$

$$x = \underline{8} \text{ Ans}$$

20

|   |    |    |   |    |   |
|---|----|----|---|----|---|
| 3 | 8  | 10 | 2 | ?  | 1 |
| 6 | 56 | 90 | 2 | 20 | 0 |

$$1^{\text{st}} \text{ Column:- } 3^2 - 3 = 6$$

$$2^{\text{nd}} \text{ " :- } 8^2 - 8 = 56$$

$$x^2 - x = 20$$

$$x = \underline{5} \text{ Ans}$$

Q1

|   |   |    |
|---|---|----|
| 9 | 4 | 20 |
| 8 | 5 | 12 |
| 7 | 6 | ?  |

I<sup>st</sup> row:-  $(9-4) \times 4 = 20$

II<sup>nd</sup> row  $(8-5) \times 4 = 12$

III<sup>rd</sup> row  $(7-6) \times 4 = \boxed{4}$  Ans

Q2

|    |    |    |    |
|----|----|----|----|
| 4  | 9  | 17 | 6  |
| 20 | 5  | 8  | 9  |
| 7  | 23 | 9  | 9  |
| ?  | 9  | 4  | 19 |

I<sup>st</sup> row:-  $(4+9+6)-17 = 2$

II<sup>nd</sup> :-  $(20+5+9)-30 = 2$

III<sup>rd</sup> :-  $9+9+7-23 = 2$

IV :-  $(x+9+4)-19 = 2$

$x = 8$  Ans

Q3

|   |               |                |
|---|---------------|----------------|
| 1 | $\frac{1}{2}$ | $\frac{3}{2}$  |
| 2 | $\frac{2}{3}$ | $\frac{8}{3}$  |
| 3 | ?             | $\frac{19}{5}$ |

I<sup>st</sup> row:-  $\left(\frac{3}{2} - 1\right) = \frac{1}{2}$

$\left(\frac{8}{3} - 2\right) = \frac{2}{3}$

$\left(\frac{19}{5} - 3\right) = \boxed{\frac{4}{5}}$  Ans

Q4

|   |   |    |    |
|---|---|----|----|
| 2 | 9 | 11 | -7 |
| 8 | 5 | 13 | -3 |
| 7 | ? | 10 | -4 |
| 6 | 4 | 10 | ?  |

I<sup>st</sup> row  $2+9 = 11$  ,  $9-2 = 7$

II<sup>nd</sup> row  $8+5 = 13$  ,  $5-8 = -3$

III<sup>rd</sup> row:  $7 + \boxed{3} = 10$  ,  $3-7 = -4$

IV<sup>th</sup> row  $6+4 = 10$  ,  $4-6 = \boxed{-2}$  Ans

Q5

|     |     |     |
|-----|-----|-----|
| 113 | 118 | ?   |
| 112 | ?   | 116 |
| ?   | 110 | 115 |

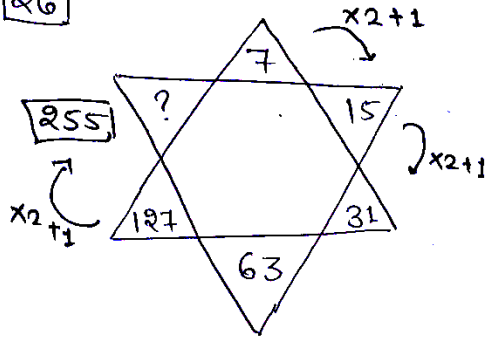
$113 + 118 + \boxed{111} = 342$

$112 + \boxed{114} + 116 = 342$

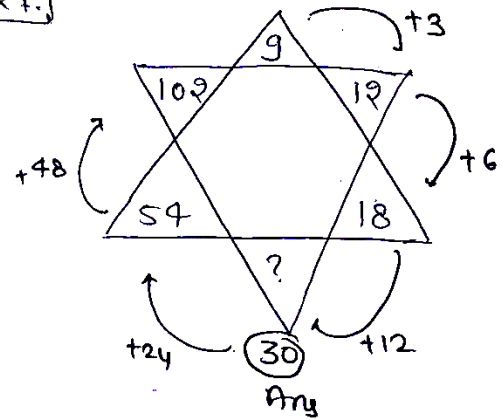
$\boxed{117} + 110 + 115 = 342$

also  $\overline{342} \quad \overline{342} \quad \overline{342}$

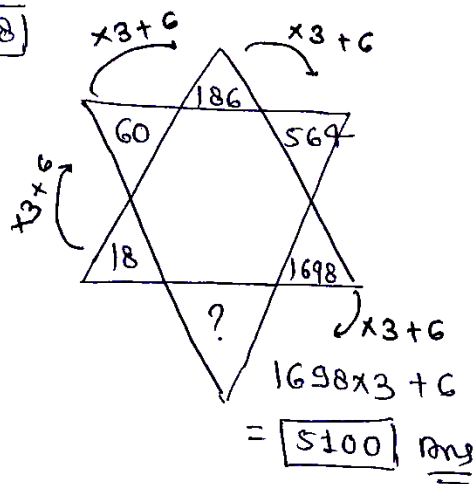
26



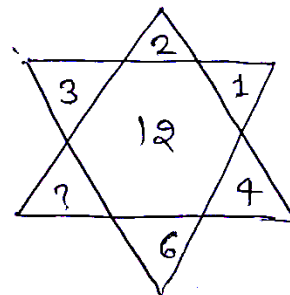
27.



28



29



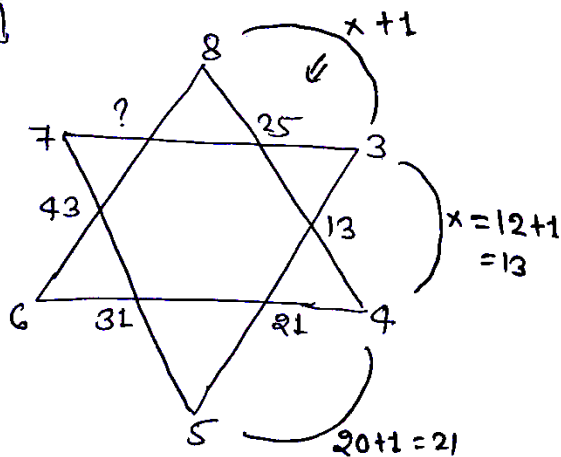
$$3 \times 4 = 12$$

$$2 \times 6 = 12$$

$$x \times 1 = 12$$

$$x = 12 \text{ Ans}$$

30

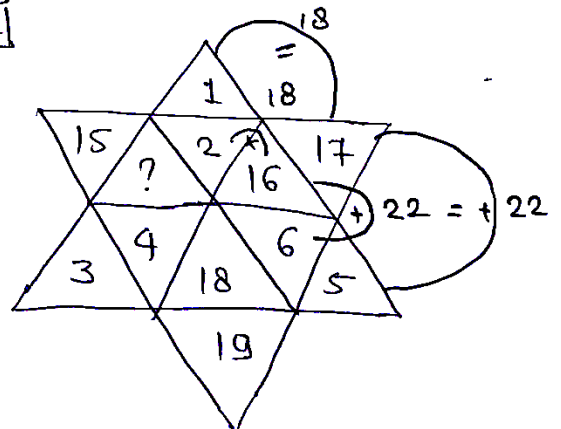


$$(8 \times 3) + 1 = 25$$

$$(5 \times 6) + 1 = 31$$

$$(7 \times 8) + 1 = 57 \text{ Ans}$$

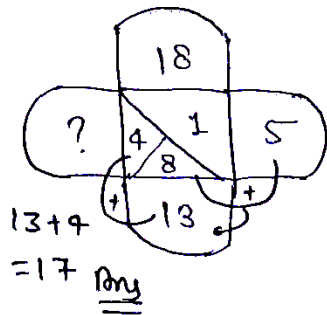
31



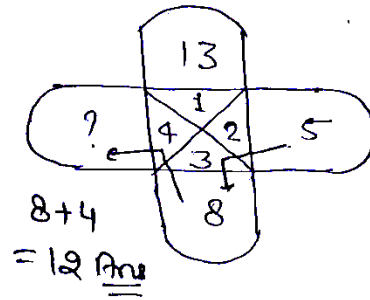
$$? + 4 = 15 + 3$$

$$? = 14 \text{ Ans}$$

32

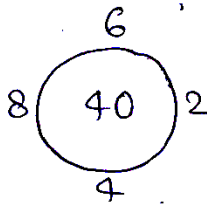


33

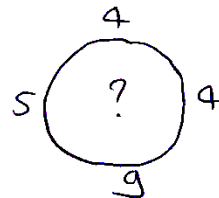
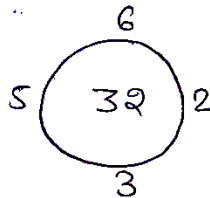


Type-2

34

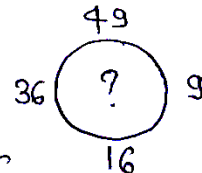
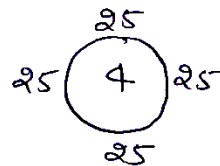
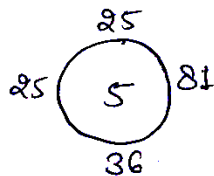
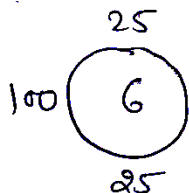


$$(6 + 2 + 4 + 8) \times 2 = 40$$



$$(5 + 4 + 4 + 9) \times 2 = 44$$
 Ans

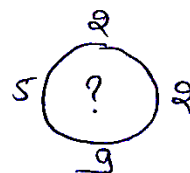
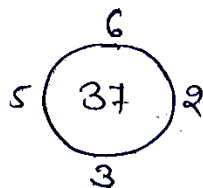
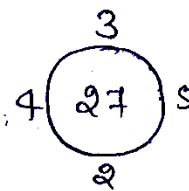
35



$$\Rightarrow \frac{\sqrt{25} + \sqrt{100} + \sqrt{25} + \sqrt{100}}{5} = 6$$

$$\Rightarrow \frac{6 + 7 + 3 + 4}{5} = 4$$
 Ans

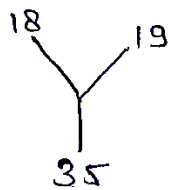
36



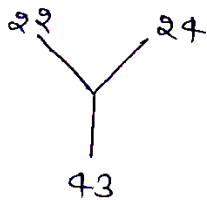
$$\frac{4^2 + 3^2 + 5^2 + 2^2}{2} = 27$$

$$\Rightarrow \frac{5^2 + 2^2 + 2^2 + 9^2}{2} = 57$$
 Ans

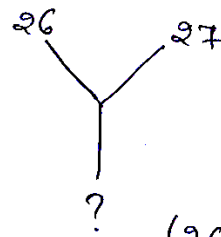
37]



$$(18+19)-2 = 35$$

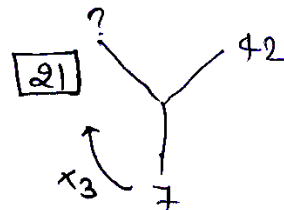
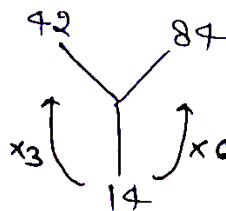
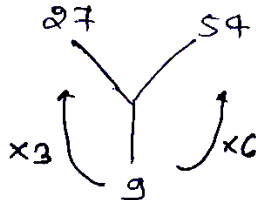


$$(22+24)-3 = 43$$

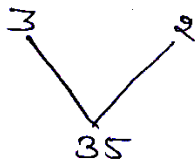


$$(26+27)-4 = 49 \text{ Ans}$$

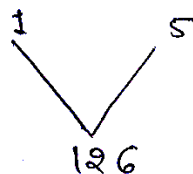
38]



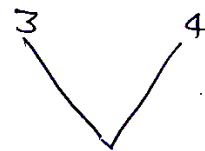
39]



$$(3^3 + 2^3) = 35$$

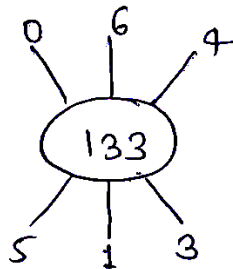


$$(1^3 + 5^3) = 126$$

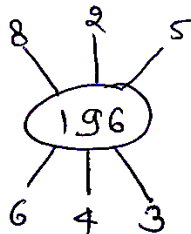


$$\begin{aligned} ? &\Rightarrow 3^3 + 4^3 \\ &= 27 + 64 \\ &= 91 \text{ Ans} \end{aligned}$$

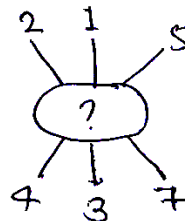
40]



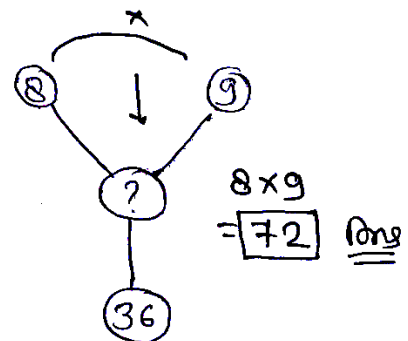
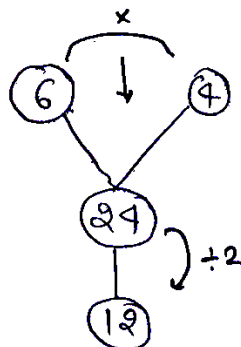
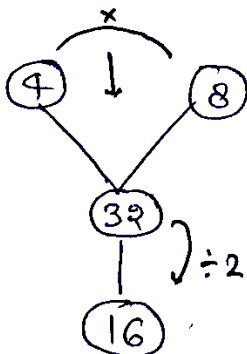
$$\begin{aligned} (0+6+4+5+1+3) &= 19 \\ 19 \times 7 &= 133 \end{aligned}$$



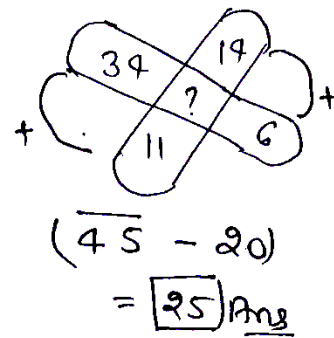
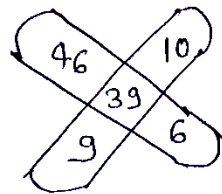
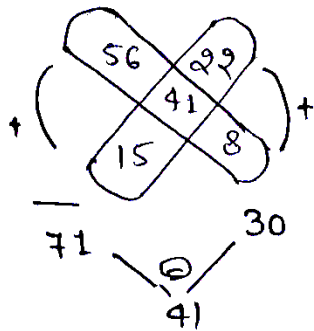
$$\begin{aligned} (2+1+5+4+3+7) &= 22 \\ 22 \times 7 &= 154 \text{ Ans} \end{aligned}$$



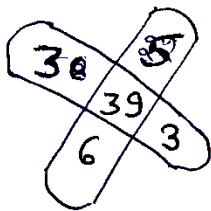
41]



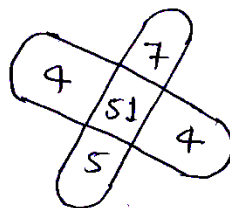
42



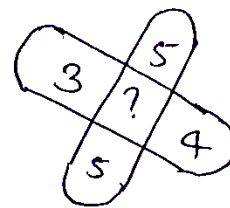
43



$$(6 \times 5) + (3 \times 3) = 30 + 9 = 39$$

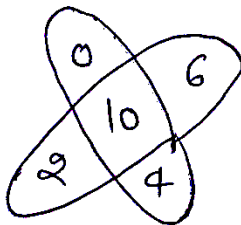


$$(5 \times 7) + (4 \times 4) = 35 + 16 = 51$$

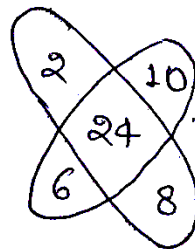


$$(5 \times 5) + (3 \times 4) = 25 + 12 = 37$$

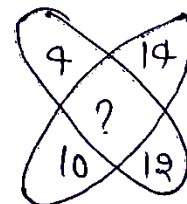
44



$$(6 + 4 + 2 + 0) = 12 - 2 = 10$$

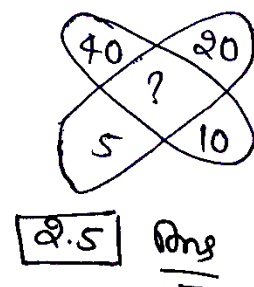
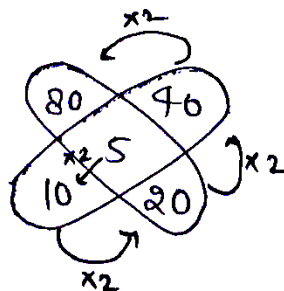
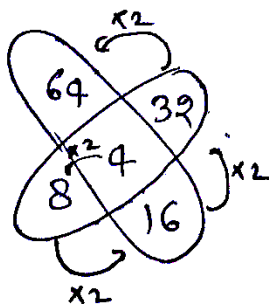


$$10 + 8 + 6 + 2 = 26 - 2 = 24$$



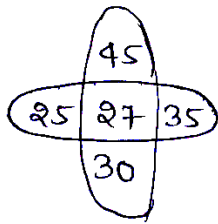
$$10 + 14 + 12 + 4 = 40 - 2 = 38$$

45



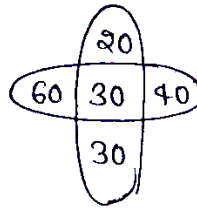
$$2.5$$

46



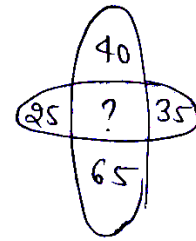
$$= \frac{25 + 45 + 35 + 30}{5}$$

$$= 27$$



$$\frac{20 + 40 + 30 + 60}{5}$$

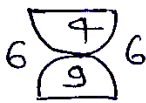
$$= 30$$



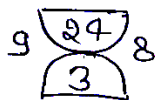
$$\frac{25 + 40 + 30 + 65}{5}$$

$$= \boxed{33} \text{ Ans}$$

47



$$6 \times 6 = 9 \times 4$$



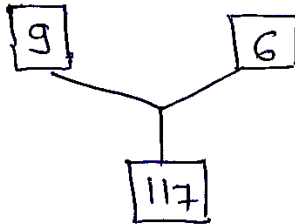
$$24 \times 3 = 8 \times 9$$



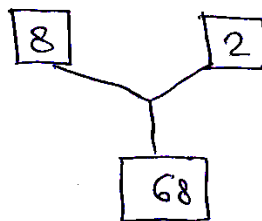
$$9 \times x = 15 \times 6$$

$$\boxed{x = 10}$$

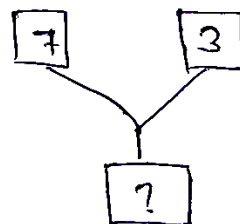
48



$$9^2 + 6^2 = 117$$

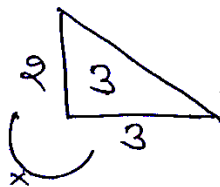


$$8^2 + 2^2 = 68$$

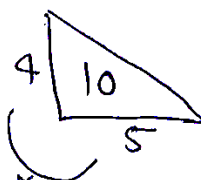


$$7^2 + 3^2 = \boxed{58} \text{ Ans}$$

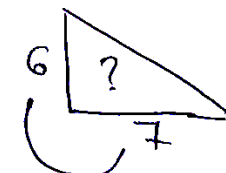
49



$$\frac{2 \times 3}{2} = 3$$



$$\frac{4 \times 5}{2} =$$



$$\frac{6 \times 7}{2} = \boxed{21} \text{ Ans}$$

# Classification Test

## 12. Classification Test

### Type-1

1. (a) Heat (b) light ☒ (c) Bulb (d) Electricity

2. (a) Wave (b) Current ☒ (c) Tide (d) storm

⇒ Except Tide, all other terms are related to both air and water.

3. (a) Wolf ☒ (b) Cat (c) Dog (d) Fox

⇒ Except Cat, all other belongs to dog family. The members of cat family are lions, tiger and leopard etc.

4. (a) Wool (b) Feather (c) Hair ☒ (d) Grass  
(Grass is Vegetation.)

5. (a) Iron (Solid) ☒ (b) Mercury (liquid) (c) silver (Solid) (d) Gold (Solid)

6. (a) Copper (metals) ☒ (b) Bronze (alloy) (c) Gold (metal) (d) Silver (metal)

7. (a) Perception (b) Discernment (c) Penetration ☒ (d) Insinuation

8. (a) Annoy (b) Distress (c) Harass ☒ (d) Ravage

9. (a) Ladder ☒ (b) Staircase  ☒ (c) Bridge  (d) Escalator 

10. ☒ (a) Diabetes (b) Swine flu (c) Chicken pox (d) Malaria

⇒ Diabetes is a deficiency disease caused by deficiency of insulin.  
All other are caused by some causative agent like virus, protozoa.

180

- 11 (a) Ink - Pen ☒ (b) Dust - Vacuum cleaner  
(c) Petrol - Car (d) Electricity - Television

- 12 (a) short - long (b) Man - Woman  
(c) Light - Heavy ☒ (d) Crime - Blame

⇒ except (d) all are antonym of the other.

- 13 (a) Ward ☒ (b) Mentor (c) Pupil (d) Student

⇒ Mentor is different from the all other. Mentor acts as a guide for ward, pupil or student.

- 14 (a) Guitar (b) Violin ☒ (c) Flute (d) Veena

⇒ Except Flute, all other are stringed musical instruments.

- 15 (a) Nose (b) Ear (c) Tongue ☒ (d) Teeth

- 16 (a) Sow ☒ (b) Peacock (c) Peahen (d) Mare

- 17 (a) Sun (b) Moon ☒ (c) sky (d) star

- 18 ☒ (a) Yellow Orange (b) Red Yellow  
(c) Red Orange (d) Yellow Green

⇒ Red, Green and Blue are called Primary colours except (a), in all others there is one primary colour.

- 19 (a) Internet Explorer (b) Chrome ☒ (c) Google (d) Firefox

- 20 (a) Pentagon (b) Circle (c) Rhombus ☒ (d) Diagonal

## Type-2

Q1 (a) 25, 36 (b) 144, 169 (c) 100, 121 ☒ (d) 9, 64  
 $\Rightarrow (5^2, 6^2) \quad (12^2, 13^2) \quad (10^2, 11^2) \quad (3^2, 8^2)$

Q2 ☒ (a) 206 (b) 125 (c) 27 (d) 8  
 $\Rightarrow (5^3) \quad (3^3) \quad (2^3)$

Q3 (a) (48, 6) (b) ☒ (21, 7) (c) (24, 3) (d) (56, 7)  
 $\Rightarrow \begin{matrix} \curvearrowright \\ \times 8 \end{matrix} \quad \begin{matrix} \curvearrowright \\ \times 8 \end{matrix} \quad \begin{matrix} \curvearrowright \\ \times 8 \end{matrix}$

Q4 (a) 6243 (b) 2643 (c) 8465 ☒ (d) 4867  
 $\begin{matrix} \downarrow & \downarrow \\ 8-7=1 \end{matrix} \quad \begin{matrix} \downarrow & \downarrow \\ 8-7=1 \end{matrix} \quad \begin{matrix} \downarrow & \downarrow \\ 12-11=1 \end{matrix} \quad \begin{matrix} \downarrow & \downarrow \\ 12-13 \neq 1 \end{matrix}$

Q5 ☒ (a) 144 (b) 125 (c) 343 (d) 64  
 $(12^2) \quad (5^3) \quad (7^3) \quad (4^3)$

Q6 (a) (47, 49) (b) ☒ (29, 31) (c) (5, 7) (d) (11, 13)  
 $\begin{matrix} \curvearrowright \\ +2 \end{matrix} \quad \begin{matrix} \curvearrowright \\ +3 \end{matrix} \quad \begin{matrix} \curvearrowright \\ +2 \end{matrix} \quad \begin{matrix} \curvearrowright \\ +2 \end{matrix}$

Q7 (a) 564327 (b) 368127 (c) 876321 ☒ (d) 742956  
 $\begin{matrix} \text{---} \\ \oplus = 27 \end{matrix} \quad \begin{matrix} \oplus \\ 27 \end{matrix} \quad \begin{matrix} \text{---} \\ \oplus = 27 \end{matrix}$

Q8 (a) 12-144 (b) 13-156 (c) 15-180 ☒ (d) 16-176  
 $12 \times 12 = 144 \quad 13 \times 12 = 156 \quad 15 \times 12 = 180 \quad 16 \times 11 = 176$

Q9 ☒ (a) 248-231 (b) 547-563 (c) 71-55 (d) 517-523  
 $\begin{matrix} \oplus \\ 17 \end{matrix} \quad \begin{matrix} \oplus \\ 16 \end{matrix} \quad \begin{matrix} \oplus \\ 16 \end{matrix} \quad \begin{matrix} \oplus \\ 16 \end{matrix}$

Q10 ☒ (a) 313 (b) 426 (c) 925 (d) 1034

$\Rightarrow$  Accept 313, the other numbers are divisible number.

- 31 (a) 2010 (b) 2011 ☒ 2012 (d) 2013  
(2012 is a leap year)

- 32 (a) 16-18  $2 \times 8 - 2 \times 9$  (b) 56-63  $7 \times 8 - 7 \times 9$  (c) 96-108  $12 \times 8 - 12 \times 9$  ☒ (d) 86-99

- 33 (a) 2016 (even) (b) 3006 (even) (c) 3016 (even) ☒ (d) 3303 (odd)

- 34 ☒ (a) 0-3-8  $\begin{array}{c} \downarrow \quad \downarrow \\ +3 \quad +5 \end{array}$  (b) 9-12-16  $\begin{array}{c} \downarrow \quad \downarrow \\ +3 \quad +4 \end{array}$  (c) 17-20-24  $\begin{array}{c} \downarrow \quad \downarrow \\ +3 \quad +4 \end{array}$  (d) 51-54-58  $\begin{array}{c} \downarrow \quad \downarrow \\ +3 \quad +4 \end{array}$

- 35 (a) 363 (b) 484 (c) 1331 ☒ (d) 5462

⇒ Except 5462, all other are multiples of 11

- 36 ☒ (a) 35 (b) 37 (Prime No.) (c) 23 (Prime No.) (d) 19 (Prime No.)

- 37 (a) 8110  $\begin{array}{c} \vee \\ \oplus = 10 \end{array}$  (b) 1234  $\begin{array}{c} \vee \\ \oplus = 10 \end{array}$  (c) 9100  $\begin{array}{c} \vee \\ \oplus = 10 \end{array}$  ☒ (d) 1189

- 38 (a) (1,0)  $1^2-1=0$  (b) (2,3)  $2^2-1=3$  (c) (3,8)  $3^2-1=8$  ☒ (d) (4,27)

- 39 ☒ (a) 543, 453 (b) 243, 432  $\begin{array}{c} \curvearrowright \\ \curvearrowleft \end{array}$  (c) 234, 342  $\begin{array}{c} \curvearrowright \\ \curvearrowleft \end{array}$  (d) 354, 543  $\begin{array}{c} \curvearrowright \\ \curvearrowleft \end{array}$

- 40 (a) 14-133  $140-7=133$  (b) 18-171  $180-9=171$  ☒ (c) 22-211 (d) 34-323  $340-17=323$

### Type-3

41) (a) X (24) ~~(b) Y~~ (25) odd (c) H (8) (d) D (4)

42) (a) A (b) I ~~(c) D~~ (consonant) (d) E

43) (a) M (b) N ~~(c) O~~ (vowel) (d) P

44) (a) PQ (b) AU Vowels (c) CD (d) RB

45) ~~(a) DI~~  $\begin{matrix} \text{D} \rightarrow \text{I} \\ +5 \end{matrix}$  (b) KQ  $\begin{matrix} \text{K} \rightarrow \text{Q} \\ +6 \end{matrix}$  (c) OU  $\begin{matrix} \text{O} \rightarrow \text{U} \\ +6 \end{matrix}$  (d) AG  $\begin{matrix} \text{A} \rightarrow \text{G} \\ +6 \end{matrix}$

46) (a) ACB  $\begin{matrix} \text{A} \rightarrow \text{C} \\ +2 \end{matrix}$   $\begin{matrix} \text{C} \rightarrow \text{B} \\ -1 \end{matrix}$  (b) GIH  $\begin{matrix} \text{G} \rightarrow \text{I} \\ +2 \end{matrix}$   $\begin{matrix} \text{I} \rightarrow \text{H} \\ -1 \end{matrix}$  (c) MON  $\begin{matrix} \text{M} \rightarrow \text{O} \\ +2 \end{matrix}$   $\begin{matrix} \text{O} \rightarrow \text{N} \\ -1 \end{matrix}$  ~~(d) POR~~

47) (a) DGJ  $\begin{matrix} \text{D} \rightarrow \text{G} \\ +2 \end{matrix}$   $\begin{matrix} \text{G} \rightarrow \text{J} \\ +2 \end{matrix}$  (b) KNQ  $\begin{matrix} \text{K} \rightarrow \text{N} \\ +2 \end{matrix}$   $\begin{matrix} \text{N} \rightarrow \text{Q} \\ +2 \end{matrix}$  (c) RUX  $\begin{matrix} \text{R} \rightarrow \text{U} \\ +2 \end{matrix}$   $\begin{matrix} \text{U} \rightarrow \text{X} \\ +2 \end{matrix}$  ~~(d) ILN~~

48) (a) FIK  $\begin{matrix} \text{F} \rightarrow \text{I} \\ +3 \end{matrix}$   $\begin{matrix} \text{I} \rightarrow \text{K} \\ +2 \end{matrix}$  (b) DGJ  $\begin{matrix} \text{D} \rightarrow \text{G} \\ +3 \end{matrix}$   $\begin{matrix} \text{G} \rightarrow \text{J} \\ +2 \end{matrix}$  (c) MPR  $\begin{matrix} \text{M} \rightarrow \text{P} \\ +3 \end{matrix}$   $\begin{matrix} \text{P} \rightarrow \text{R} \\ +2 \end{matrix}$  ~~(d) KND~~

49) ~~(a) MTP~~  $\begin{matrix} \text{M} \rightarrow \text{T} \\ +7 \end{matrix}$   $\begin{matrix} \text{T} \rightarrow \text{P} \\ -14 \end{matrix}$  (b) SLE  $\begin{matrix} \text{S} \rightarrow \text{L} \\ -7 \end{matrix}$   $\begin{matrix} \text{L} \rightarrow \text{E} \\ -7 \end{matrix}$  (c) RKD  $\begin{matrix} \text{R} \rightarrow \text{K} \\ -7 \end{matrix}$   $\begin{matrix} \text{K} \rightarrow \text{D} \\ -7 \end{matrix}$  (d) UNG  $\begin{matrix} \text{U} \rightarrow \text{N} \\ -7 \end{matrix}$   $\begin{matrix} \text{N} \rightarrow \text{G} \\ -7 \end{matrix}$

50) (a) XYZ  $\begin{matrix} \text{X} \rightarrow \text{Z} \\ +2 \end{matrix}$   $\begin{matrix} \text{Z} \rightarrow \text{Y} \\ -1 \end{matrix}$  (b) MON  $\begin{matrix} \text{M} \rightarrow \text{O} \\ +2 \end{matrix}$   $\begin{matrix} \text{O} \rightarrow \text{N} \\ -1 \end{matrix}$  (c) PRQ  $\begin{matrix} \text{P} \rightarrow \text{R} \\ +2 \end{matrix}$   $\begin{matrix} \text{R} \rightarrow \text{Q} \\ -1 \end{matrix}$  ~~(d) EAC~~  $\begin{matrix} \text{E} \rightarrow \text{A} \\ -4 \end{matrix}$   $\begin{matrix} \text{A} \rightarrow \text{C} \\ +2 \end{matrix}$

51 (a) RNMP  $\begin{matrix} \text{R} & \text{N} & \text{M} & \text{P} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ -4 & -1 & +3 & \end{matrix}$  (b) JEEH  $\begin{matrix} \text{J} & \text{E} & \text{E} & \text{H} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ -4 & -1 & +3 & \end{matrix}$  (c) ☒ RPOQ  $\begin{matrix} \text{R} & \text{P} & \text{O} & \text{Q} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ -2 & -1 & +2 & \end{matrix}$  (d) HDCE  $\begin{matrix} \text{H} & \text{D} & \text{C} & \text{E} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ -4 & -1 & +3 & \end{matrix}$

52 (a) PQSU  $\begin{matrix} \text{P} & \text{Q} & \text{S} & \text{U} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ +1 & +2 & +2 & \end{matrix}$  (b) ☒ BCQN  $\begin{matrix} \text{B} & \text{C} & \text{Q} & \text{N} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ +1 & +1 & -3 & \end{matrix}$  (c) ABDE  $\begin{matrix} \text{A} & \text{B} & \text{D} & \text{E} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ +1 & +2 & +2 & \end{matrix}$  (d) MNPR  $\begin{matrix} \text{M} & \text{N} & \text{P} & \text{R} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ +1 & +2 & +2 & \end{matrix}$

53 (a) CBCE  $\begin{matrix} \text{C} & \text{B} & \text{C} & \text{E} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ -1 & +1 & \end{matrix}$  (b) THIJ  $\begin{matrix} \text{T} & \text{H} & \text{I} & \text{J} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ -1 & +1 & \end{matrix}$  (c) ONOP  $\begin{matrix} \text{O} & \text{N} & \text{O} & \text{P} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ -1 & +1 & \end{matrix}$  (d) ☒ UTUW  $\begin{matrix} \text{U} & \text{T} & \text{U} & \text{W} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ -1 & +2 & \end{matrix}$

54 (a) XRMKL (b) ☒ XRMIF  $\begin{matrix} \text{X} & \text{R} & \text{M} & \text{I} & \text{F} \\ & & & \downarrow & \\ & & & \text{Vowel} & \end{matrix}$  (c) XRNLJ (d) XSNLJ

55 (a) AbcdE (b) IfghO (c) ☒ ApqrL (d) Vi mnE

Except (c) in all other the first and the last letters given in Capital are vowels

56 (a) Man = 13114  $\begin{matrix} \text{M} & \text{a} & \text{n} \\ \downarrow & \downarrow & \downarrow \\ 13 & 1 & 4 \end{matrix}$  (b) Ram = 18113  $\begin{matrix} \text{R} & \text{a} & \text{m} \\ \downarrow & \downarrow & \downarrow \\ 18 & 1 & 3 \end{matrix}$  (c) Jug = 10217  $\begin{matrix} \text{J} & \text{u} & \text{g} \\ \downarrow & \downarrow & \downarrow \\ 10 & 2 & 7 \end{matrix}$  (d) ☒ Hub = 82110  $\begin{matrix} \text{H} & \text{u} & \text{b} \\ \downarrow & \downarrow & \downarrow \\ 8 & 2 & 110 \end{matrix}$

57 (a) ccdffgg  $\begin{matrix} \text{c} & \text{c} & \text{d} & \text{f} & \text{f} & \text{g} & \text{g} \\ & & \downarrow & & \downarrow & & \\ & & +2 & & +2 & & \end{matrix}$  (b) hhikll  $\begin{matrix} \text{h} & \text{h} & \text{i} & \text{k} & \text{l} & \text{l} \\ & & \downarrow & & \downarrow & \\ & & +2 & & +2 & \end{matrix}$  (c) ☒ mmnopp  $\begin{matrix} \text{m} & \text{m} & \text{n} & \text{o} & \text{p} & \text{p} \\ & & \downarrow & & \downarrow & \\ & & +1 & & +1 & \end{matrix}$  (d) qqrtuu  $\begin{matrix} \text{q} & \text{q} & \text{r} & \text{t} & \text{u} & \text{u} \\ & & \downarrow & & \downarrow & \\ & & +2 & & +2 & \end{matrix}$

58 (a) MLNO  $\begin{matrix} \text{M} & \text{L} & \text{N} & \text{O} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{O} & \text{M} & \text{N} & \text{L} \end{matrix}$  (b) KJPQ  $\begin{matrix} \text{K} & \text{J} & \text{P} & \text{Q} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{Q} & \text{K} & \text{P} & \text{J} \end{matrix}$  (c) IHRS  $\begin{matrix} \text{I} & \text{H} & \text{R} & \text{S} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{S} & \text{I} & \text{R} & \text{H} \end{matrix}$  (d) ☒ GFTV  $\begin{matrix} \text{G} & \text{F} & \text{T} & \text{V} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{V} & \text{G} & \text{T} & \text{F} \end{matrix}$

59 (a) ABYZ  $\begin{matrix} \text{A} & \text{B} & \text{Y} & \text{Z} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{Z} & \text{A} & \text{Y} & \text{B} \end{matrix}$  (b) CDWX  $\begin{matrix} \text{C} & \text{D} & \text{W} & \text{X} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{X} & \text{C} & \text{W} & \text{D} \end{matrix}$  (c) EFUV  $\begin{matrix} \text{E} & \text{F} & \text{U} & \text{V} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{V} & \text{E} & \text{U} & \text{F} \end{matrix}$  (d) ☒ GHTV  $\begin{matrix} \text{G} & \text{H} & \text{T} & \text{V} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{V} & \text{G} & \text{T} & \text{H} \end{matrix}$

60 (a) BYCX  $\begin{matrix} \text{B} & \text{Y} & \text{C} & \text{X} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{X} & \text{B} & \text{C} & \text{Y} \end{matrix}$  (b) EVFU  $\begin{matrix} \text{E} & \text{V} & \text{F} & \text{U} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{U} & \text{E} & \text{F} & \text{V} \end{matrix}$  (c) GTHS  $\begin{matrix} \text{G} & \text{T} & \text{H} & \text{S} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{S} & \text{G} & \text{T} & \text{H} \end{matrix}$  (d) ☒ LOIR  $\begin{matrix} \text{L} & \text{O} & \text{I} & \text{R} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \text{R} & \text{L} & \text{O} & \text{I} \end{matrix}$

# Type-4

61 (a) 676. (b) 454 (c) 343 ~~(d) 564~~

$$\Rightarrow 232 + 111 = 343, 454 + 111 = 565, 565 + 111 = 676$$

$$343 + 111 = 454$$

62 Find out the incorrect term in the series.

2, 5, 10, 3, 6, 18, 4, 7, 30

$\underbrace{2 \rightarrow 5}_{\times 2.5}$   $\underbrace{5 \rightarrow 10}_{\times 2}$   $\underbrace{10 \rightarrow 3}_{\div 3.33}$   $\underbrace{3 \rightarrow 6}_{\times 2}$   $\underbrace{6 \rightarrow 18}_{\times 3}$   $\underbrace{18 \rightarrow 4}_{\div 4.5}$   $\underbrace{4 \rightarrow 7}_{\times 1.75}$   $\underbrace{7 \rightarrow 30}_{\times 4.28}$

Incorrect term: 3

63 Find out the incorrect term in the series?

12, 28, 30, 39, 48

$\underbrace{12 \rightarrow 28}_{+16}$   $\underbrace{28 \rightarrow 30}_{+2}$   $\underbrace{30 \rightarrow 39}_{+9}$   $\underbrace{39 \rightarrow 48}_{+9}$

Incorrect term: 28

64 Which number is wrong in the given series?

225, 169, 144, 100, 64, 48

$15^2, 13^2, 12^2, 10^2, 8^2$

Incorrect term: 48

65 3, 9, 27, 81, 243, 730

$\underbrace{3 \rightarrow 9}_{\times 3}$   $\underbrace{9 \rightarrow 27}_{\times 3}$   $\underbrace{27 \rightarrow 81}_{\times 3}$   $\underbrace{81 \rightarrow 243}_{\times 3}$   $\underbrace{243 \rightarrow 730}_{\times 3}$

Incorrect term: 730

66 2, 10, 30, 68, 120, 222

2, 10, 30, 68, 130, 222

$\underbrace{2 \rightarrow 10}_{+8}$   $\underbrace{10 \rightarrow 30}_{+20}$   $\underbrace{30 \rightarrow 68}_{+38}$   $\underbrace{68 \rightarrow 130}_{+62}$   $\underbrace{130 \rightarrow 222}_{+92}$

$\underbrace{2 \rightarrow 30}_{+12}$   $\underbrace{30 \rightarrow 68}_{+18}$   $\underbrace{68 \rightarrow 130}_{+24}$   $\underbrace{130 \rightarrow 222}_{+30}$

$\underbrace{2 \rightarrow 68}_{+6}$   $\underbrace{68 \rightarrow 130}_{+6}$   $\underbrace{130 \rightarrow 222}_{+6}$

Incorrect term: 120

67 12439, 23549, 34659, 45769, 57689

$\underbrace{12439 \rightarrow 23549}_{+11110}$   $\underbrace{23549 \rightarrow 34659}_{+11110}$   $\underbrace{34659 \rightarrow 45769}_{+11110}$   $\underbrace{45769 \rightarrow 57689}_{+11110}$

Incorrect term: 57689

# Coding Decoding

### 13. Coding - Decoding

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|
| A | B | C | D | E | F | G | H | I | J  | K  | L  | M  | N  | O  | P  | Q  |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| R  | S  | T  | U  | V  | W  | X  | Y  | Z  |
| 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |

Opposite

Trick to Learn

A-Z → A to Z

B-Y → Boy

C-X → Complex

D-W → Dodh Wala

E-V → Even

F-U → Father Uncle

G-T → G.T Road

H-S → Honey Singh

I-R → Indian Railway

J-Q → Jaipur Queen

K-P → Kala Pani

L-O → Love

M-N → Moon

Type-1

(a) Fixed Coding:-

7. In a code language the following alphabets are coded in a particular way:

|   |   |   |   |   |   |   |   |   |   |    |   |
|---|---|---|---|---|---|---|---|---|---|----|---|
| A | C | D | E | G | H | K | L | M | N | O  | S |
| ^ | 9 | ≠ | h | j | F | ← | 7 | P | U | // | V |

Which word can be decoded from the following?

7 // P ^ 9 h

↓ ↓ ↓ ↓ ↓ ↓

HOMAGE

ans

2. If 'PENCIL' is coded as ? @, = ; 7 and 'PAPER' is coded as ? 9 ? @ 5 how will you code 'CLIP' ?

⇒

|             |           |      |         |
|-------------|-----------|------|---------|
| PENCIL      | PAPER     | then | CLIP    |
| ↓ ↓ ↓ ↓ ↓ ↓ | ↓ ↓ ↓ ↓ ↓ |      | ↓ ↓ ↓ ↓ |
| ? @ , = ; 7 | ? 9 ? @ 5 |      | = 7 ; ? |

Dns

3. In a certain code, P is #, A is %, C is φ and E is @. How is 'PACE' written in that code?

Soln

|       |         |
|-------|---------|
| P → # | P A C E |
| A → % | # % φ @ |
| C → φ |         |
| E → @ |         |

Dns

4. In a certain code 'R' is %, 'E' is '#', 'D' is @ and 'A' is Δ. How is 'DARE' written in that code

Soln

|       |         |
|-------|---------|
| R → % | D A R E |
| E → # | ↓ ↓ ↓ ↓ |
| D → @ | @ Δ % # |
| A → Δ |         |

5. In a certain code MADE is written as '\$ % + #' and 'Some' is 'x ÷ \$ #' then How is written 'DOES' is written in that code?

Soln

|          |          |      |         |
|----------|----------|------|---------|
| MADE     | SOME     | then | DOES    |
| \$ % + # | x ÷ \$ # |      | ↓ ↓ ↓ ↓ |
|          |          |      | + ÷ # x |

Dns

(b) One to One :-

[1] In a certain code TEACHER is written as VGCEJGT. How is CHILDREN written in that code?

Soln

|    |    |   |   |   |    |   |    |
|----|----|---|---|---|----|---|----|
|    | 20 | 5 | 1 | 3 | 8  | 5 | 18 |
|    | T  | E | A | C | H  | E | R  |
| +2 | ↓  | ↓ | ↓ | ↓ | ↓  | ↓ | ↓  |
|    | V  | G | C | E | J  | G | T  |
|    | 22 | 7 | 3 | 5 | 10 | 7 | 20 |

CHILDREN  
 ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓  
[EJKNFTGP] Ans

[2]

KANCHAN → NEQGKEQ then JAYANTI = ??

Soln

|    |    |   |    |   |    |   |    |
|----|----|---|----|---|----|---|----|
|    | 11 | 1 | 14 | 3 | 8  | 1 | 14 |
|    | K  | A | N  | C | H  | A | N  |
| +3 | ↓  | ↓ | ↓  | ↓ | ↓  | ↓ | ↓  |
|    | N  | E | Q  | G | K  | E | Q  |
|    | 14 | 5 | 17 | 7 | 11 | 5 | 17 |

JAYANTI  
 ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓  
MEBEQXL Ans

[3] IF DEAR → FGCT then READ = ??

Soln

|    |   |   |   |    |
|----|---|---|---|----|
|    | 4 | 5 | 1 | 18 |
|    | D | E | A | R  |
| +2 | ↓ | ↓ | ↓ | ↓  |
|    | F | G | C | T  |
|    | 6 | 7 | 3 | 20 |

READ  
 ↓ ↓ ↓ ↓  
TGCF Ans

[4] IF MISTAKE → JLPWXOB then INTIMATE = ??

Soln

|    |    |    |    |    |    |    |   |
|----|----|----|----|----|----|----|---|
|    | 13 | 9  | 19 | 20 | 1  | 11 | 5 |
|    | M  | I  | S  | T  | A  | K  | E |
| -3 | ↓  | ↓  | ↓  | ↓  | ↓  | ↓  | ↓ |
|    | J  | L  | P  | W  | X  | O  | B |
|    | 10 | 12 | 16 | 23 | 24 | 15 | 2 |

INTIMATE  
 ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓  
FQQ LJDOH

Ans

(c) Cross Coding:-

[1] IF TIME → JUFN then BOTH → ??

Soln

$\begin{matrix} 20 & 9 & 13 & 5 \\ T & I & M & E \\ \swarrow & \searrow & \swarrow & \searrow \\ J & U & F & N \\ 10 & 21 & 6 & 14 \end{matrix}$

$\begin{matrix} B & O & T & H \\ \swarrow & \searrow & \swarrow & \searrow \\ P & C & I & U \end{matrix}$  Ans

[2] IF DIVE → EIVD then FEAR = ??

Soln:-

$\begin{matrix} D & I & V & E \\ \swarrow & \searrow & \swarrow & \searrow \\ E & I & V & D \end{matrix}$

$\begin{matrix} F & E & A & R \\ \swarrow & \searrow & \swarrow & \searrow \\ R & E & A & R \end{matrix}$  Ans

[3] If COMPUTER → VSFQONND then SAMACHAR → ??

Soln:-

$\begin{matrix} 3 & 15 & 13 & 16 & 21 & 20 & 5 & 18 \\ C & O & M & P & U & T & E & R \\ \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow \\ V & S & F & Q & O & N & N & D \\ 22 & 19 & 6 & 17 & 15 & 14 & 14 & 4 \\ +1 & -1 & +1 & -1 & -1 & +1 & -1 & +1 \end{matrix}$

$\begin{matrix} 19 & 1 & 13 & 3 & 8 & 1 & 18 \\ S & A & M & A & C & H & A & R \\ \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow \\ D & G & B & Q & Z & N & Z & T \end{matrix}$  Ans

[4] If KINGDOM → JMCLJHP then QUANTUM → ?

Soln:-

$\begin{matrix} 11 & 9 & 14 & 7 & 4 & 15 & 13 \\ K & I & N & G & D & O & M \\ \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow \\ J & M & C & L & J & H & P \\ 10 & 13 & 3 & 12 & 10 & 8 & 16 \end{matrix}$

$\begin{matrix} Q & U & A & N & T & U & M \\ \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \searrow \\ P & Z & S & L & V & O & V \end{matrix}$

Ans

## Type-2

1.

If 'white' is called 'blue', 'blue' is called 'red', 'red' is called 'yellow', 'yellow' is called 'green', 'green' is called 'black', 'black' is called 'violet', 'violet' is called 'orange', what would be the colour of human blood?

Soln  $\hookrightarrow$  Red  $\longrightarrow$  yellow (Given)  
Ans

2.

If 'book' is called 'watch', 'watch' is called 'bag', 'bag' is called 'dictionary' and 'dictionary' is called 'window', which is used to carry the books?

Soln :-

$\hookrightarrow$  'bag'  $\longrightarrow$  'dictionary' Ans

3.

If 'wall' is called 'window', 'window' is called 'door', 'door' is called 'floor', 'floor' is called 'roof' and 'roof' is called 'ventilator' what will a person

stand on?

Soln :-  $\hookrightarrow$  Floor  $\longrightarrow$  roof Ans

### Type-3

1] If REQUEST is written as S2R52TU, then how will ACID be written?

Soln

REQUEST  
 $\begin{matrix} +1 & & +1 & & +1 & +1 \\ \downarrow & & \downarrow & & \downarrow & \downarrow \end{matrix}$   
 S2R52TU

ACID  
 $\begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 13 & 5 & \end{matrix}$  Ans

A E I O U  
 1 2 3 4 5

2] In a certain code, the word DEAL is coded as 4-5-1-12. Following the same rule of coding, what should be the code for the word LADY

Soln

DEAL  
 $\begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \\ 4 & 5 & 1 & 12 \end{matrix}$  (its position)

LADY  
 $\begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \\ 12 & 1 & 4 & 25 \end{matrix}$  Ans

3] If A = 2, M = 26, Z = 52 then BET = ?

Soln :-  $1 \times 2 = 2$      $\hookrightarrow 13 \times 2 = 26$      $\hookrightarrow 26 \times 2 = 52$      $\hookrightarrow \begin{matrix} \downarrow & \downarrow & \downarrow \\ 4 & 10 & 40 \end{matrix}$   
 $= 54$  Ans

4] If A = 26, SUN = 27, then CAT = ?

Soln :-  $\begin{matrix} \downarrow \\ Z = 26 \end{matrix}$      $\begin{matrix} \downarrow & \downarrow & \downarrow \\ H & F & M \\ 8 & 6 & 13 \end{matrix} = 27$      $\begin{matrix} \downarrow & \downarrow & \downarrow \\ X & Z & G \\ 24 & 26 & 7 \end{matrix} = 57$  Ans

5] If in a certain code, BAT = 23 and CAT = 24, then how will you code BALL?

Soln    BAT = 23     $2 + 1 + 12 + 12$   
 $2 + 1 + 20 \rightarrow$     27 Ans

[6] If GO = 32, SHE = 49, then SOME = ??

Soln:-

$$\begin{array}{c} \downarrow \downarrow \\ T \quad L \\ 20 + 12 = 32 \end{array}$$

$$\begin{array}{c} \downarrow \downarrow \downarrow \downarrow \\ H \quad L \quad N \quad V \\ 8 + 12 + 14 + 22 \\ = 56 \text{ Ans} \end{array}$$

[7] If AT = 20, BAT = 40, then CAT will be ?

Soln:-

$$\begin{array}{c} \text{AT} = 20 \\ 2 \times 20 = 40 \end{array}$$

$$\begin{array}{c} \text{CAT} \\ 3 \times 20 = 60 \text{ Ans} \end{array}$$

[8] If ZIP = 198 and ZAP = 246, then VIP = ??

Soln

$$\begin{array}{c} \text{ZIP} \\ \downarrow \downarrow \downarrow \\ A \quad R \quad K \\ 1 \quad 18 \quad 11 \end{array} \quad \begin{array}{c} 2 \times \\ (2 + 19 + 12) \times 6 = 198 \end{array}$$

$$\begin{array}{c} \text{VIP} \\ \downarrow \downarrow \downarrow \\ E \quad R \quad K \\ (6 + 19 + 12) \times 6 = 222 \text{ Ans} \end{array}$$

[9] If DEER = 12215 and HIGH = 5645, how will you Code HEEL → ??

Soln

$$\begin{array}{c} \text{DEER} \\ 4 \quad 5 \quad 5 \quad 18 \\ -3 \curvearrowright \\ 1 \quad 2 \quad 2 \quad 15 \end{array}$$

$$\begin{array}{c} \text{HEEL} \\ 8 \quad 5 \quad 5 \quad 12 \\ -3 \curvearrowright \\ 5 \quad 2 \quad 2 \quad 9 \text{ Ans} \end{array}$$

[10] If WORK is coded as 4-12-9-16 then WOMEN → ??

Soln:-

$$\begin{array}{c} \text{WORK} \\ \downarrow \downarrow \downarrow \downarrow \\ O \quad L \quad I \quad P \\ 4-12-9-16 \end{array}$$

$$\begin{array}{c} \text{WOMEN} \\ \downarrow \downarrow \downarrow \downarrow \downarrow \\ O \quad L \quad N \quad V \quad M \\ \underline{4-12-14-22-13} \text{ Ans} \end{array}$$

### Type-4

IF

more money in market  $\rightarrow$  zo li aa to —①

share in market Profit  $\rightarrow$  vo to je li —②

making more profit now  $\rightarrow$  su je zo ka —③

now the market gains  $\rightarrow$  do li yo su —④

(i) vo  $\rightarrow$  ? share any

(ii) making  $\rightarrow$  ? 'ka'

(iii) gains  $\rightarrow$  ? do or yo

(iv) more gains share  $\rightarrow$  ? zo do/yo vo

(v) 'to ka li aa'  $\rightarrow$  ? in making market money.

Sol<sup>n</sup>:- from ①, ② and ④  $\rightarrow$  market  $\Rightarrow$  li

from ② & ③  $\rightarrow$  Profit  $\Rightarrow$  je

from ③ & ④  $\rightarrow$  now  $\Rightarrow$  su

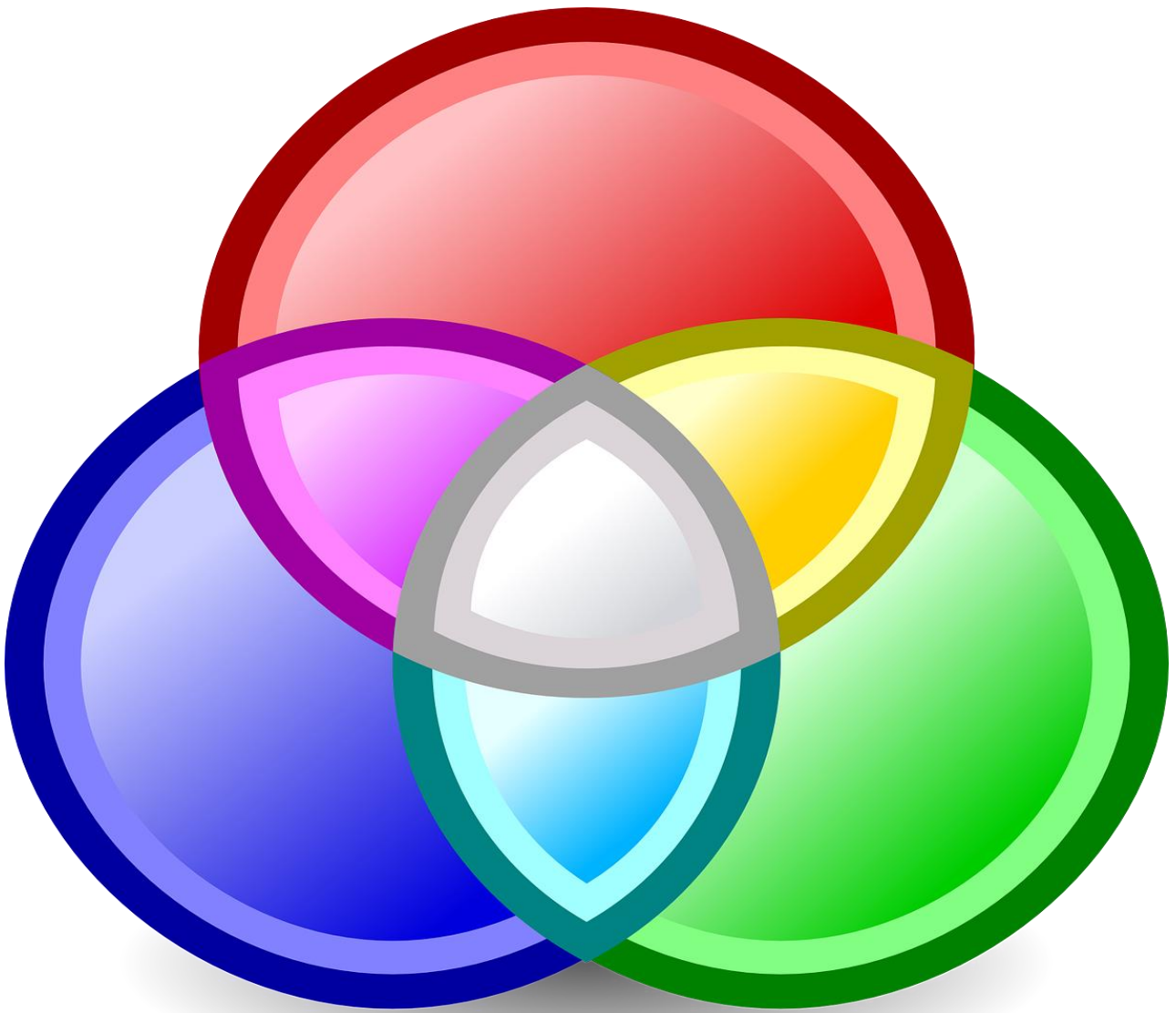
from ① & ③  $\rightarrow$  more  $\Rightarrow$  zo

from ③  $\rightarrow$  making  $\Rightarrow$  ka

from ① & ②  $\rightarrow$  in  $\Rightarrow$  to

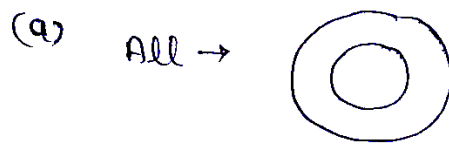
from ②  $\rightarrow$  share  $\Rightarrow$  vo

gains  $\Rightarrow$  do/yo

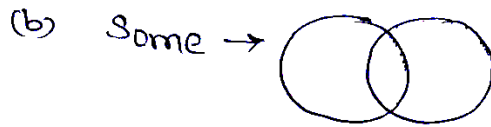


# Ven Diagram

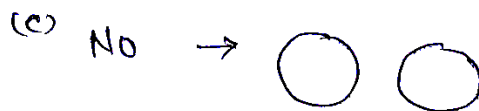
## 14. Ven Diagram



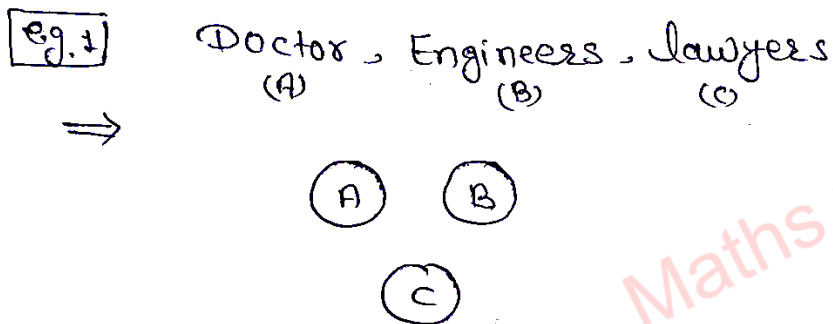
Example:  
Apples, fruits



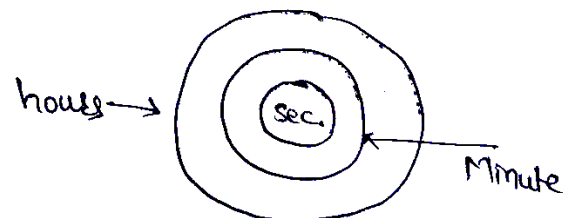
Women, Teachers



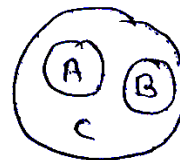
Chair, Table



**eg.2** Second, Minute, hour



**eg.3** Table, chair, Furniture  
(A) (B) (C)

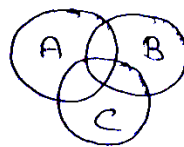


**eg.4** Dogs, pets, cats  
(A) (B) (C)



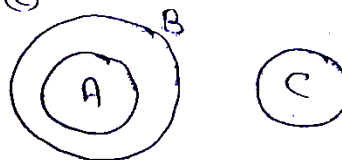
e.g.6

Clerk, Government Employees, Education  
(A) (B) (C)



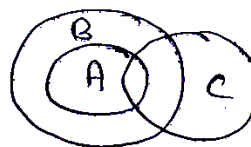
e.g.7

Engineers, Human Beings, Rats  
(A) (B) (C)



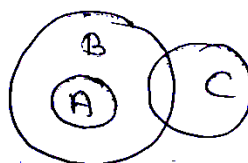
e.g.8

Female, Mothers, Doctors  
(B) (A) (C)



e.g.9

Male, Father, children  
(B) (A) (C)



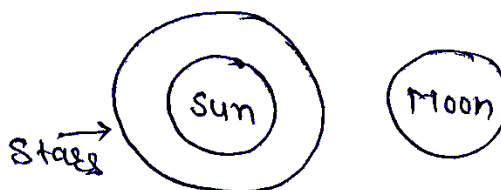
e.g.10

Professors, Authors, children  
(A) (B) (C)



e.g.11

Sun, Moon, stars

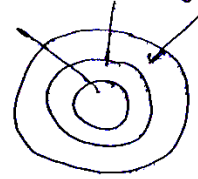


## Examples

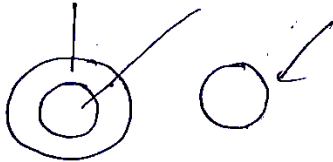
1. Vegetable, Potato, cabbage



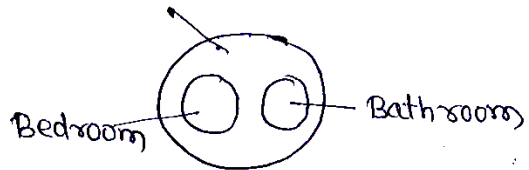
2. Day, week, year



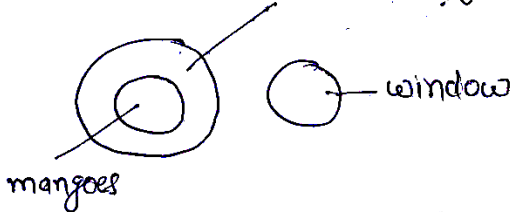
3. Criminal, Thief, Judge



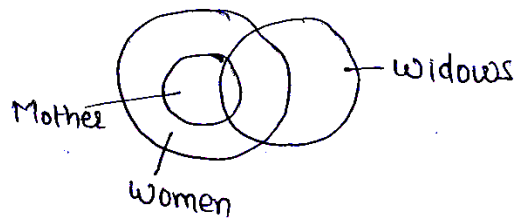
4. House, Bedroom, Bathroom



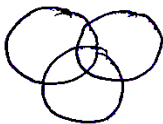
5. Window, fruit, mangoes



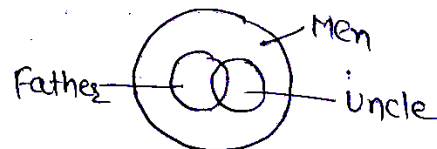
6. Widows, women, Mothers



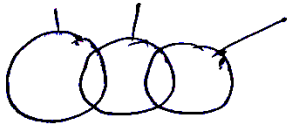
7. Authors, Teachers, Men



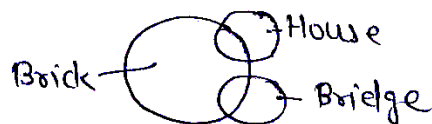
8. Men, uncle, Father



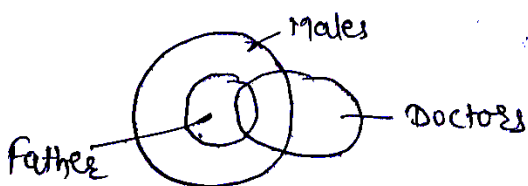
9. Men, Musicians, Women



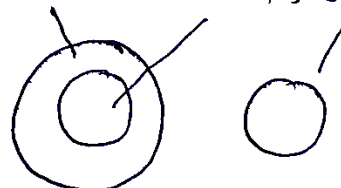
10. House, Brick, Bridge



11. Males, Fathers, Doctors

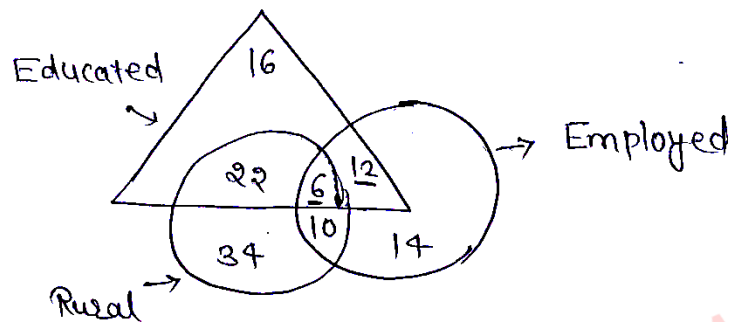


12. Planet, Earth, Sun



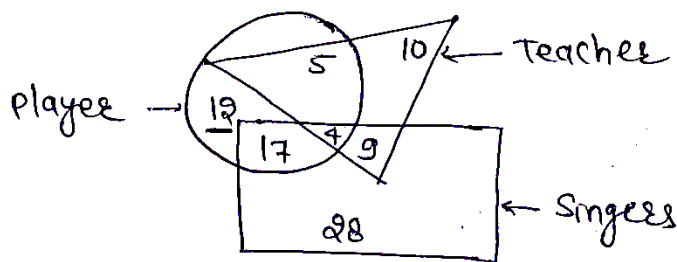
## Type-2

**eg.1** How many Educated people are employed?



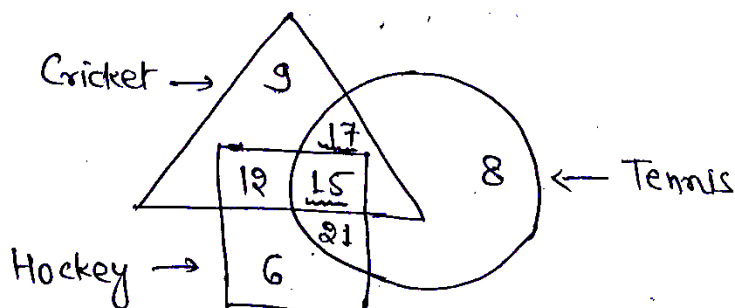
Soln  $12 + 6 = 18$  Ans

**eg.2** How many players are neither Teachers nor Singers?



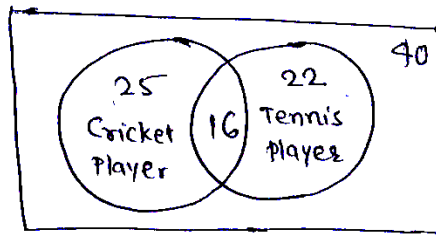
Soln  $12$  Ans

**eg.3** How many people like cricket and tennis both?



Soln :-  $17 + 15 = 32$  Ans

eg.4



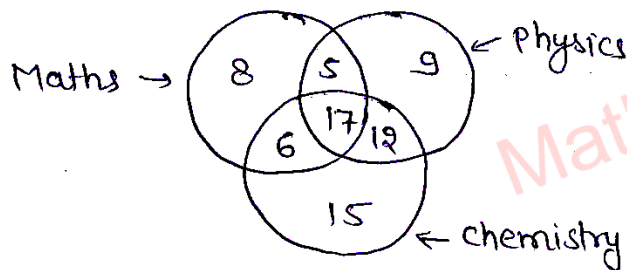
Find out the number of students who play only Cricket.

Soln  $25 - 16 = 9$  Ans

eg.5

In the given figure, how many people study only 2 subjects?

Soln :-



Soln :-

Only maths & physics = 5

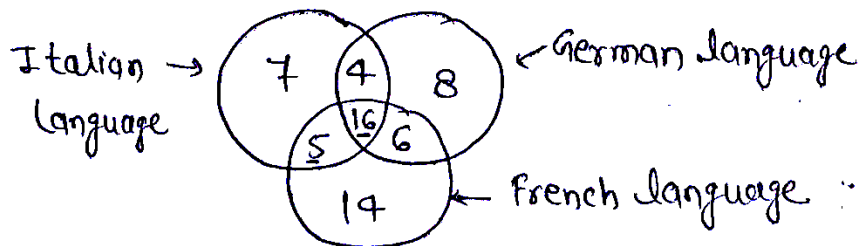
Only physics & chemistry = 12

only maths & chemistry = 6

+ 23 Ans

eg.6

How many people speak Italian and French language?



Soln :-

$5 + 16 = 21$  Ans

# Blood Relation

## 15. Blood Relation

How to make family tree.

1. A is the father of B

⇒



2. A is the mother of B

⇒



3. A is the son of B.

⇒

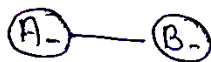


4. A is the daughter of B

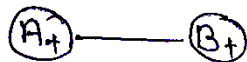
⇒



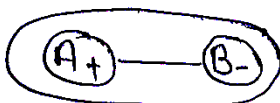
5. A is sister of B, B is sister of A



6. A and B are Brothers



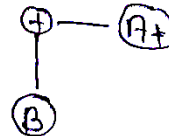
7. A is the Husband of B



+ → for male

- → for female

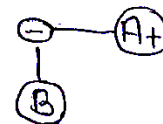
8. A is paternal uncle of B



9. A is the paternal Grand father of B



10. A is maternal uncle of B



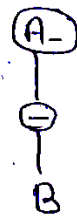
11. A is maternal grand father of B



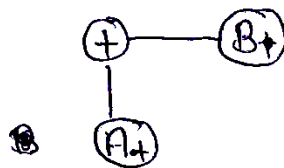
12] A is the paternal Grand mother of B.



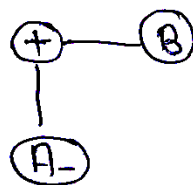
13] A is maternal Grand mother of B



14] A is the Nephew of B



15] A is the niece of B

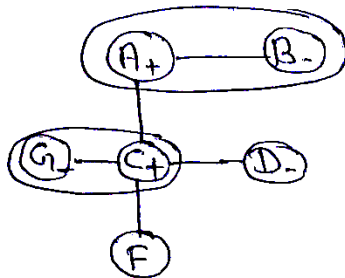


Maths with Gajanand

## Type-1     Puzzle form

**E.g. 1** In a family, A is the Husband of B, C is a Son of A, D is a daughter of B. F is a daughter of G. C is the father of F. Then how is G related to B.

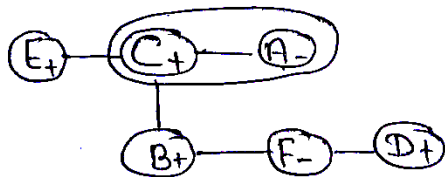
Sol<sup>n</sup>



G is the daughter in law of B

**E.g. 2** There are six members of the family A, B, C, D, E and F are travelling together in which B is a son of C but C is not the mother of B. E is the Brother of C, D is a Brother of F, F is a daughter of A and (A and C) are married Couple.

Sol<sup>n</sup>



(i) How many male members are there?

⇒ 4

(ii) How many children A have?

⇒ 3

(iii) Who is the mother of B?

⇒ A

(iv) Who is the brother of E?

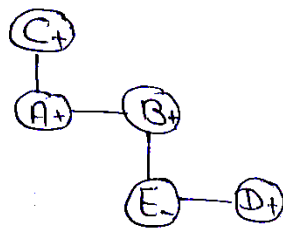
⇒ C

(v) Who is the wife of C?

⇒ A

**[e.g.3]** In a family A is the brother of B, C is the father of A, D is the brother of E and E is a daughter of B then the uncle of D is ?

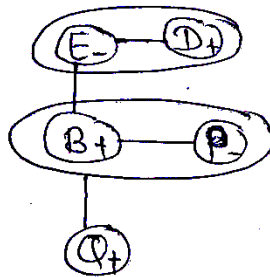
Soln:-



(A) Ans

**[e.g.4]** In a family B is the husband of P, Q is the only grand son of E, who is the wife of D. E is the mother-in-law of P.

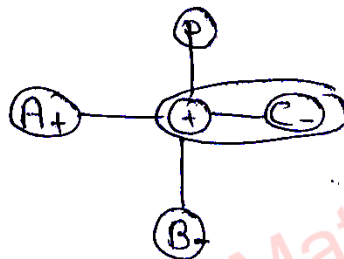
Soln



How is B related to D  $\Rightarrow$  Son Ans

**[e.g.5]** In a family A is the uncle of B who is the daughter of C, C is a daughter-in-law of P, then, how is A related to P ?

Soln:-



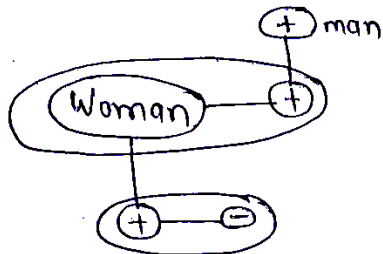
$\Rightarrow$  A is the son of P

## Type-2

### Pointing to or indicate to

**eg.1** Pointing to man in a photo-graph a woman says  
 "He is the father of my only daughter-in-law's father-  
 -in-law." How is man related to woman?

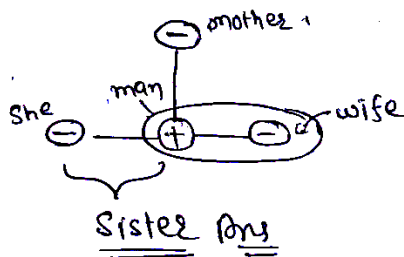
Soln:-



Father-in-law     Ans

**eg.2** Pointing to a woman in a photograph a man says  
 "she is the only daughter of my wife's mother-in-law."  
 How is the woman related to the man?

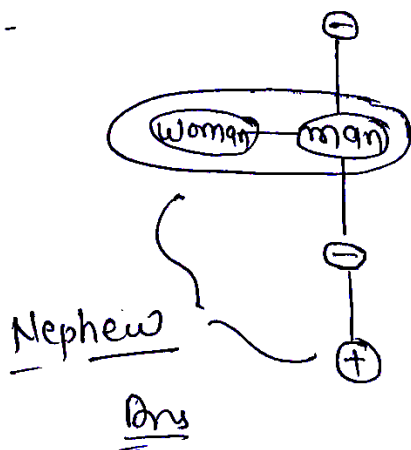
Soln:-



Sister     Ans

**eg.3** Pointing to a photograph a woman says "He is the  
 only son of the daughter of my mother-in-law's only  
 son. How is the man related to woman?"

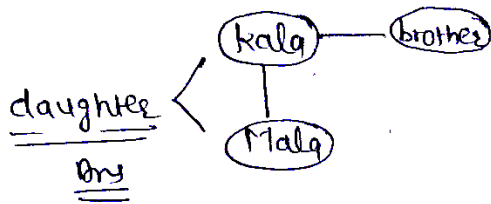
Soln:-



Nephew     Ans

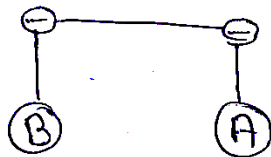
**eg 4** Pointing to Mala, Kala said, "She is my brother's only sister's daughter." How is Mala related to Kala?

Sol<sup>n</sup>:-



**eg 5** Pointing towards A, B said "your mother is the younger sister of my mother." How is A related to B?

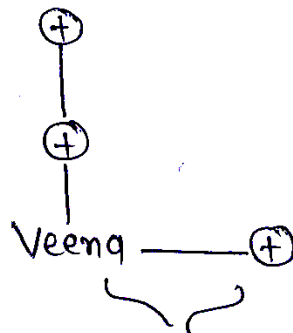
Sol<sup>n</sup>:-



Brother Ans

**eg 6** Pointing towards a boy Veena said, "He is the son of only son of my grandfather." How is that boy related to Veena?

Sol<sup>n</sup>:-



Brother Ans

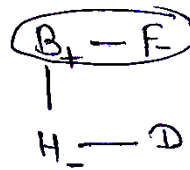
### Type-3

#### Coded Relationship

- I. ' $P \times Q$ ' means 'p is the father of Q'
- II. ' $P \div Q$ ' means 'p is a daughter of Q'
- III. ' $P + Q$ ' means 'p is a sister of Q'
- IV. ' $P - Q$ ' means 'p is the husband of Q'

(1)  $B \times H + D \div F$

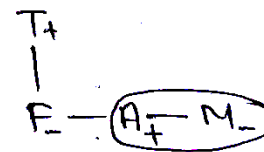
- (i) How is F related to H ?  
 $\Rightarrow$  mother



- (ii) How is B related to F ?  
 $\Rightarrow$  Husband

(2)  $F \div T \times A - M$

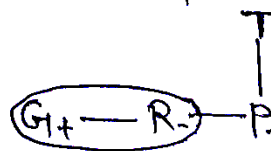
- (i) How is A related to F ?  
 $\Rightarrow$  Brother



- (ii) How is T related to M ?  
 $\Rightarrow$  Father-in-law

(3)  $G - R + P \div T$

- (i) How is G related to T ?  
 $\Rightarrow$  Son-in-law



**Q.2** If 'A+B' means 'A' is father of 'B' [SSC 2017]

'A-B' means 'A' is mother of 'B'

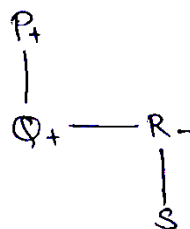
'A\*B' means 'A' is brother of 'B'

and 'A%B' means 'A' is sister of 'B'

then how is Q related to S in 'P+Q\*R-S' ?

Soln

Q is maternal uncle of S



**Q.3**

A+B means B is the father of A

A-B means B is the mother of A

AxB means B is the brother of A

A÷B means B is the sister of A

A\*B means B is the son of A

A#B means B is the daughter of A

U-T

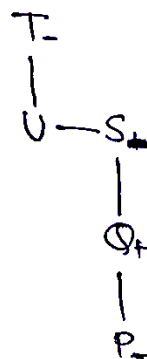
U-T#S\*Q#P

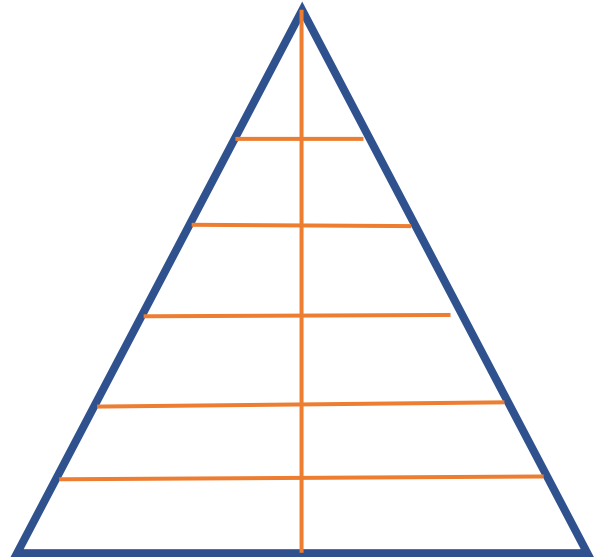
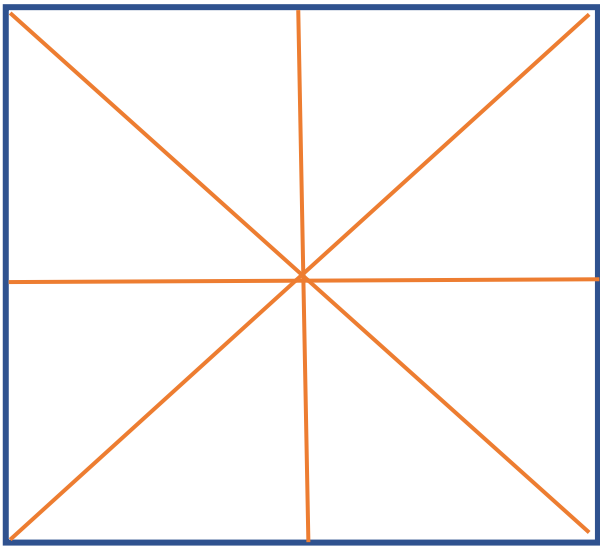
Soln:-

How is P related to U in the given expression?

⇒ Grand-daughter

Ans



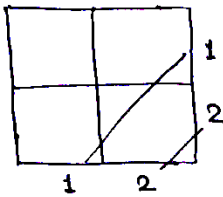


# Counting Figure

## 16. Counting Figures

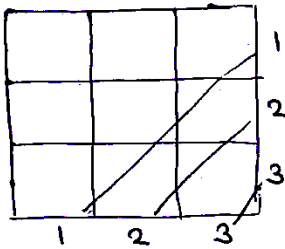
Square

1.



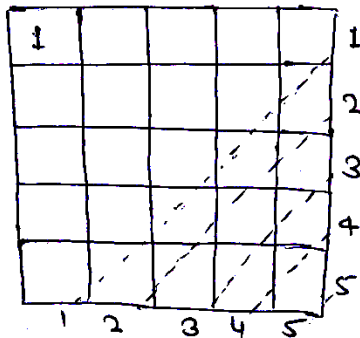
$$\begin{array}{r} 1 \times 1 = 1 \\ 2 \times 2 = 4 \\ \hline 5 \end{array} \quad \underline{\text{Ans}}$$

2.



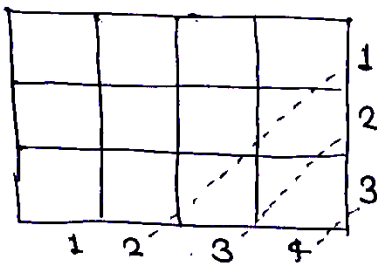
$$\begin{array}{r} 1 \times 1 = 1 \\ 2 \times 2 = 4 \\ 3 \times 3 = 9 \\ \hline 14 \end{array} \quad \underline{\text{Ans}}$$

3.



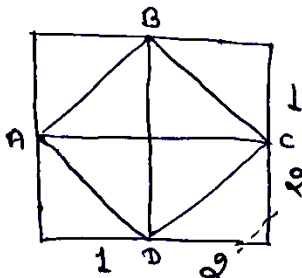
$$\begin{array}{r} 1 \times 1 = 1 \\ 2 \times 2 = 4 \\ 3 \times 3 = 9 \\ 4 \times 4 = 16 \\ 5 \times 5 = 25 \\ \hline 55 \end{array} \quad \underline{\text{Ans}}$$

4.



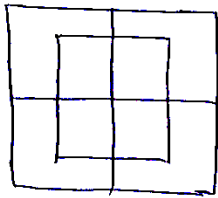
$$\begin{array}{r} 1 \times 2 = 2 \\ 2 \times 3 = 6 \\ 3 \times 4 = 12 \\ \hline 20 \end{array} \quad \underline{\text{Ans}}$$

5.



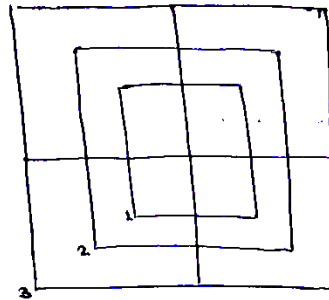
$$\begin{array}{r} 1 \times 1 = 1 \\ 2 \times 2 = 4 \\ \hline 5 \end{array} + 1 = 6 \quad \underline{\text{Ans}}$$

6.



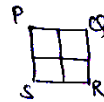
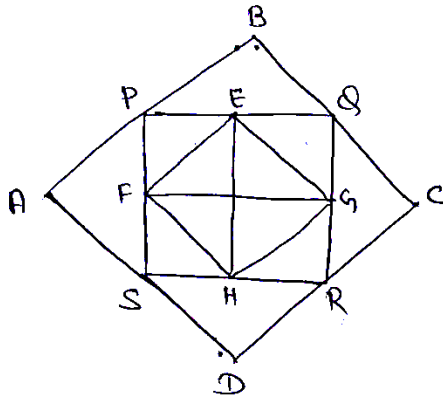
$$5 \times 2 = 10$$

7.



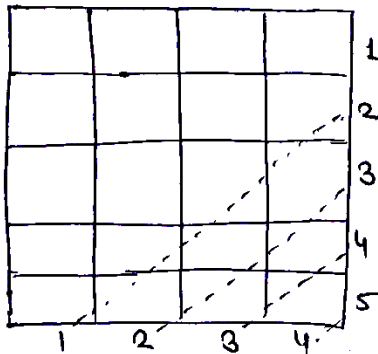
$$5 \times 3 = 15 \text{ Ans}$$

12.



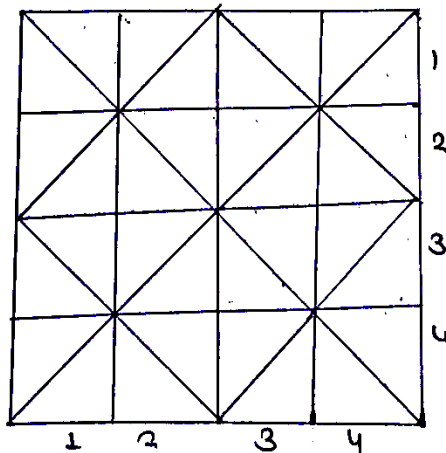
$$\begin{array}{rcl} & \rightarrow & 5 \\ +1 & \leftarrow & EFGH \\ +1 & \leftarrow & ABCD \\ \hline & & 7 \text{ Ans} \end{array}$$

13.



$$\begin{array}{rcl} 1 \times 2 & = & 2 \\ 2 \times 3 & = & 6 \\ 3 \times 4 & = & 12 \\ 4 \times 5 & = & 20 \\ \hline & & 40 \text{ Ans} \end{array}$$

14.



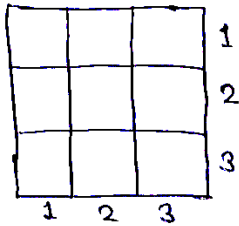
$$\begin{array}{rcl} 1 \times 1 & = & 1 \\ 2 \times 2 & = & 4 \\ 3 \times 3 & = & 9 \\ 4 \times 4 & = & 16 \\ \hline 30 + 5 & = & 35 \text{ Ans} \end{array}$$



$$\text{Diamond} = 5$$

## Rectangle

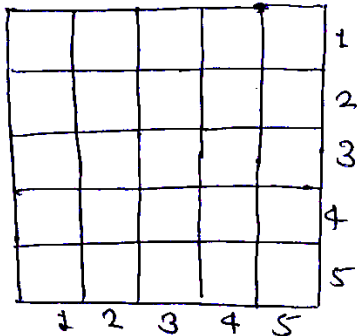
1.



$$(1)^3 + (2)^3 + (3)^3$$

$$1 + 8 + 27 = 36 \text{ Ans}$$

2.



$$(1)^3 + (2)^3 + (4)^3 + (5)^3$$

$$= 1 + 8 + 64 + 125$$

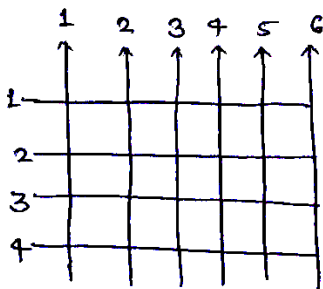
$$= 225 \text{ Ans}$$

4.

$$\text{No. of Rectangle} = n_1 C_2 \times n_2 C_2$$

$n_1 = \text{No. of Horizontal Line}$   
 $n_2 = \text{No. of Vertical line}$

3.



$$R = n_1 C_2 \times n_2 C_2$$

$$= {}^4 C_2 \times {}^6 C_2$$

$$= \frac{4 \times 3}{2} \times \frac{6 \times 5}{2}$$

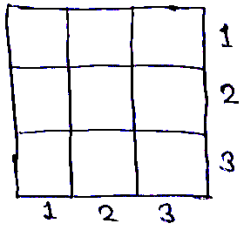
$$= 6 \times 15$$

$$= 90 \text{ Ans}$$

$$\left\{ \because n C_r = \frac{n!}{r!(n-r)!} \right\}$$

## Rectangle

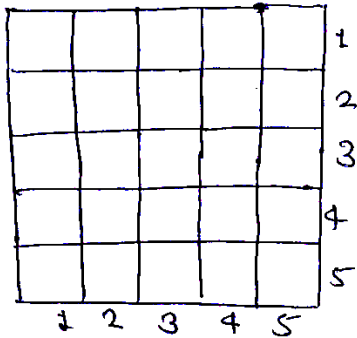
1.



$$(1)^3 + (2)^3 + (3)^3$$

$$1 + 8 + 27 = 36 \text{ Ans}$$

2.



$$(1)^3 + (2)^3 + (4)^3 + (5)^3$$

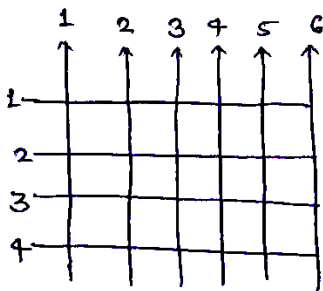
$$= 1 + 8 + 64 + 125$$

$$= 225 \text{ Ans}$$

4.

$$\left[ \begin{aligned} \text{No. of Rectangle} &= n_1 C_2 \times n_2 C_2 \\ n_1 &= \text{No. of Horizontal Line} \\ n_2 &= \text{No. of Vertical line} \end{aligned} \right]$$

3.



$$R = n_1 C_2 \times n_2 C_2$$

$$= {}^4 C_2 \times {}^6 C_2$$

$$= \frac{4 \times 3}{2} \times \frac{6 \times 5}{2}$$

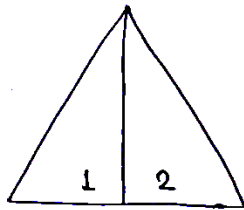
$$= 6 \times 15$$

$$= 90 \text{ Ans}$$

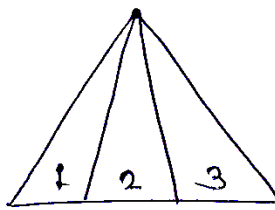
$$\left\{ \because n C_r = \frac{n!}{r!(n-r)!} \right\}$$

# Triangle

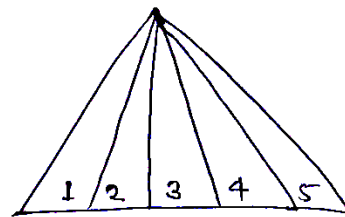
## Type-1



Trick =  $1+2 = 3$  Ans

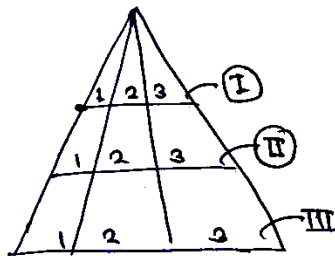


$1+2+3 = 6$  Ans



$1+2+3+4+5 = 15$  Ans

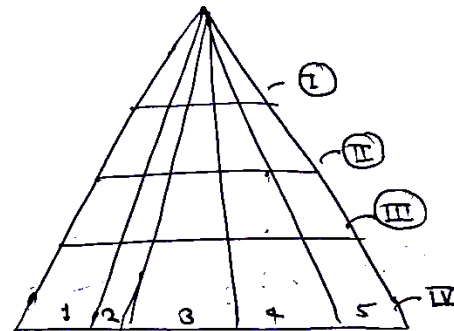
## Type-2



I  $\rightarrow 1+2+3 = 6$

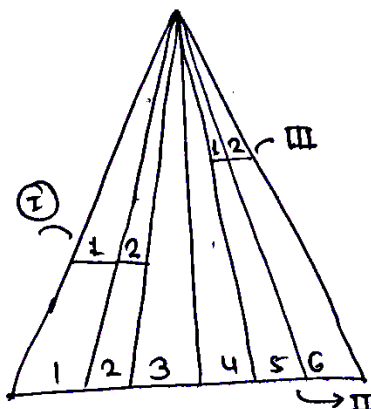
II  $\rightarrow 1+2+3 = 6$

III  $\rightarrow 1+2+3 = 6$   
18 Ans



$1+2+3+4+5 = 15$

total =  $15 \times 4 = 60$  Ans



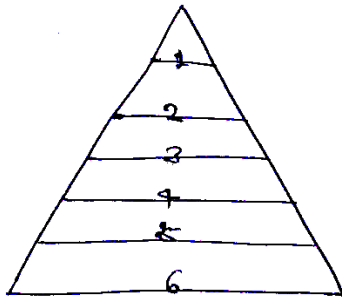
I  $\rightarrow 1+2 = 3$

II  $\rightarrow 1+2+3+4+5+6 = 21$

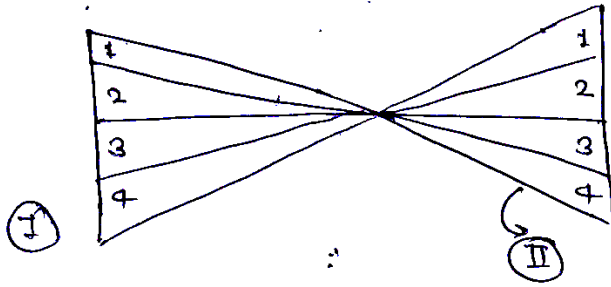
III  $\rightarrow 1+2 = 3$

Total triangle = I + II + III

= 27 Ans



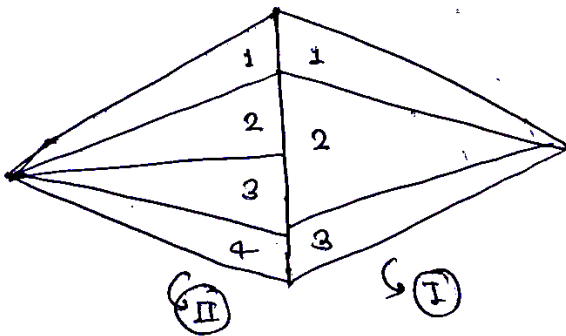
Triangle = 6



$$\textcircled{\text{I}} \rightarrow 1+2+3+4 = 10$$

$$\textcircled{\text{II}} \rightarrow 10$$

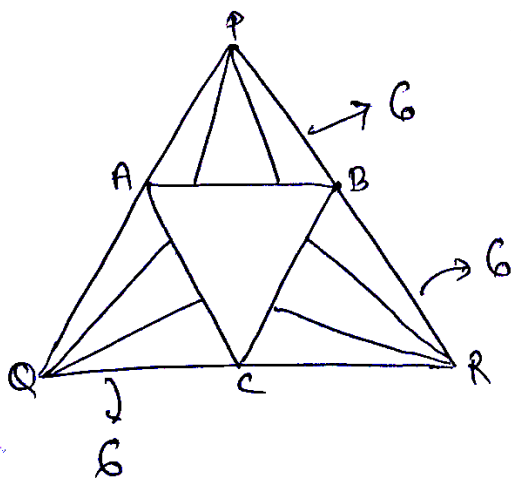
Total = 20 Ans



$$\textcircled{\text{I}} \rightarrow 1+2+3 = 6$$

$$\textcircled{\text{II}} \rightarrow 1+2+3+4 = 10$$

Total = 16

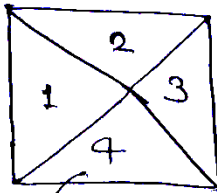


$$6 + 6 + 6 + \triangle ABC + \triangle PQR$$

$\Rightarrow$  20 triangle

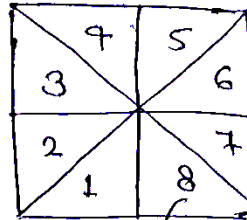
### Type-3

{SSC CGL-2020}



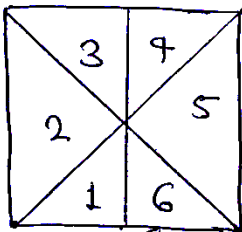
$$4 \times 2 = \boxed{8}$$

triangle

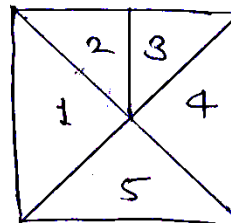


$$8 \times 2 = \boxed{16}$$

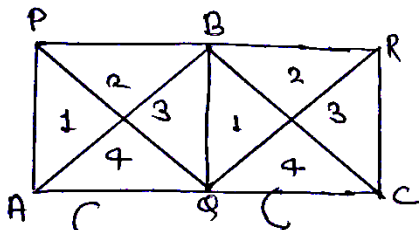
triangle Done



$$6 \times 2 = \boxed{12}$$

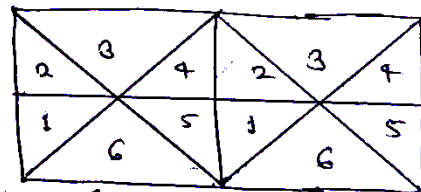


$$5 \times 2 = \boxed{10}$$



$$8 + 8 + \Delta ABC + \Delta PQR = \boxed{18}$$

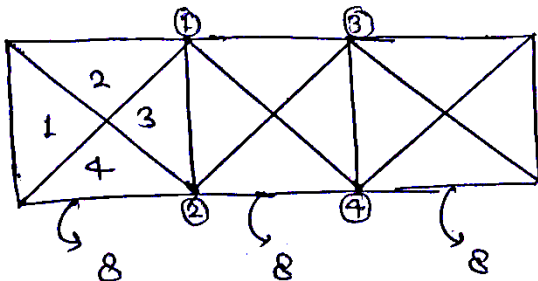
Done



$$6 \times 2 = 12$$

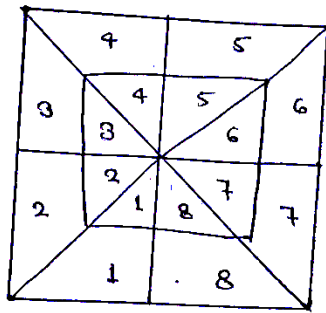
$$6 \times 2 = 12$$

$$12 + 12 + 4 = \boxed{28}$$



$$8 + 8 + 8 + 4 = \boxed{28}$$

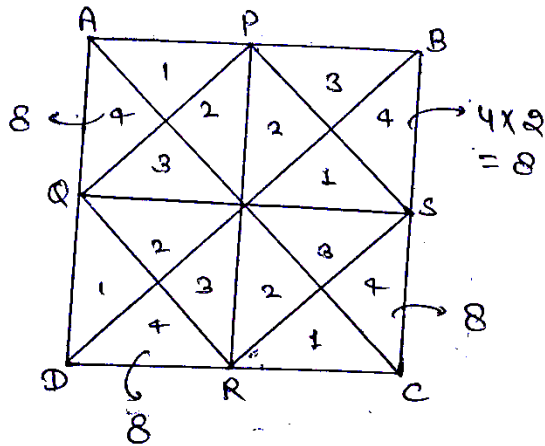
Maths with Gajanand



$$8 \times 2 = 16$$

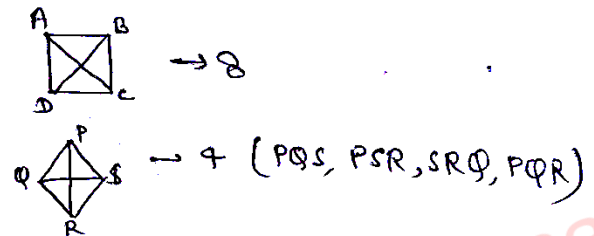
$$8 \times 2 = 16$$

$$\text{total} = \boxed{32 \text{ triangle}}$$

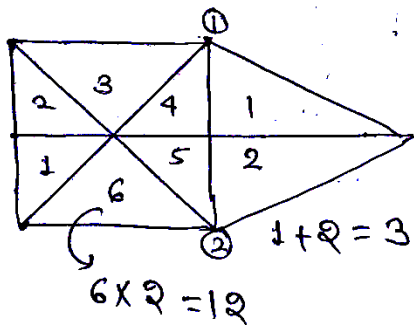


$$8 + 8 + 8 + 8 + 8 + 4$$

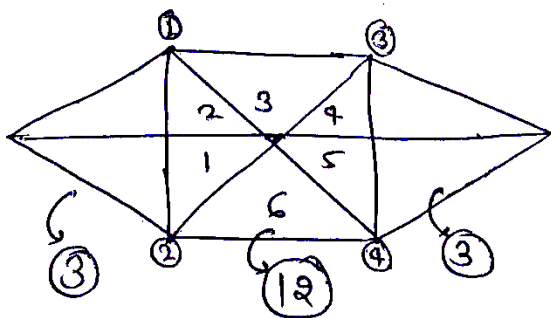
$$= \boxed{44} \text{ ans}$$



Mix type

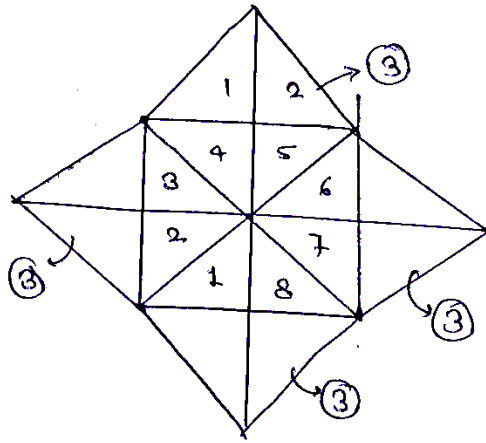


$$12 + 3 + 2 = \boxed{17} \text{ ans}$$

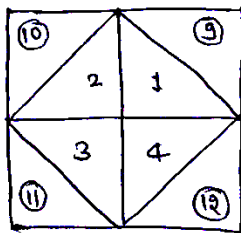


$$3 + 12 + 3 + 4$$

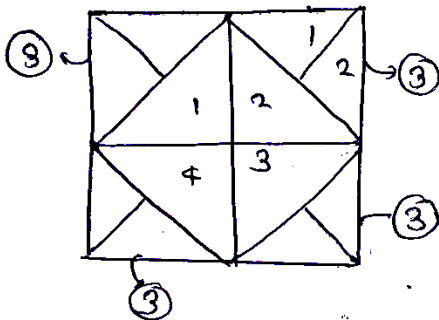
$$= \boxed{22} \text{ Triangle}$$



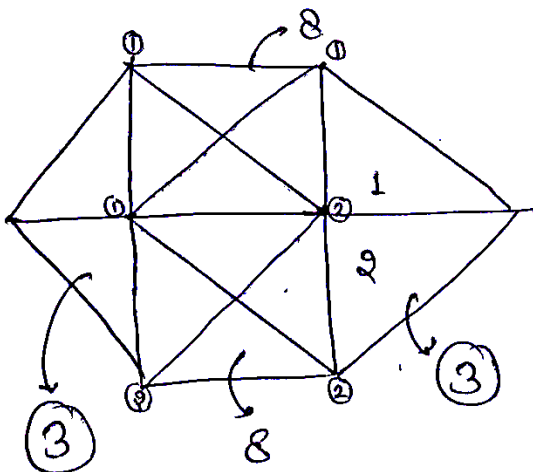
$$\begin{aligned}
 8 \times 2 &= 16 \\
 3 \times 4 &= 12 \\
 \text{Common} &= 8 \\
 \text{triangle} &= \boxed{36}
 \end{aligned}$$



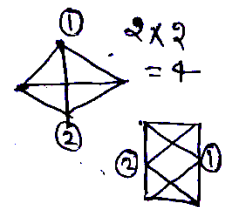
$$\begin{aligned}
 4 \times 2 &= 8 \\
 + 4 & \\
 \text{triangle} &:- \boxed{12}
 \end{aligned}$$

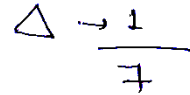
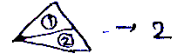
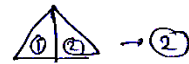
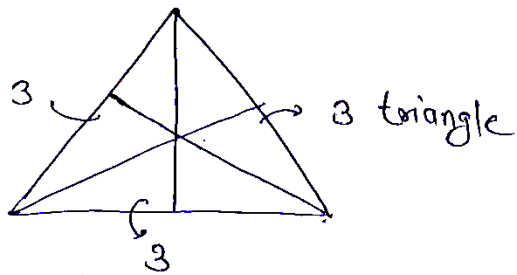


$$\begin{aligned}
 4 \times 2 &= 8 \\
 3 \times 4 &= 12 \\
 \boxed{20} & \text{ triangle}
 \end{aligned}$$

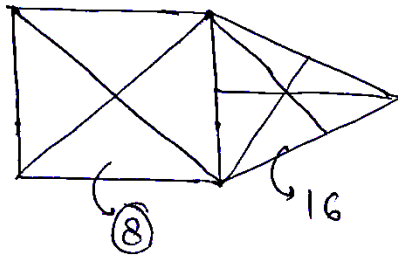


$$\begin{aligned}
 8 \times 2 &= 16 \\
 3 \times 2 &= 6 \\
 \text{Common} &= 2 \\
 &= 4 \\
 \boxed{28} & \text{ ans}
 \end{aligned}$$

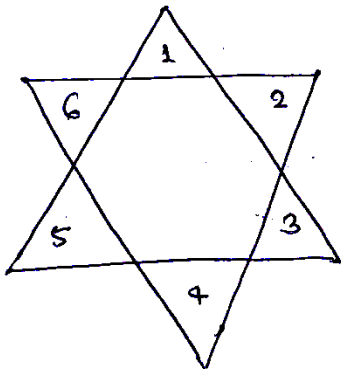
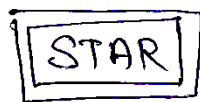




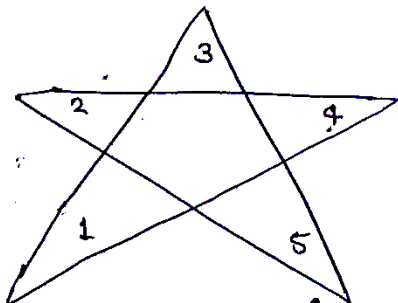
$$3 + 3 + 3 + 7 = \boxed{16} \text{ triangle}$$



$$8 + 16 = \boxed{24} \text{ triangle}$$



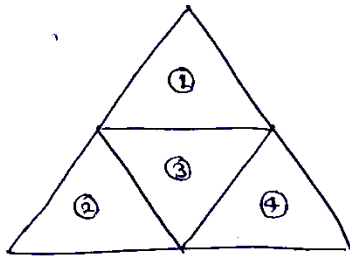
$$6 + \overset{\text{Big}}{1} 2 = \boxed{8} \text{ triangle}$$



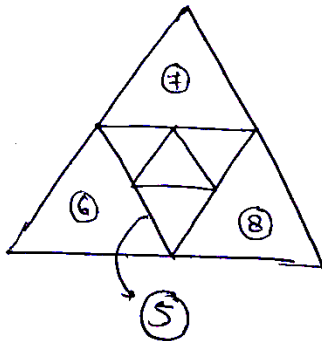
$$5 + 5 = \boxed{10}$$

Trick  $5 \times 2 = 10$  (Like Type-3)

# Type-4



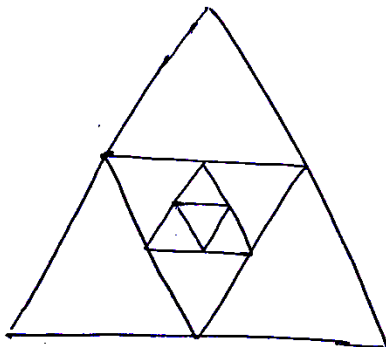
Complete  $\Delta$   
 $4 + 1 = 5$  triangle



Big  $\Delta$   
 $8 + 1 = 9$  triangle  
**OR**

There are 2 smaller  $\Delta$ 's and one larger  $\Delta$ .

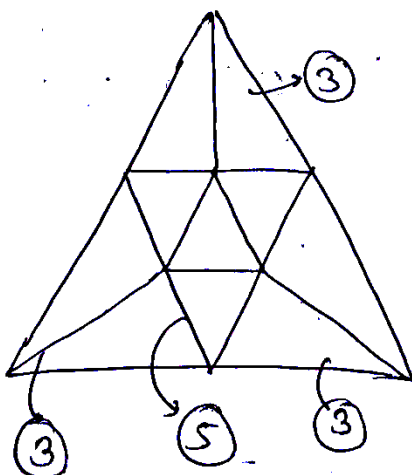
Total =  $\underbrace{4 + 4}_{2 \text{ small}} + \underbrace{1}_{\text{larger}} = 9$



There are 3 smaller  $\Delta$ s

Total =  $\underbrace{4 + 4 + 4}_{3 \text{ small}} + \underbrace{1}_{\text{larger}}$

Total = 13 triangle

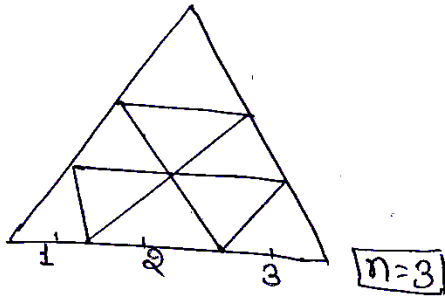


$5 + 3 + 3 + 3 + 1$   
 $\underbrace{\hspace{1cm}}_{\text{larger}}$

= 15 triangle

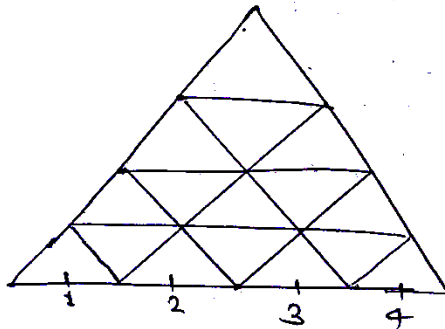
## # Type-5

Formula -  $\frac{1}{8} [n(n+2)(2n+1)]$  if  $n \rightarrow \text{even}$   
 $\frac{1}{8} [n(n+2)(2n+1) - 1]$  if  $n \rightarrow \text{odd}$

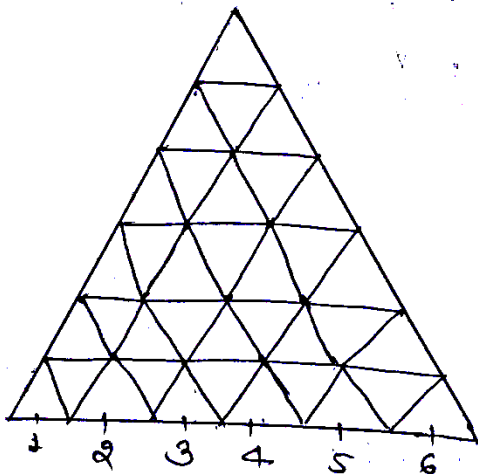


$$\frac{1}{8} [3 \times 5 \times 7 - 1]$$

$$\frac{104}{8} = 13 \text{ Triangle}$$



$$\frac{1}{8} (4 \times 6 \times 9) = 27 \text{ triangle}$$

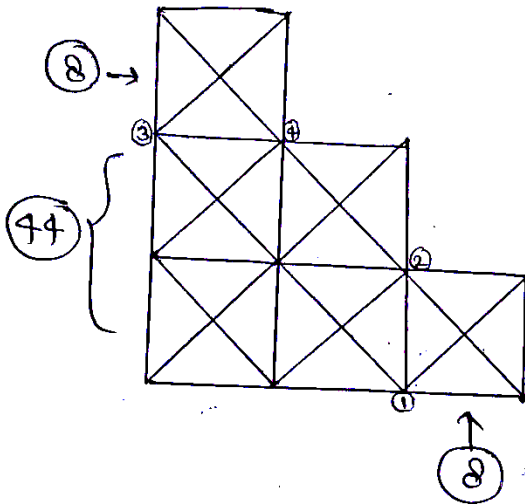


$$\frac{1}{8} (6 \times 8 \times 13) = 78 \text{ triangle}$$

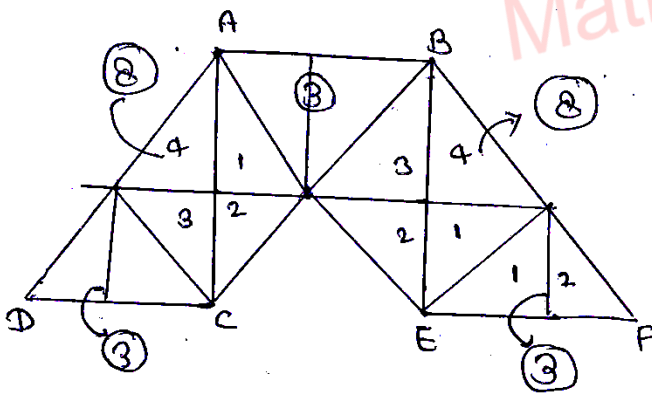
| $n$ | $\Delta$ |
|-----|----------|
| 1   | 1        |
| 2   | 5        |
| 3   | 13       |
| 4   | 27       |
| 5   | 48       |
| 6   | 78       |
| 7   | 118      |
| 8   | 170      |

Imp

# # Type-6

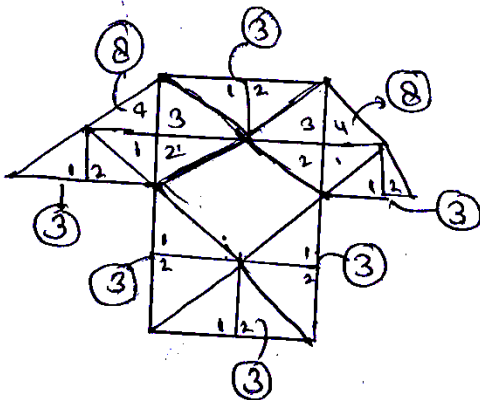


$$\begin{array}{r}
 44 \\
 8 \times 2 = 16 \\
 \text{other} = 9 \\
 \hline
 69 \text{ Ans}
 \end{array}$$



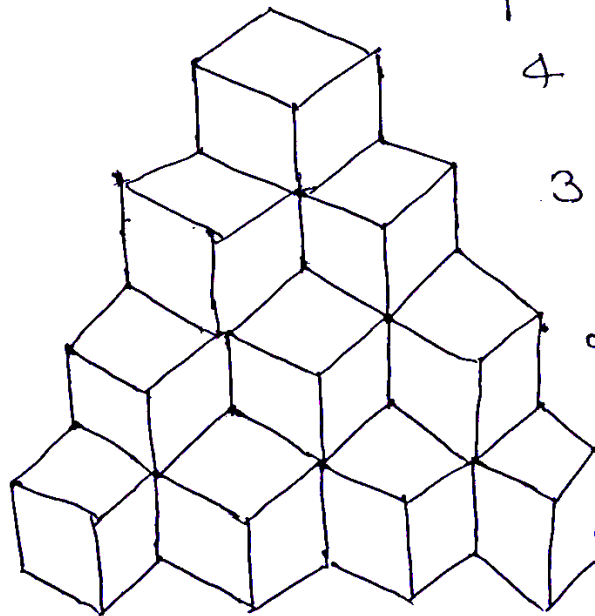
$$\begin{array}{r}
 8 \times 2 = 16 \\
 3 \times 3 = 9 \\
 \text{other} = 2 \times 2 = 4 \\
 \hline
 29 \text{ Ans}
 \end{array}$$

other 4  
 $\triangle ACD, \triangle ABC$   
 $\triangle BEF, \triangle ABE$



$$\begin{array}{r}
 8 \times 2 = 16 \\
 3 \times 6 = 18 \\
 \hline
 47 \\
 41 \text{ Ans}
 \end{array}$$

## Cube



Counting start



$$4 \times 1 = 4$$

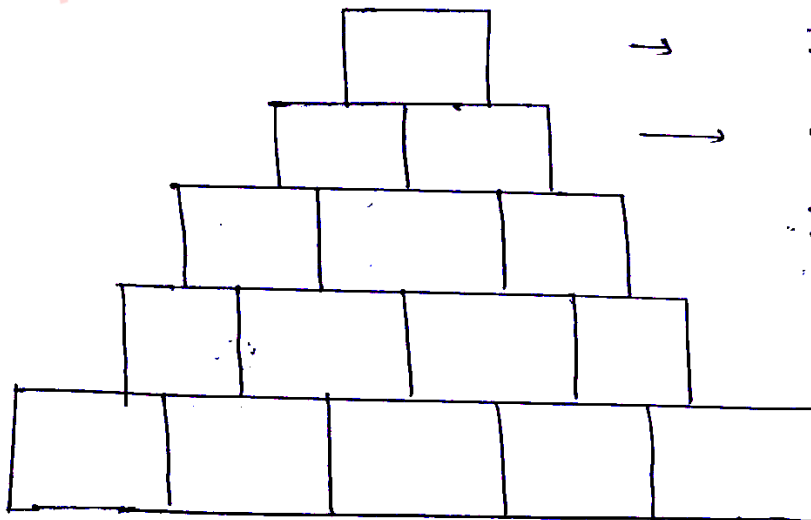
$$3 \times 2 = 6$$

$$2 \times 3 = 6$$

$$1 \times 4 = 4$$

$$\boxed{20} \text{ Ans}$$

Maths with Gajanand



$$1^2 \Rightarrow 1$$

$$2^2 \Rightarrow 4$$

$$3^2 \Rightarrow 9$$

$$4^2 \Rightarrow 16$$

$$5^2 \Rightarrow 25$$

$$\boxed{55} \text{ Ans}$$

# Analogy & Similarity

## 17. Analogy & Similarity

### Type-1

1. Fox : Cunning :: Rabbit : ?

- (a) Courageous (b) Dangerous ☒ Timid (d) Ferocious

2. Flexible : Rigid :: Confidence : ?

- ☒ Diffidence (b) Indifference (c) Cowardice (d) Scare

3. Sorrow : Goy :: <sup>(खुशी)</sup> Mirth : ?

- (a) Joy ☒ Laugh (c) Frown (d) Kleep

4. Love : Hate :: Proud : ?

- (a) Sorrowful (b) Miserable ☒ Humble (d) Conceited

5. Light : Darkness :: Knowledge : ?

- ☒ Ignorance (b) Intelligence (c) Brightness (d) Greatness

⇒ Light wards off Darkness. Similarly, knowledge wards off Ignorance.

6. Stethoscope : Heartbeat :: ? : temperature

- (a) Heat (b) Mercury (c) scale ☒ Thermometer

7. Spring : Summer ::

☒ Sunday : Monday

(b) Thursday : Wednesday

(c) Tuesday : Friday

(d) Friday : Monday

⇒ Summer comes after Spring. Similarly, Monday comes after Sunday.

8] Country : President :: State : ?

(a) Chief Minister

(b) Prime Minister

(c) Speaker

☒ (d) Governor

⇒ President is the head of union Executive.

9] Mirage : Desert :: ?

(a) Sky : Illusion

☒ (b) Rainbow : Sky

(c) Rain : Rainbow

(d) Image : Mirror

10] Fish : Scales :: Bear : ?

(a) Feathers

(b) Leaves

☒ (c) Fur

(d) Skin

[The body of fish remains covered with scales externally. Similarly, the body of bear remains covered with (fur)]

11] Writer : Pen :: ?

(a) Needle : Tailor

(b) Artist : Brush

☒ (c) Painter : Canvas

(d) Teacher : Class

12] Book : Author :: Statue : ?

(a) ~~Mason~~ Mason

(b) Painter

☒ (c) Sculptor

(d) Calligrapher

13] Life : Death :: Hope : ?

(a) Weep <sup>opposite</sup>

(b) Pain

☒ (c) Despair

(d) Sadness

14] Son : Nephew :: Daughter : ?

☒ (a) Niece

(b) Sister-in-law

(c) Sister

(d) Mother

15] Ring : Ear :: ? : Wrist

- (a) Pendant (b) chain ☒ (c) Bangle (d) Ribbon

16] Wood : Paper :: Steel : ?

- (a) Metal ☒ (b) Nail (c) Glass (d) Iron

17] Jewellery : Gold :: Furniture : ?

- ☒ (a) Wood (b) Paint (c) Table (d) Tree

18] Author : Novel :: Choreographer : ?

- (a) Make-up ☒ (b) Dance (c) Music (d) Picture

19] Food : Hunger :: Sleep : ?

- ☒ (a) Weariness (b) Night (c) Health (d) Dream

20] Doctor : Hospital :: Chef : ?

- (a) knife ☒ (b) kitchen (c) Food (d) Cook

21] Hen : Egg :: Tree : ?

- (a) Bark ☒ (b) seed (c) leaves (d) Fruit

22] Window : Pane :: Book : ?

- (a) Novel (b) Glass (c) Cover ☒ (d) Page

23] Prologue : Epilogue :: Dawn : ?

- (a) Dusk (b) Evening ☒ (c) Twilight (d) Morning

- 24] Sheep : Mutton :: Deer : ?  
 (a) Meat (b) flesh ☒ (c) Venison (d) Veal
- 25] Soil : Erosion :: Forest : ?  
 (a) Pollution (b) Degradation (c) Dense (d) Deforestation
- 26] Demographer : People :: Philatelist : ?  
 (a) Fossils ☒ (b) stamps (c) Photography (d) Music
- 27] Play : Actor :: Concert : ?  
 (a) Percussion (b) Symphony ☒ (c) Musician (d) Piano
- 28] Whale : Mammal :: Turtle : ?  
 (a) Amphibian ☒ (b) Reptile (c) Fish (d) Mollusc
- 29] Player : Team :: Minister : ?  
 (a) Army (b) Troupe ☒ (c) Council (d) Crowd
- 30] Lips : Smile :: Eyes : ?  
 (a) Black ☒ (b) Wink (c) Two (d) Spectacles
- 31] Ammeter : Current :: ? : ?  
 (a) Scale : Speed (b) Seismograph : Density  
 (c) Barometer : Mass ☒ (d) Anemometer : Wind
- 32] Pressure : Barometer :: ? : Odometer  
 (a) Humidity ☒ (b) Distance  
 (c) Thickness (d) Wind

## Type-2

1.  $1 : 8 :: 27 : ?$

- (a) 37 (b) 47 (c) 57 ~~(d) 64~~

$\Rightarrow (1)^3 : (2)^3 :: (3)^3 : (4)^3 \Rightarrow \boxed{64}$

2.  $24 : 126 :: 48 : ?$

$\Rightarrow 5^2 - 1 \quad 5^2 + 1 \quad 7^2 - 1 \quad 7^2 + 1 \Rightarrow \boxed{344} \text{ Ans}$

3.  $2 : 7 :: 6 : ?$

$\Rightarrow (2)^2 + 3 = 7, \quad (6)^2 + 3 = \boxed{39} \text{ Ans}$

4.  $47 : 65 :: 59 : ?$  ~~(a) 95~~ (b) 110 (c) 106 (d) 118

$\Rightarrow 4+7 = 6+5 = 11 \quad 5+9 = 5+9 = 14$

5.  $20 : 30 :: ? : 72$

- ~~(a) 56~~ (b) 59 (c) 68 (d) 61

$\Rightarrow 4 \times 5 = 20, 5 \times 6 = 30$  Similarly  $7 \times 8 = 56, 8 \times 9 = 72$

6.  $342 : 453 :: 831 : ?$

- (a) 942 (b) 943 (c) 941 (d) 842

$\Rightarrow$

7.  $11 : 119 : 15 : ?$

- (a) 154 (b) 223 (c) 233 (d) 152

$\Rightarrow (11)^2 - 2 = 119, \quad (15)^2 - 2 = 225 - 2 = \boxed{223} \text{ Ans}$

8)  $21 : 3 :: 574 : ?$

- (a) 23      (b) 82      (c) 97      (d) 113

$\Rightarrow \frac{21}{3} = 3$  similarly  $\frac{574}{?} = \boxed{82}$  Ans

9)  $212 : 436 :: 560 : ?$

- (a) 786      (b) 682      (c) 784      (d) 688

$\Rightarrow 436 - 212 = 224$        $560 + 224 = \boxed{784}$  Ans

10)  $7 : 48 :: 12 : ?$

- (a) 143      (b) 84      (c) 121      (d) 112

$\Rightarrow (7)^2 - 1 = 48$  ,  $(12)^2 - 1 = \boxed{143}$  Ans

11)  $64 : ? :: 72 : 53$

- (a) 44      (b) 54      (c) 52      (d) 70

$\Rightarrow$  (b) 54       $7+2=9$        $7 \ominus 2 = 1$   
 $5+4=9$        $5+3=8$   
 $6+4=10$        $\ominus = 1$

12)  $11 : 132 :: ?$

- (a) 10 : 100      (b) 9 : 90      (c) 13 : 169      (d) 15 : 250

$\Rightarrow (11)^2 + 11 = 132$  ,  $(9)^2 + 9 = 90$

13)  $20 : 11 :: 102 : ?$

- (a) 52      (b) 49      (c) 61      (d) 96

$\Rightarrow \frac{20+2}{2} = 11$  ,  $\frac{102+2}{2} = \boxed{52}$

14  $6415 : 5304 :: 7896 : ?$

- (a) 6705 (b) 6905 (c) 6907 (d) ~~6785~~

$\Rightarrow$

|                                                                                     |                                                                                                         |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| $\boxed{-1} \left( \begin{array}{c} 6415 \\ \downarrow \\ 5304 \end{array} \right)$ | $\boxed{-1} \left( \begin{array}{c} 7896 \\ \downarrow \\ \boxed{6785} \end{array} \right) \text{ Ans}$ |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|

15  $9536 : 6203 :: 5873 : ?$

- (a) ~~2540~~ (b) 2343 (c) 2353 (d) 2531

$\Rightarrow$

|                                                                                     |                                                                                             |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| $\boxed{-3} \left( \begin{array}{c} 9536 \\ \downarrow \\ 6203 \end{array} \right)$ | $\boxed{-3} \left( \begin{array}{c} 5873 \\ \downarrow \\ \boxed{2540} \end{array} \right)$ |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

16  $121 : 12 :: 25 : ?$

- (a) 1 (b) 2 (c) 6 (d) 7

$\Rightarrow$   $(11)^2 \xrightarrow{+1} 12$  Similarly  $(5)^2 \xrightarrow{+1} \boxed{6}$

17  $8 : 23 :: 48 : ?$

- (a) 90 (b) 138 (c) 168 (d) 112

$\Rightarrow$   $8 : 23 :: 48 : \boxed{23 \times 6} = \boxed{138}$

18  $2.25 : \frac{13}{4} :: 3.75 : ?$

- (a)  $\frac{17}{4}$  (b)  ~~$\frac{19}{4}$~~  (c)  $\frac{15}{4}$  (d)  $\frac{11}{4}$

$\Rightarrow$   $2.25 + 1 = \frac{13}{4}$  ,  $3.75 + 1 = \boxed{\frac{19}{4}}$

19  $5 : 130 :: 6 : ?$

$\Rightarrow$   $5^3 + 5 = 130$  ,  $(6)^3 + 6 = \boxed{222}$

20)  $6 : -3 :: -18 : ?$

- (a) 3 (b) 9 (c) -9 (d) -6

$\Rightarrow \frac{6}{-3} = -2$  Similarly  $\frac{-18}{-2} = \boxed{9}$

21)  $292 : 146 :: 582 : ?$

- (a) 272 (b) 286 (c) 291 (d) 292

$\Rightarrow \frac{292}{2} \rightarrow 146$  ,  $\frac{582}{2} \rightarrow \boxed{291}$

22)  $\frac{2}{3} : \frac{19}{29} :: \frac{8}{7} : ?$

- (a)  $\frac{89}{79}$  (b)  $\frac{79}{79}$  (c)  $\frac{79}{69}$  (d)  $\frac{80}{70}$

$2 \times 9 + 1 = 19 \Rightarrow \frac{19}{29}$  Similarly  $8 \times 9 + 7 = 79 \Rightarrow \frac{79}{69}$   
 $3 \times 9 + 2 = 29$   $7 \times 9 + 6 = 69$

23)  $2, 4, 2 : 4, 16, 4 :: 1, 1, 1 : ?$

- (a) 2, 4, 2 (b) 1, 1, 1 (c) 2, 2, 2 (d) 2, 13, 2

$\Rightarrow \begin{array}{c} 2, 4, 2 \rightarrow 4, 16, 4 \\ \begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \times 2 & \times 4 & \times 2 \end{array} \end{array}$  Similarly  $\begin{array}{c} 1, 1, 1 \rightarrow 1, 1, 1 \\ \begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \times 1 & \times 1 & \times 1 \end{array} \end{array}$

24)  $1, 2, 4, 7 : 3, 4, 6, 9 :: ? : 2, 3, 5, 8$

- (a) 0, 1, 3, 6 (b) 2, 4, 5, 8 (c) 1, 3, 4, 7 (d) 3, 5, 6, 8

$\Rightarrow \begin{array}{c} 1, 2, 4, 7 \rightarrow 3, 4, 6, 9 \\ \begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ +2 & +2 & +2 \end{array} \end{array}$  ,  $\boxed{0, 1, 3, 6} \rightarrow 2, 3, 5, 8$

### Type-3

1. AZCX : BYDW :: HQJO : ?

- (a) GRFP (b) IPKM (c) ☒ IPKN (d) GRJP

⇒

|                   |                   |
|-------------------|-------------------|
| A Z C X           | H Q J O           |
| +1   -1   +1   -1 | +1   -1   +1   -1 |
| B Y D W           | I P K N           |

2.  $\frac{ABC}{F} : \frac{BCD}{I} :: \frac{CDE}{L} : ?$

- (a) ☒  $\frac{DEF}{O}$  (b)  $\frac{DEF}{N}$  (c)  $\frac{EDF}{O}$  (d)  $\frac{DEF}{M}$

⇒

|                      |                      |
|----------------------|----------------------|
| F                  I | L                  O |
| -----                | -----                |
| +3                   | +3                   |

3. EXCUSE : CXEESU :: ERODES : ?

- (a) OREESD (b) ☒ ORESED (c) REOSDE (d) ERODSE

⇒

|             |             |
|-------------|-------------|
| E X C U S E | E R O D E S |
| X       U   | O       S   |
| C X E E S U | O R E S E D |

4. AB : L :: BC : ?

- (a) X (b) V (c) Y (d) W

⇒

|         |                                                                           |
|---------|---------------------------------------------------------------------------|
| A B     | B C                                                                       |
| ↓   ↓   | ↓   ↓                                                                     |
| 1 2 → L | 2 3 → <span style="border: 1px solid black; padding: 0 5px;">W</span> Ans |

5. JPC : GMZ :: PUV : ?

- (a) MRS (b) MSR (c) MQR (d) ☒ RMS

⇒

|               |               |
|---------------|---------------|
| J P C : G M Z | P U V : M R S |
| -----         | -----         |
| -3            | -3            |

6.  $\sqrt{AFI} = M : \sqrt{ADD} = L :: \sqrt{ABA} = ?$

- (a) I (b) N (c) O (d) ~~K~~

$\Rightarrow \sqrt{AFI} = \sqrt{169} = 13$  ,  $\sqrt{ABA} = \sqrt{121} = 11 \Rightarrow \boxed{K}$

7.  $\sqrt{AFI} : 13 :: \sqrt{ADD} : ?$

- (a) ~~12~~ (b) 22 (c) 21 (d) 24

$\Rightarrow \sqrt{169} = 13$  ,  $\sqrt{144} = \boxed{12}$

8.  $Y^2 : 4 :: V^2 : ?$

- (a) 9 (b) ~~25~~ (c) 49 (d) 16

$\Rightarrow Y \rightarrow B$   
 $\downarrow$   
 $2^2 \rightarrow 4$   
 $V \rightarrow E$   
 $\downarrow$   
 $5^2 \rightarrow \boxed{25}$

9. RUST : 9687 :: TSUR : ?

- (a) 7896 (b) ~~7869~~ (c) 7689 (d) 6789

$\Rightarrow$

|              |              |              |              |  |              |              |              |              |
|--------------|--------------|--------------|--------------|--|--------------|--------------|--------------|--------------|
| R            | U            | S            | T            |  | T            | S            | U            | R            |
| $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ |  | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ |
| 9            | 6            | 8            | 7            |  | 7            | 8            | 6            | 9            |

10. D x H : 4 x 8 , as M x Q : ?

- (a) 14 x 18 (b) 12 x 17 (c) 12 x 16 (d) ~~13 x 17~~

$\Rightarrow$

|              |   |              |  |              |   |              |
|--------------|---|--------------|--|--------------|---|--------------|
| D            | x | H            |  | M            | x | Q            |
| $\downarrow$ |   | $\downarrow$ |  | $\downarrow$ |   | $\downarrow$ |
| 4            |   | 8            |  | $\downarrow$ |   | $\downarrow$ |
|              |   |              |  | $\boxed{13}$ | x | $\boxed{17}$ |

# Arithmetical Problem

## 18. Arithmetical Problem

Type: Age related Question

**Que.]** Ashok's mother was 3 times as old as Ashok 5 year ago. After 5 year she will be twice as old as Ashok. How old is Ashok today?

**Soln:-** Basic:-

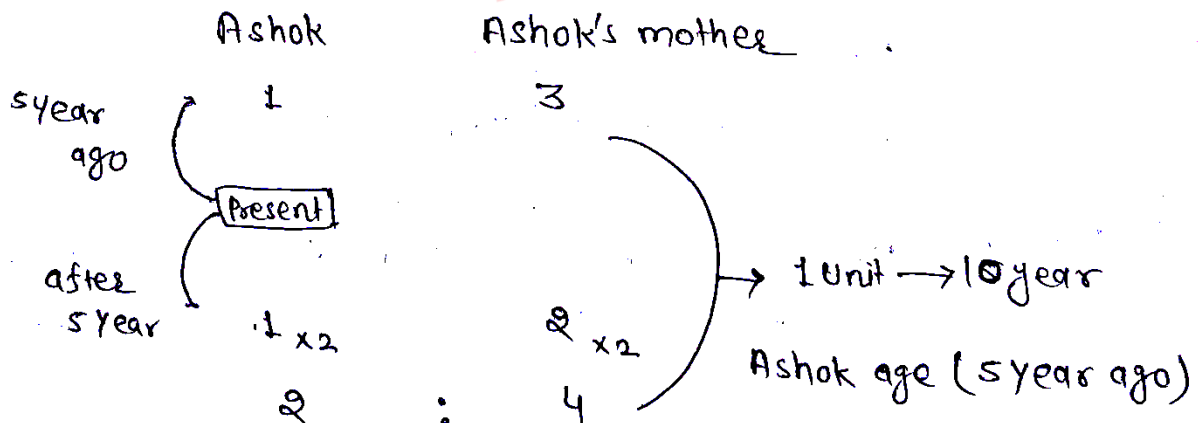
Let Present age of Ashok is  $x$  year and his mother is  $y$  years

then 5 year ago                      after 5 year

$$\left. \begin{array}{l} 3(x-5) = (y-5) \\ \Rightarrow 3x - y = 10 \end{array} \right\} \begin{array}{l} 2(x+5) = (y+5) \\ \Rightarrow 2x - y = -5 \end{array}$$

Solve  $x = 15 \text{ years}$

**Shortcut**



after 5 year  $\Rightarrow 10 + 5$

$\Rightarrow$   $15$

**Que. 2** A man is 3 years older than his wife and four times as old as his son. If the son becomes 15 years old after 3 years, what is the present age of the wife?

Soln:- Let the present age of son =  $x$

So age of father =  $4x$

after 3 years son becomes 15 years

so  $x + 3 = 15 \Rightarrow \boxed{x = 12}$

$\boxed{48}$

wife age =  $48 - 3 = \boxed{45}$  Ans

**Que. 3** The age of a father is twice that of his son's at present age. After 5 years the sum of their ages will be 85. How old are they now?

Soln:-

Son  
 $x$

father  
 $2x$

after 5 years:-  $x + 5 + 2x + 5 = 85$

son  $\rightarrow \boxed{x = 25}$  Year

father  $\rightarrow 2x = \boxed{50}$  Years Ans

**Que. 4** Nareesh's age is 4 years less than twice the age of his brother. Which of the following represents the equation to find his age?

(a)  $2x + 4$

(b)  $4x + 4$

(c)  $x - 4$

(d)  $2x - 4$

Soln

Let brother age is  $x$  years

then Nareesh's age =  $\boxed{2x - 4}$  Ans

**Que.5** A person's present age is two-fifth of the age of his mother. After 8 years, he will be one-half of the age of his mother. How old is the mother at present?

Sol:- Let mother's present age =  $x$  year

then

$$\begin{array}{l|l} \text{Person's age} = \frac{2x}{5} & \text{after 8 year} \\ & \left( \frac{2x}{5} + 8 \right) = \frac{(x+8)}{2} \\ & \frac{2x+40}{5} = \frac{x+8}{2} \\ & \boxed{x = 40 \text{ year}} \quad \text{Ans} \end{array}$$

**Que.6** Amit was born 5 years before Rakesh. Rakesh is 3 years younger to Anil. If Amit is now 17 years old, how old is Anil?

Sol:- Amit = 17 years

Rakesh =  $17 - 5 = 12$  years

Anil =  $12 + 3 = \boxed{15 \text{ years}}$

**Que.7** A father said to his son, "I was as old as you are at the present at time of your birth". If the father's age is 52 years now, then the son's age 8 years back was?

Sol:- Let present age of son =  $x$  years

" of father = 52 years

So

$$x + x = 52$$

$$2x = 52$$

$$x = 26 \text{ years}$$

$$\text{Son's age 8 years back} = 26 - 8 = \boxed{18 \text{ Years}}$$

**Q8** Adam who is 20 years old is 4 times as old as Mary. What will be Mary's age when Adam is twice as old as her.

Soln:- Adam age = 20 year  
 Mary age =  $\frac{20}{4} = 5 \text{ year}$

After 10 years

Adam = 30 year | Mary = 15 year Ans

**Que 9**

A was twice as old as B, two years ago. If the difference in their ages be 2 years, find A's age.

Soln:-

|             | A    | B   |
|-------------|------|-----|
| 2 years ago | $2x$ | $x$ |

$$2x - x = 2 \text{ Years}$$

$$\boxed{x = 2}$$

$$\text{A's age now} = 2x + 2 = \boxed{6 \text{ years}}$$

**Que 10** Mahesh is '60' years old, Ram is '5' years Junior to Mahesh and '4' years senior to Raju. The youngest brother of Raju is Babu and he is '6' years Junior to him. what is the age difference between Mahesh and Babu?

Soln:-

|               |   |                                      |
|---------------|---|--------------------------------------|
| Mahesh age    | = | 60 years                             |
| Ram           | = | $60 - 5 = 55 \text{ years}$          |
| Raju          | = | $55 - 4 = 51 \text{ years}$          |
| Babu          | = | $51 - 6 = 45 \text{ years}$          |
| Mahesh - Babu | = | $60 - 45 = \boxed{15 \text{ years}}$ |

Type :- Miscellaneous

**Que.1** Mr. and Mrs. Gopal have two daughter and each daughters has one brother. How many persons are there in the family?

- (a) 8                      (b) 6                      (c) 7                      ~~(d) 5~~

**Que.2** In a family Mr. Prakash has his wife and his two married brothers of whom one has two children and another has no issue. How many members are there in the family?

- (a) 12                      ~~(b) 8~~                      (c) 6                      (d) 10

**Que.3** In a group of equal number of cows and herdsmen the number of legs was 28 less than four times the number of heads. The number of herdsmen was.

Sol<sup>n</sup> let the number of cows =  $x$

Therefore " herdsmen =  $x$

then  $2x + 4x = 2x \times 4 - 28$

$$2x = 28$$

$$\boxed{x = 14} \quad \underline{\text{Ans}}$$

**Que.4** In a zoo, there are Rabbits and Pigeons. If heads are counted, there are 200 and if legs are counted, there are 580. How many pigeons are there

Sol<sup>n</sup> let Rabbits =  $x \rightarrow \text{leg} = 4x$   
Pigeons =  $y \rightarrow \text{leg} = 2y$

$$x + y = 200$$

$$4x + 2y = 580$$

$\rangle$  solve  $\boxed{y = 110} \quad \underline{\text{Ans}}$

**Ques 5** If 4 cats can kill 4 rats in 4 minutes, how many minutes will it take 8 cats to kill 8 rats?

Soln:-

$$\frac{4 \times 4}{4} = \frac{8 \times t}{8} \quad \left[ \frac{n_1 \times t_1}{w_1} = \frac{n_2 \times t_2}{w_2} \right]$$

$$t = 4 \text{ minute}$$

**Ques 6** In a double track rail route, two trains start simultaneously from the opposite stations, 200km apart. The train A runs westwards at a speed of 50 km/hr. and the opposite train B runs eastwards at 40 km/hr. After 3 hours, what will be the distance between these two trains?

Soln:- relative speed of A & B = 90 km/hr

$$\begin{aligned} \text{Distance covered in 3 hour} &= 90 \times 3 \\ &= 270 \text{ km} \end{aligned}$$

$$\begin{aligned} \text{Distance between the two trains} &= 270 - 200 \\ &= 70 \text{ km} \end{aligned}$$

**Ques 7** Excluding stoppages, the speed of a bus is 54 km/hr and including stoppages, it is 45 km/hr. For how many minutes does the bus stop per hour?

Soln:-

$$\begin{aligned} \text{Stoppage per hour} &= \frac{54 - 45}{54} \times 60 \\ &= 10 \text{ minutes} \end{aligned}$$

**Que. 8** A boat can travel with a speed of 30 km/hr in still water. If the speed of the stream is 6 km/hr, find the time taken by the boat to go 108 km downstream.

Soln:- Speed of boat downstream =  $30 + 6$   
 $= 36 \text{ km/hr}$

time =  $\frac{108}{36} = \boxed{3 \text{ hours}}$

**Que. 9** Two persons A and B get the same salary. Their basic pay are different. The allowances are 65% and 80% of the basic pay respectively. What is the ratio of the basic pay?

Soln:- Let Basic pay of A =  $x$   
 " " B =  $y$

then  $x \times \frac{165}{100} = y \times \frac{180}{100}$

$\frac{x}{y} = \frac{180}{165} = \frac{12}{11} \quad \boxed{12 : 11}$

**Que. 10** The price of an article has been reduced by 25%. In order to restore the original price, the new price must be increased by

Soln:-  $25\% \rightarrow \frac{1}{4}$



Increased =  $\frac{1}{5} \times 100 = \boxed{20\%}$

**Que.11** If in a business, Alok gains 75% more profit than Akash, then by what percentage profit of Akash is less than the profit of Alok?

Soln:-  $75\% \rightarrow \frac{3}{4}$

Alok      Akash  
7          4

$\frac{3}{7} \times 100 = \boxed{42.85\%}$

**Que.12** The ratio of the number of girls to the number of boys in a town is 90%. If the total number of boys and girls in the town is 190, how many girls are in the town?

Soln:-  $90\% \rightarrow \frac{\text{Girl}}{\text{Boys}} = \frac{90}{100}$       so the no. of girl =  $\boxed{90}$

**Que.13** Ann, Bill and Ken shared some stamps in the ratio 2:3:4. After a game the ratio became 5:2:2. If Ann won 21 stamps how many did Ken lose?

Soln  
 $21 \leftarrow \textcircled{3} \begin{pmatrix} 2 : 3 : 4 \\ 5 : 2 : 2 \end{pmatrix} \rightarrow 2 \rightarrow 7 \times 2 = \boxed{14}$   
 $7 \leftarrow 1$

**Que.14** The height of three towers are in the ratio 5:6:7. If a spider takes 15 minutes to climb the smallest tower, how much time will it take to climb the highest one?

Soln:-  
 $5 : 6 : 7$   
 $5 \rightarrow 15 \text{ minute}$   
 $1 \rightarrow 3 \text{ minute}$   
 $7 \rightarrow 7 \times 3 \text{ minute}$   
 $\boxed{21 \text{ minute}}$

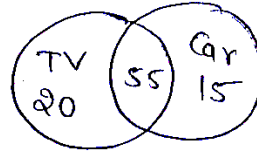
**Que.15** In a survey, 70% of those surveyed owned a car and 75% of those surveyed owned a TV. If 55% owned both a car and a TV, what percent of those surveyed did not own either a car or a TV?

Soln:-

$$\Rightarrow 100 - (20 + 55 + 15)$$

$$\Rightarrow 100 - 90$$

$$\Rightarrow \boxed{10\%}$$

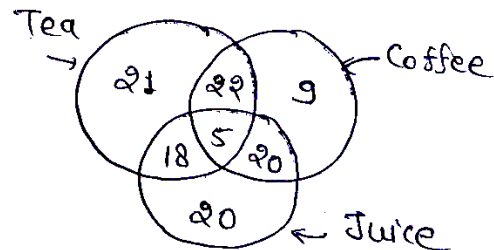


**Que.16** An official meeting is attended by 130 department employees. of them, 66 drink tea, 56 drink coffee and 63 drink juice, 27 can drink either tea or coffee, 25 can drink coffee or juice and 23 can drink juice and tea. 5 employees can drink any of the three. How many drink only tea?

Soln:-

$$\text{Total} = 130$$

21 employees drink only tea.



**Que.17** A bus leaves Delhi with half the number of woman as men. At Meerut, ten men get down and five women get down and five in. Now there are equal number of men and women. How many passengers boarded the bus initially at Delhi?

Soln:-

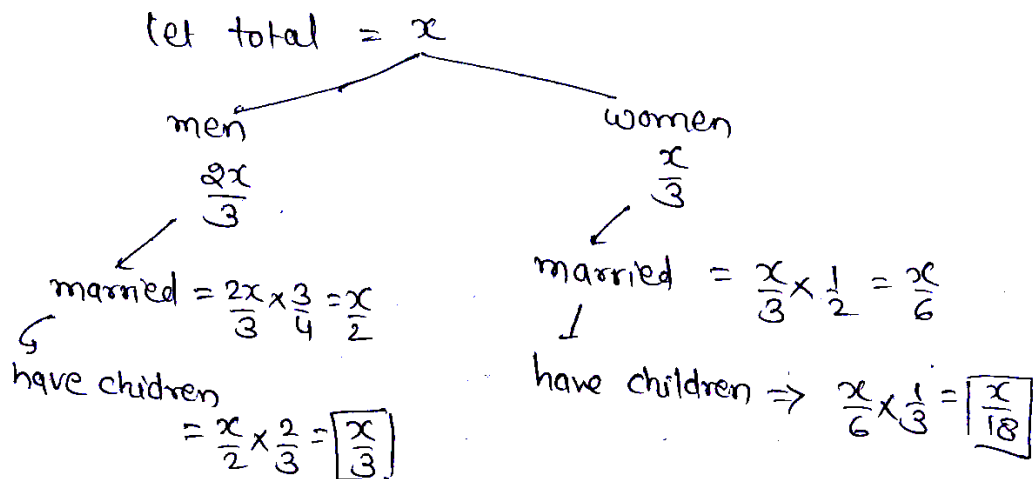
$$\begin{array}{lcl}
 \text{men} & & \text{women} \\
 2x & & x \\
 -10 & & +5 \\
 \hline
 2x-10 & = & x+5 \\
 \Rightarrow & & \Rightarrow \boxed{15} \\
 \boxed{30} & & 
 \end{array}$$

Total = 30 + 15 =  $\boxed{45}$  Ans

$\boxed{x = 15}$

**Que.18** In a certain office,  $\frac{1}{3}$  of the workers are women,  $\frac{1}{2}$  of the someone married and  $\frac{1}{3}$  of the married women have children. If  $\frac{3}{4}$  of the men are married men have and  $\frac{2}{3}$  of the married men have children, then what part of workers are without children?

Soln:-



Number of workers who have children =  $\frac{x}{3} + \frac{x}{18} = \frac{7x}{18}$

No. of workers without children =  $x - \frac{7x}{18} = \boxed{\frac{11x}{18}}$

**Que.19** A book has 300 pages and each pages has 20 lines of 10 words each, How many words are there in the book altogether?

Soln:-  $300 \times 20 \times 10 = \boxed{60000}$

**Que.20** The sum of a number and its reciprocal is  $\frac{17}{4}$ , find the numbers.

Soln:-  $x + \frac{1}{x} = \frac{17}{4}$   
 $= 4 + \frac{1}{4} \Rightarrow 4, \frac{1}{4} \underline{\text{Ans}}$

**Que. 21** If a number is greater than 5 but less than 9 and greater than 7 but less than 11, the number is  
(a) 5 (b) 6 (c) 7 (d) 8

**Que. 22** In a class composed of  $x$  girls,  $y$  boys what part of the class is composed of girls?

Soln  $\frac{x}{x+y}$

**Que. 23** In a row, 25 trees are planted at equal distance from each other. The distance from each other. The distance between 1<sup>st</sup> and 25<sup>th</sup> tree is 30m. What is the distance between 3<sup>rd</sup> and 15<sup>th</sup> tree?

Soln:- The distance between the two tree =  $\frac{30}{24} = \frac{5}{4} \text{ m}$

The distance between 3<sup>rd</sup> and 15<sup>th</sup> tree =  $\frac{5}{4} \times 12 = \boxed{15 \text{ m}}$

**Que. 24** A man climbing up a wall of 24 meters high. He climbs 16m on one day but slipped back by 3m 40cms in the evening. How far had the man reached on that day?

Soln:- Then man climbed =  $16 - 3.40$   
=  $\boxed{12.60 \text{ m}}$  Ans

Maths with Gajanand

## 19. Arrangement of Words in Logical Order

Type - 1

[Alphabetical order of words based]

Eg 1 (a) Spine (b) Spinal (c) Spinner (d) Spinet  
(e) Spindle

Soln (b) Spinal  
(e) Spindle  
(a) Spine  
(d) Spinet  
(c) Spinner

Eg 2 (a) Praise (b) Practical (c) Prank (d) Prayer  
(e) Practise

Soln :- (b) Practical  
(e) Practise  
(a) Praise  
(c) Prank  
(d) Prayer

Eg 3 (a) Hobby (b) Holiday (c) Hoarse (d) Hollow  
(e) Hobble

Soln :- (c) Hoarse, (e) Hobble (a) Hobby (b) Holiday (d) Hollow

eg 3 SSC 2017

Arrange the given words in the sequence in which they occur in the dictionary.

1. Yangtze      2. Yakking      3. Yakuzas      4. Yobbery  
5. Yobbish

- (a) 24531      (b) ~~2~~3145      (c) 54231      (d) 54321

Sol<sup>n</sup>

2. Yakking  
3. Yakuzas  
1. Yangtze  
4. Yobbery  
5. Yobbish

Maths with Gajanand

eg 4

1. Ropped      2. Roster      3. Roasted      4. Road  
5. Roller

- (a) 35412      (b) 45312      (c) 34512      (d) ~~4~~3512

Sol<sup>n</sup>

4. Road  
3. Roasted  
5. Roller  
1. Ropped  
2. Roster

Type-2

[Letter Word Problems]

In each of the following questions, find out how many such pairs of letters are there in the given word each of which has as many letters between them in the word as in the English alphabet.

[ऐसे कितने अक्षरों के जोड़े बनते हैं जिसमें प्रत्येक में उतने ही अक्षर बीच में हैं जितने कि अंग्रेजी की वर्णमाला है।]

Que 1. Necessary

- (a) 0      ☒ (b) 1      (c) 2      (d) 3

Soln:-

NECESSARY      (1)

Que 2 CREATIVE

- (a) 0      (b) 1      (c) 2      ☒ (d) 3

Soln:-

CREATIVE      (3)

Que 3 CREDIBLE

- (a) 0      ☒ (b) 4      (c) 2      (d) 3

Soln:-

CREDIBLE      (4)

Que 4

PRISON      (4)

### Type-3

### Letter-Series

Total - 26

First Half:- A B C D E F G H I J K L M

Second Half:- N O P Q R S T U V W X Y Z

**Que.1** Which letter is exactly midway between G and Q  
 (a) K (b) L (c) M (d) N

Soln:-

$$\begin{array}{ccc} G & Q & \\ 7 & + & 17 \end{array} \Rightarrow \frac{24}{2} = 12 \quad \boxed{L}$$

**Que.2** Which letter is midway between the eighteenth letter from the left and tenth letter from the right end of the alphabet series.

(a) No letter (b) K (c) Q (d) R

Soln Left  $\rightarrow 18^{\text{th}}$   
 Right  $\rightarrow 10^{\text{th}}$   
 Left  $\rightarrow (27-10) = 17^{\text{th}}$

midway  $\frac{18+17}{2} = \frac{35}{2}$  No letter

OR

$18^{\text{th}} \rightarrow R$

$17^{\text{th}} \rightarrow Q$

No letter

**Que.3** Which letter in the alphabet is as far from G as T is from M?

Soln:-

M 13 T 20  
 $+7$

G 7  $+7$   $\rightarrow$   $\boxed{14} \rightarrow \boxed{N}$  Ans

## 20. Word Formation

From the given alternatives select the word which cannot be formed using the letter of the given word.

[नीचे दिए गए निम्न विकल्पों में से वह शब्द चुनिए जो दिए गए शब्द के अक्षरों से नहीं बनाया जा सकता है।]

1 IMPASSIONABLE

(a) IMPASSABLE

(b) IMPOSSIBLE

☒ (c) IMPASSIVE

(d) IMPASSION

2 INDETERMINATE

(a) DETERMINE

☒ (b) RITINUE

(c) REMIND

(d) RETINA

3 CATERING

(a) ARGENTIC

(b) CREATING

☒ (c) RETIRING

(d) REACTING

4 HONESTY

(a) NEST

☒ (b) HONED

(c) HOST

(d) HONEY

5 ~~COLLABORATION~~ EMBOSSEMENT

(a) BOSS

(b) TOMB

(c) STEM

☒ (d) MATS

Form the given alternatives select the word which can be formed using the letters of the given word.

[ निम्न विकल्पों में से वह शब्द चुनिए, जो दिए गए शब्द के अक्षरों से बनाया जा सकता है ]

[6] OPERATION

(a) CAPTION

(b) ROTATION

☒ (c) OPTION

(d) NATION

[7] EXAMINATION

(a) EXAMINE

☒ (b) NAME

(c) MINOR

(d) GAME

[8] EXPERIENCE

☒ (a) EXPIRE

(b) PERCISE

(c) EMPIRE

(d) EXPENSE

[9] CONTROVERSY

(a) RIVER

☒ (b) STORY

(c) OTHER

(d) YOURS

[10] INFORMATION

(a) FARMER

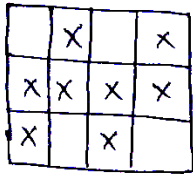
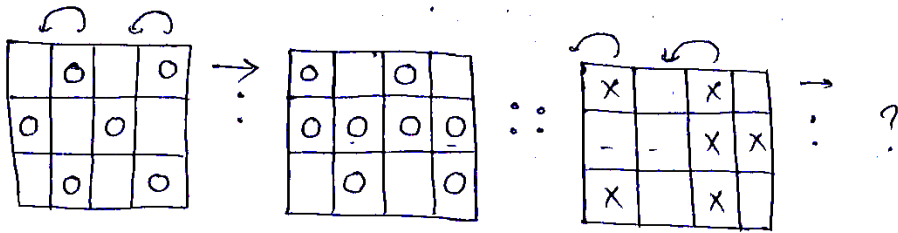
☒ (b) MOTION

(c) FIREMAN

(d) NAME

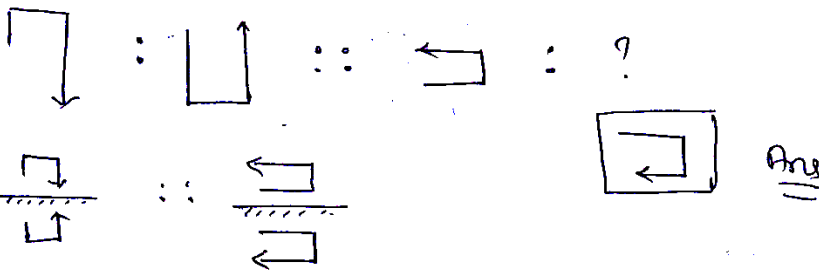
## 21. Analogy & Similarity

1

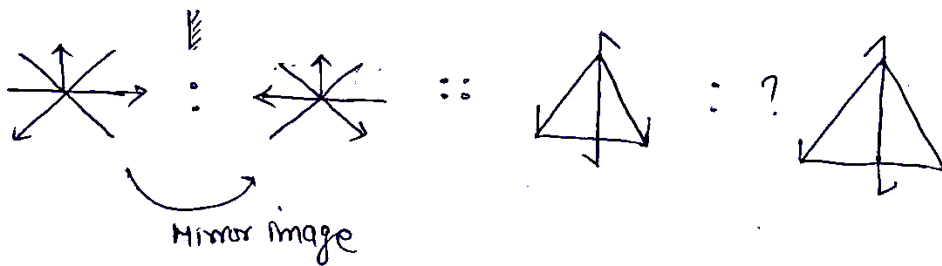


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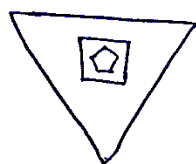
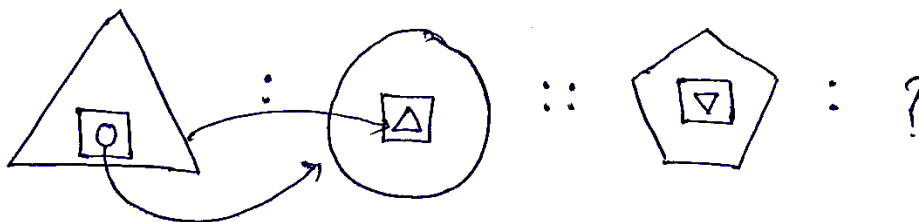
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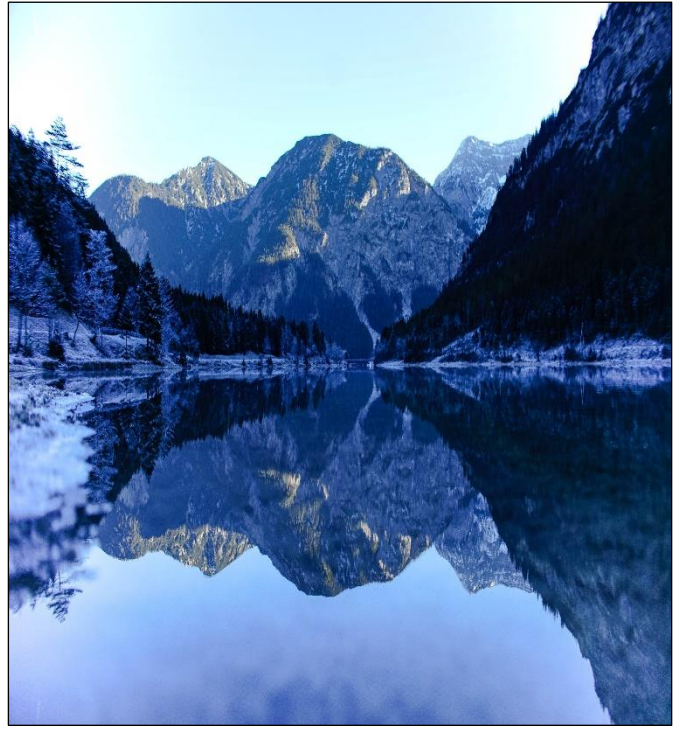


3



4





# Mirror & Water Image

# MIRROR AND WATER IMAGES

## 22. Mirror Image

Type-1 :- Letter mirror image :-

|     | M.I |     | M.I |     | M.I |
|-----|-----|-----|-----|-----|-----|
| (A) | A   | (I) | I   | (Q) | Q   |
| B   | B   | J   | J   | R   | R   |
| C   | C   | K   | K   | S   | S   |
| D   | D   | L   | L   | (T) | T   |
| E   | E   | (M) | M   | (U) | U   |
| F   | F   | N   | N   | (V) | V   |
| G   | G   | (O) | O   | (W) | W   |
| (H) | H   | P   | P   | (X) | X   |
|     |     |     |     | (Y) | Y   |
|     |     |     |     | Z   | Z   |

[No changes :- A, H, I, M, O, T, U, V, W, X, Y]

Example :-

MOBILE    ‖    ELIBOM  
 SUNDAY    ‖    YAUHNUZ  
 HOME    ‖    EMOH  
 NATIONAL    ‖    LAUOITAN

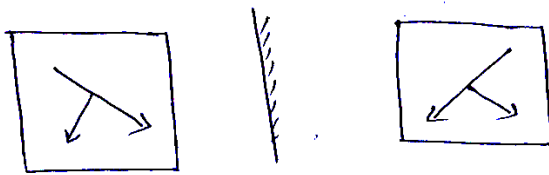
Type-2    Number Mirror Image :-

143    ‖    341  
 7890    ‖    0987  
 462    ‖    264

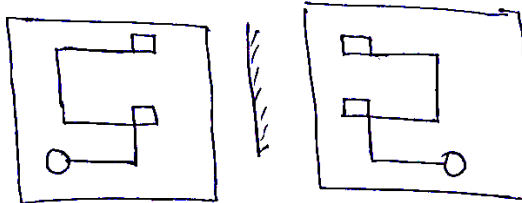
Maths with Gajanand

### Type-3) Geometrical/shape Mirror Image

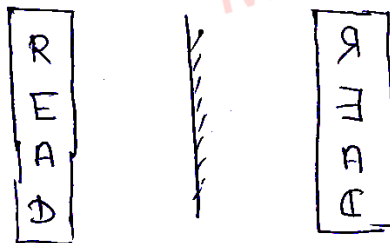
Eg.1



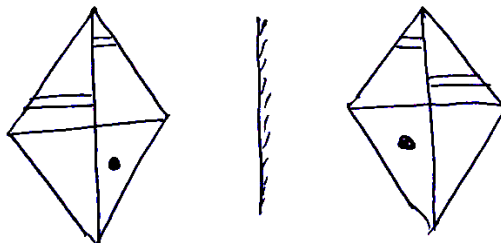
Eg.2



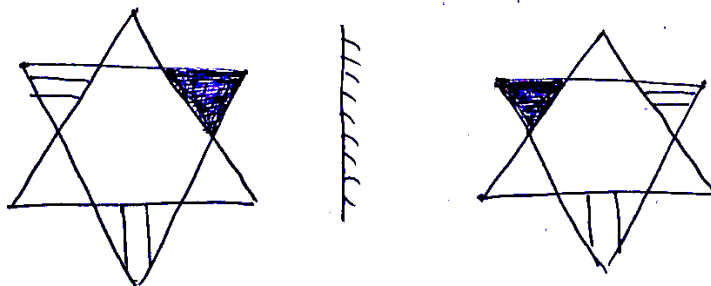
Eg.3



Eg.4



Eg.5



## WATER IMAGE

Type-1

W.I  
 A B C D E F G H I J K L M N O P  
 H B C D E F G H I J K L M N O P  
 Q R S T U V W X Y Z  
 Q R S T U V W X Y Z

No changes :- B, C, D, E, H, I, K, O, X

Example :-

APPLE  
 -----  
 VBBGE

DANIEL  
 -----  
 DANIIEΓ

WINCHESTER  
 -----  
 MIICHE2IEB

Type-2 Number water Image :-

example :-

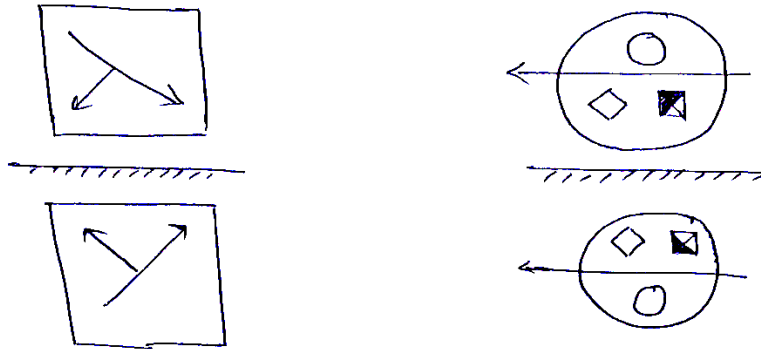
654  
 -----  
 e24

70239  
 -----  
 30532

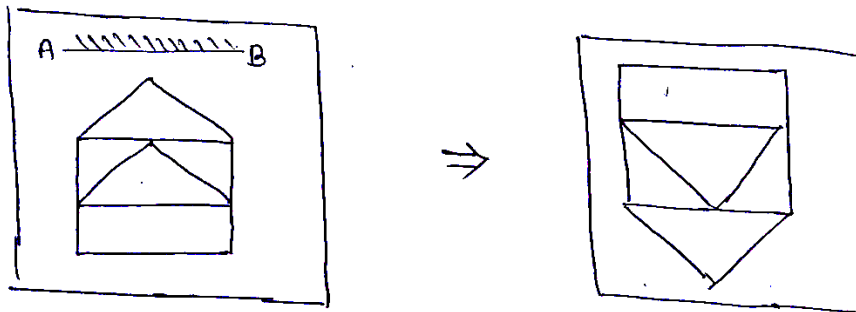
94  
 01  
 -----  
 04  
 24

## Type-3 Geometrical / shape water Image:-

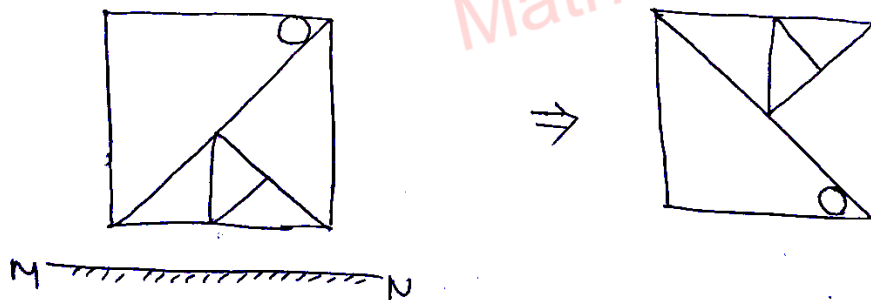
eg.1



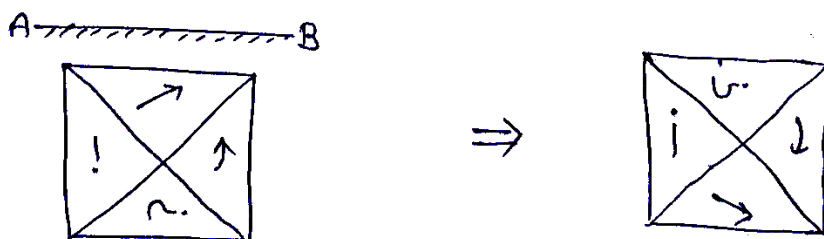
eg.2



eg.3

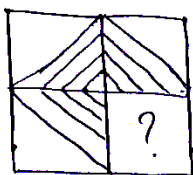


eg.4

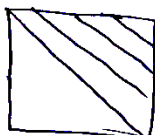


## 24. Completion of Figure

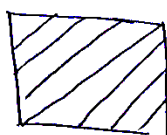
Que.



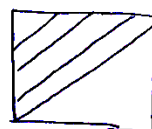
(a)



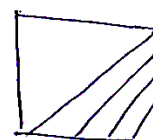
(b)



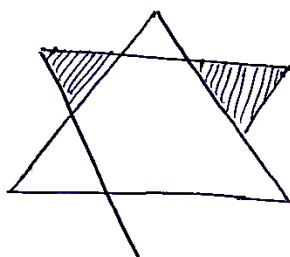
~~(c)~~



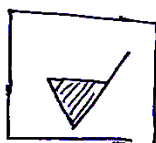
(d)



Que.



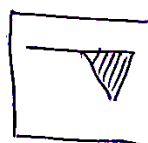
~~(a)~~



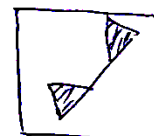
(b)



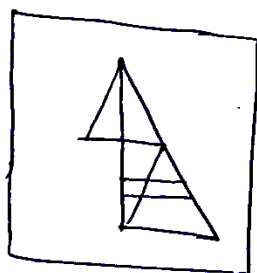
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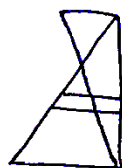
(d)



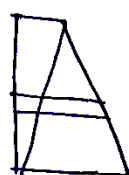
Que.



(a)



(b)



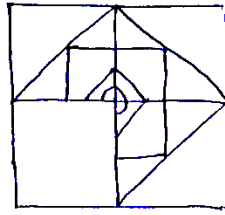
~~(c)~~



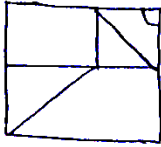
(d)



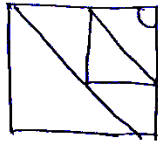
Que. 4



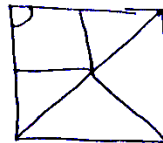
(a)



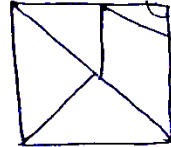
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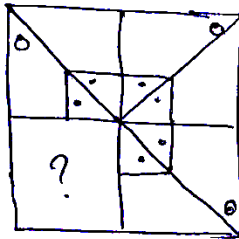
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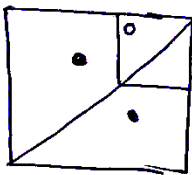
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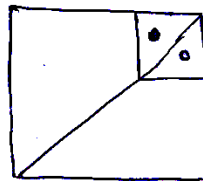
Que. 5



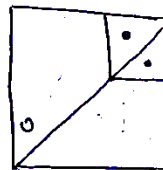
(a)



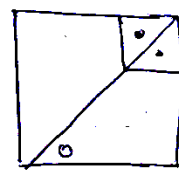
(b)



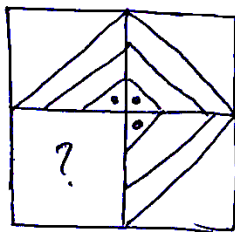
(c)



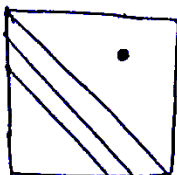
(d)



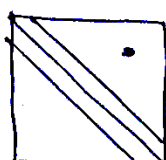
Que. 6



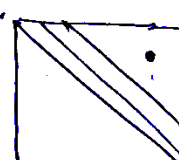
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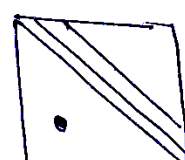
(b)



(c)

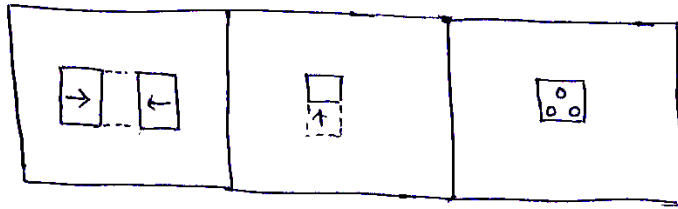


(d)

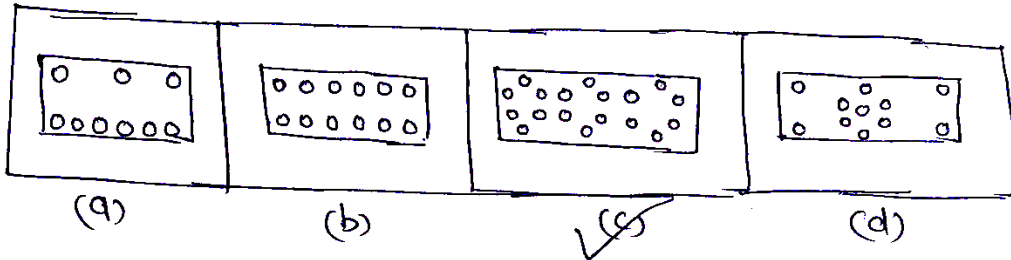


## 25. Paper cutting & Folding

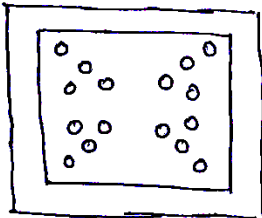
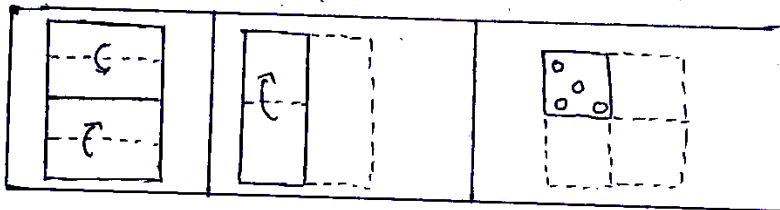
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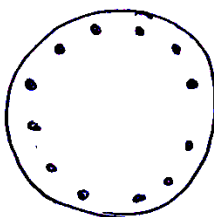
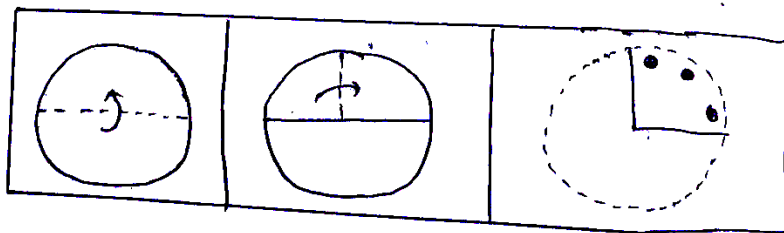
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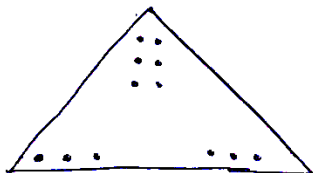
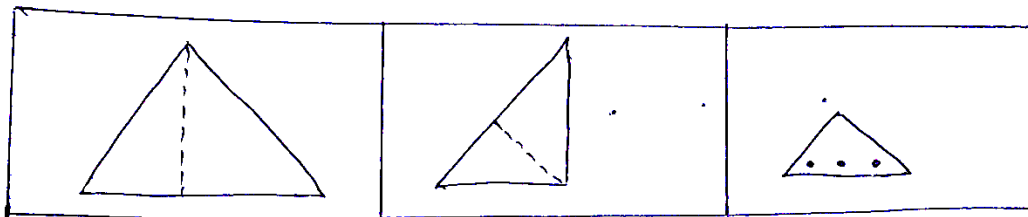
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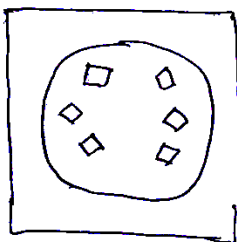
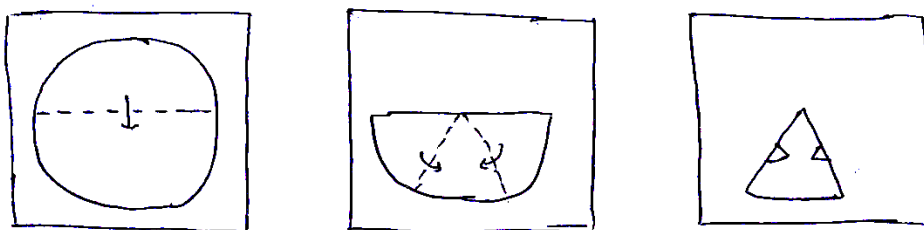
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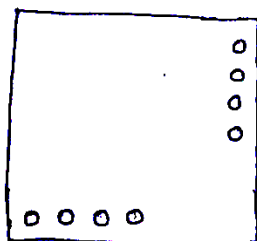
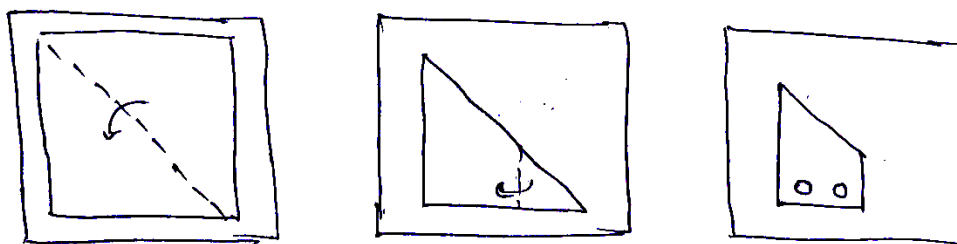


5



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6



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